

Who Killed the Green Fairy? Temperance, Language, and Wine in the Swiss Absinthe Ban of 1908

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Abstract

On July 5th, 1908, Swiss voters wrote a ban on absinthe into their federal constitution, the first national prohibition of the drink enacted by popular vote. The temperance movement led the campaign. Deputies and historians pointed to the wine industry, which stood to gain if drinkers switched to wine, but its part has not, to our knowledge, been measured. We measure it across 25 cantons at both stages: the petition that put the ban on the ballot, and the vote. Signatures per hundred eligible voters rose with the Protestant share, our measure of organized temperance, while wine is not detectable there. At the vote, the raw returns hide wine's part: the wine-growing cantons tended to be French-speaking, and French-speaking cantons gave the ban less support. Holding the French-speaking share fixed, a one-point higher share of national wine revenue goes with a 0.620-point higher yes-share. Both are significant at five percent under all four inference procedures. Under the same specification, wine is not detectable on six unrelated ballots and is positive, at ten percent by permutation test, on the 1906 food-law referendum. Temperance carried the petition; the wine interest appears at the ballot, as a beneficiary rather than a leader.

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1 Introduction

In 1910 Gantner's caricature for the Swiss satirical revue *Le Guguss* laid the green fairy out dead. A blue cross is driven into her chest, a black-robed figure wearing the emblem of the Blue Cross temperance society stands over her with a Bible, and the wreaths of Neuchâtel and Geneva, the only two cantons to vote against the ban, lie at her feet ([Gantner, 1910](#), reproduced in [Montagne, 2013](#), p. 509). The ban's opponents had named their killer.

On July 5th, 1908, Swiss voters wrote a ban on absinthe into their federal constitution ([Swissvotes, 1908](#)), the first national prohibition of the drink enacted by popular vote, mid-stream in a wave of absinthe's earliest bans running from 1906 to 1915 ([Royaume de Belgique, 1906](#); [Arnold, 1989](#), p. 116; [Lachenmeier et al., 2006](#), p. 367). The majority was itself a measure of the moment: 63.5 percent overall, and a local majority in all but two cantons ([Swissvotes, 1908](#); [Schweizerischer Bundesrat, 1908](#)). Both chambers carried it against the Federal Council's own recommendation to reject ([Schweizerischer Bundesrat, 1907b](#), p. 351).

Breadth of support, however, is not identity of authorship. That a majority said yes tells us little about which of the coalition's members, the temperance churches, the medical establishment, or the drink trades that competed with absinthe, moved votes, where, and by how much.

To contemporaries, little about the outcome required explanation. Absinthe was cheap, by the campaign's own reckoning ([Studer, 2024](#), p. 126), and it carried a diagnosis of its own: absinthism, held to be a pathology distinct from ordinary drunkenness, attributed to the essence of wormwood.¹ Nor was the instrument novel: the Confederation had curbed the rivals of the "natural" drinks since the alcohol monopoly of 1885–87 ([Swissvotes, 1885, 1887](#); Section 2.2), and the pro-ban press set the initiative in that line of standing federal restrictions ([Journal de Genève, 1908](#), 3 July 1908, p. 1). The public-health premise does not bear the weight the conventional account places on it, and the era itself knew the case was contested (Section 2.3). Modern analysis finds nothing in pre-ban absinthe at obtainable doses that distinguishes it from chronic alcohol abuse ([Lachenmeier et al., 2008](#), p. 3073; [Padosch et al., 2006](#), p. 11).

A yes-or-no vote buries the fact that both camps pointed beyond absinthe to other drinks. The Alcohol Administration's own expert forecasted that other spirituous beverages "would take the place of the prohibited absinthe" ([Milliet, 1907](#), p. 365), and on the temperance side an abstainers' paper, the Alkoholgegnerbund's *Freiheit*, had conceded in its No. 11

¹By the campaign years, it was attributed more narrowly to its ketone thujone.

of 1905, before the federal initiative was launched, that absinthe was “by no means the most harmful drink for us Swiss” and that “there are worse things to fight,” while urging its readers to fix on “what we can already achieve now” (Trechsel, 1989, p. 153). A policy aimed at one drink while both camps pointed to others is an insufficient explanation for the ban’s majority support across all but two cantons.

We study the two canton-level records the initiative left, the petition that put the ban on the ballot and the vote itself, across the twenty-five cantons: a cross-section estimated by least squares with the French-speaking share held fixed, each estimate read under four small-sample inference procedures, among them a Freedman–Lane permutation test. Signing was voluntary, so the petition, measured in signatures per hundred eligible voters, records organized mobilization, and the vote the choice of the electorate at large (Section 3.1.1). The petition was filed with 167,814 valid signatures (Schweizerischer Bundesrat, 1907b, p. 353), the largest signature count then recorded (Trechsel, 1989, p. 152). Switzerland’s federal structure produces variation across the cantons in wine production, confessional composition, language and direct exposure to absinthe manufacture, and the canton is the unit at which the vote was decided under the double majority, of voters nationally and of cantons.

We find that, once language is held fixed, a canton’s stake in the national wine harvest predicts its support for the ban: a one-point higher share of the national wine harvest’s revenue goes with a 0.620-point higher yes-share (standard error 0.232), and the estimate holds under each of the four inference procedures. The conditioning is not a specification choice made for convenience: language overlaps both the wine geography and the absinthe geography, the cross-section reverses sign without it, and the cultural divide therefore masks the within-bloc relation rather than explaining it away.

The petition’s results differ. Signatures per hundred eligible voters rise with the Protestant share, our measure of standing temperance organization (0.174, standard error 0.065), and the wine share is not detectable there. This stage-separation pattern is the central refinement our cross-canton evidence brings to existing readings of the 1908 vote: the perspective that it was mostly a cultural, moral, or health-related decision is partly supported, but it ignores how closely wine production overlaps the linguistic cleavage.

We then ask what distinguishes this vote from other federal ballots of the period. Under the same specification the wine coefficient is not detectable on six ballots whose subjects had nothing to do with drink, while the French-speaking share often is. It returns on the food act of 1905, confirmed by referendum in June 1906, where wine was directly at stake: the shared specification gives a positive coefficient at the ten percent line (.083), and

the Act's own legislative record points at the wine interest directly, the wine merchants' associations having asked, as Section 5.2.1 sets out, for an outright ban on artificial wine ([Schweizerische Bundesversammlung, 1905a](#), pp. 393, 1050–1051).

One feature of the record is left standing by a purely cultural account. The absinthe-producing cantons voted lower in the raw returns, and the only two cantons to reject the ban, Neuchâtel and Geneva, were producing cantons, as [Heimberg \(2000, p. 105, n. 12\)](#) had already noted, and the country's largest producers ([Swissvotes, 1908](#); [Milliet, 1907](#), pp. 360–361). The Council's December 1907 message carries an annex by E. W. Milliet, director of the Alcohol Administration, listing the forty absinthe houses known to it, with each one's five-year purchases of monopoly spirit and its exports ([Milliet, 1907](#), pp. 360–361), which we check house by house against the period records as far as they allow (Section 3.1.4; Appendix A.3). With four producer cantons, three of them French-speaking, the producer indicator is not detectable, and we do not read that as absence of resistance. We do not offer it as proof. We offer it as the reason a third factor had to be measured rather than asserted.

The role of the wine interest has been asserted in passing, doubted in passing, and left open by careful authors for want of evidence; to our knowledge no one has estimated it. Three readings of the 1908 vote can be distinguished in the existing literature. The first treats wine-industry interests as a material contributing factor. [Cahannes \(1981, pp. 44–45\)](#), in the closest existing analysis of the Swiss case, concludes that the initiative passed partly because absinthe competed with white wine and the winegrowers therefore supported it (Section 2.4). [Padosch et al. \(2006, p. 4\)](#) name the winegrowers among those calling for the ban in France, and [Prestwich \(1979, p. 319\)](#) characterizes the banned drink as “less a villain than a convenient victim” of a coalition that combined hygienic concern with material interest. The second reading treats the temperance movement and the cultural cleavage as effectively sufficient: much of the medical and cultural-history literature takes the Lanfray case ([Studer, 2024, p. 178](#)) and the Protestant revival mobilization ([Trechsel, 1989, pp. 29–30](#)) as adequate explanations, and the Federal Council itself characterized the petition as “a hurried, little-thought-through legislative attempt by the bearers of the temperance movement” ([Schweizerischer Bundesrat, 1907b, p. 351](#)). The third treats the two as jointly necessary, with the relative weight of each unsettled in the absence of quantitative evidence. [Cahannes \(1981, p. 44\)](#) is a careful proponent of it, qualifying her own wine attribution with the observation that “the motives underlying this initiative were without doubt a concern not only for public order but also for the physical and mental health of the population,” and [Studer \(2024, p. 18\)](#) treats the campaign as similarly multi-causal. The analysis here builds on the third, which we argue is under-studied.

Two modern-era treatments read the geography of the result from the raw returns. [Heimberg \(2000, p. 99\)](#) holds that the prohibition “did not really touch major economic interests” and was decided by German-speaking regions little affected by absinthe consumption, and [Auderset \(2021, pp. 152–153\)](#) reads the returns the same way. Both readings are fair on the raw returns and neither was offered as a test. Studer’s history names the pressure of the French wine lobby among the causes of the 1915 ban but treats the Swiss vote narratively, with no canton-level analysis ([Studer, 2024, p. 18, ch. 7](#)); the wine calculation appears elsewhere as authorial supposition, the growers having “deduced that absinthe could easily be sacrificed” ([Adams, 2004, p. 197](#)). Popular histories state the calculation with less caution, placing wealthy wine producers behind the scenes of the bans ([Hicks, 2010](#)) or counting the wine industry, “go figure,” among absinthe’s biggest enemies ([Clifford, 2023](#)). Neither ties the claim to the Swiss vote, and neither gives a source for it.

Popular votes on prohibition elsewhere in the same era give the nearest comparisons. In Canada’s plebiscite of 1898, [Dostie and Dupré \(2012, pp. 511–512\)](#) find beer and spirit production not detectable. In the American statewide votes of 1907–1919 that [Lewis \(2008, pp. 389, 393–394\)](#) studies, on laws that, as he describes them, prohibited wine with the other drinks, all but four sparing sacramental wine, breweries are detectable and grape-growers are not detectable in the dry share. Ours was a vote on one spirit that left wine on sale. To our knowledge this is the first quantitative analysis of the 1908 Swiss absinthe vote, and it joins a short list: outside the United States, [Dostie and Dupré \(2016, pp. 493–494\)](#) could find only one earlier empirical study of a prohibition referendum, their own on the Canadian vote, and added a second on New Zealand’s votes of 1911–1922.

We constrain our analysis to what the evidence can answer without conspiratorial allegations. Where every written account is an interested account, testimony cannot arbitrate among motives: the government that recommended rejection also sold absinthe’s makers their spirit, through its own alcohol monopoly (Sections [2.2](#) and [2.5](#)), and the opposition charged, on one faction’s stated ambitions, that the ban was a first step toward total prohibition ([Gantner, 1908](#)). It’s superficial to argue merely that markets play a role when historical accounts witness that the prices of spirits and wine are considerations. The tractable questions are how market interests were involved, and by extension, whether their involvement is legible in the cross-canton pattern, granting from the outset that the temperance movement led the campaign and that the cultural cleavage is real. Both can be true. Put as a logical condition, we argue that neither of two statements holds, because each ignores the other: that cultural attitudes and popular consensus drove support for the ban and markets played no important part; and that markets drove it and culture played

no important part. A movement can lead a coalition whose success serves an interest that never took the podium. Writing the temperance movement's own history, [Trechsel \(1989, p. 153\)](#) records the order of events: the abstainers "climbed aboard as a free rider on a movement that started from a winegrowing village." What we measure is the political weight of a producer interest at the ballot. We make no claim that wine sales rose once absinthe was gone.

The comparative political economy of regulation supplies the mechanism such a reading needs. Regulation tends to reward organized, concentrated interests at the expense of diffuse ones ([Stigler, 1971](#); [Peltzman, 1976](#)). A prohibition that reaches one drink and not the rest is tempting on that logic. [Room \(1984, fol. 3\)](#) puts the mechanism plainly: drawing a line between a bad drink and a residual good one "splits the industry interests and potentially avoids a head-on attack on the most powerful." For France, [Prestwich \(1979, p. 309\)](#) documents a wine industry long resentful of absinthe as a newer competitor, which had used political influence to secure wine's tax-favored standing as a hygienic drink ([Prestwich, 1997, p. 1258](#); [Studer, 2024, p. 5](#)), while cautioning that it was a beneficiary rather than the campaign's prime mover. Switzerland has a precedent of its own: when the Confederation rebuilt its alcohol taxes in 1885–87, the Federal Council meant the abolition of the cantonal entry duties to cheapen wine, cider and beer ([Schweizerischer Bundesrat, 1884, p. 473](#)), as [Section 2.2](#) sets out. The food laws of 1897 to 1906 extended that preference to wine against its imitations, and the absinthe article of 1908 removed one more rival of the fermented drinks, a protection made attractive to a wine canton by the price disadvantage the Valais government itself conceded for its own wine ([Schweizerischer Bundesrat, 1907b, p. 345](#)).

[Section 2](#) supplies the half-century of record behind the vote, since the differential treatment of the drink classes is older than the initiative. [Sections 3–5](#) construct the measures and test them.

2 Background

Read without the half-century before it, the vote of 1908 invites two presumptions: that the state had left the drink market alone, and that absinthe was the first prominent competitor wine had faced. Neither is so. Wine had faced cheap substitutes since the vine diseases of the 1860s through 1880s, with oidium preceding the more infamous phylloxera blight ([Nye, 2007, p. 58](#)), a *first wave* that precedes the absinthe ban by two decades ([Section 2.1](#)). Nor was the state a bystander. In 1885–87 the federal government rebuilt a tax system

that had charged drink by the container into one that charged by alcohol content, and in the rebuilding it gave the fermented drinks a constitutional shield (Section 2.2). By the 1900s, protection of the “natural” drinks, wine among them, was *de facto* standing federal policy, and the food laws of 1897 to 1906 had extended it to wine against its imitations. The case made against absinthe was medical, older than the campaign, and disputed among physicians at the time (Section 2.3). Unlike those earlier measures, the ban had temperance backing on a scale none of them had. Temperance bodies organized the petition and the campaign; in the chambers, deputies on both sides spoke openly of the vineyard (Section 2.4). The article as enacted banned the distilled liqueur and spared vermouth, the wine flavored with the same herb (Section 2.5). Throughout, we keep apart what the actors believed and what the record can sustain. Two questions run through the section. *Why would wine be involved in a health vote about one spirit?* And what distinguishes that vote from the other legislation of the period aimed at wine’s competitors?

2.1 The drink and its market

Phylloxera, an insect of the aphid family indigenous to America, first appeared in southern France in 1863. By 1890 it had destroyed 40 percent of French vineyards, and output recovered only as growers replanted on resistant American roots (Banerjee et al., 2010, pp. 714–717).

The insect was first found in Switzerland in 1874, in the vine greenhouses of an estate at Pregny near Geneva; the canton, which had taken the necessary powers from its Grand Council two years before, expropriated the owners and ordered the infected vineyards destroyed (Furrer, 1889, s.v. Reblaus, p. 646). By 1877–78 the federal government had undertaken by federal decree to meet not less than a third of cantonal eradication costs, and in September 1878 the international phylloxera convention was concluded at Bern (Furrer, 1889, pp. 653–654). How far Swiss vineyards were themselves hit is not something we try to settle; the cantonal area record for these decades is coarse, and nothing in what follows turns on it.

Swiss wine production was meanwhile trending downward. The Federal Council’s November 1884 message reckoned that Swiss wine yields since 1876 had stood on average a quarter to a third below the mean and had barely reached a million hectoliters (Schweizerischer Bundesrat, 1884, p. 381). The message names the phylloxera once, and it is France’s vineyards the insect has devastated; from there the Council drew its implications to the Swiss drinker:

“If one further considers that in France, the principal wine country, wine production has fallen considerably as a consequence of the devastation by the phylloxera; that in consequence the price of wine has risen while its quality has fallen; and moreover that our purchasing power has also diminished through the growing industrial crisis, is it then still a wonder that our population, accustomed in the good years to a heightened enjoyment of life, has gone over to other drinks?” ([Schweizerischer Bundesrat, 1884](#), pp. 381–382)

Replacing that shortfall in private consumption, the Council reckoned, would have taken “a large quantity of additional import and additional production of spirits” before one could say the population was drinking as much as it had in the good years ([Schweizerischer Bundesrat, 1884](#), p. 382). Opening its section on distilled drinks, it put the same point as a matter of record: these circumstances “encouraged domestic manufacture to supply the increased demand itself as far as possible” ([Schweizerischer Bundesrat, 1884](#), p. 382). The opening that cheap spirits filled is therefore not our reconstruction. It is the federal executive’s own reading, two decades before the absinthe vote.

It’s no secret that this is a textbook setting for a substitution effect: the price of wine rises, and demand shifts toward the substitutes, cheap spirits among them. Wine and beer, the Council wrote, “as a daily fortification, have been made almost inaccessible to the worker, while a schnapps injurious to health is everywhere to be had at derisory prices,” and it called this the “economic, sanitary and moral fruits” of the small-distillery industry, which we take up in Section 2.2 ([Schweizerischer Bundesrat, 1884](#), p. 456; Appendix A.1). It had put the fiscal half of that case earlier: if legislation treats a necessary foodstuff “like a luxury article” and “through corresponding taxes” makes it inaccessible to the worker, the Council asked, where then does the fault lie when the need for a fortifying drink drives him to the cheaper schnapps ([Schweizerischer Bundesrat, 1884](#), p. 445).

The French record puts numbers to the price path the Swiss messages describe only in words. The wholesale price of industrial alcohol fell from 166 francs per hectoliter in 1854 to 28 francs in 1901, “despite increases in taxes on alcoholic consumption” ([Marrus, 1974](#), pp. 128–129; Appendix A.1), and once phylloxera had deprived French makers of the wine alcohol absinthe had been built on, the drink could be bought “for ten centimes, less than a glass of wine” ([Prestwich, 1979](#), p. 302).

What the market story alone omits is that none of this played out on neutral ground. The federal government had been legislating on exactly this margin for two decades before the absinthe vote, and each time it wrote the law, wine, beer, and cider came out on the protected side.

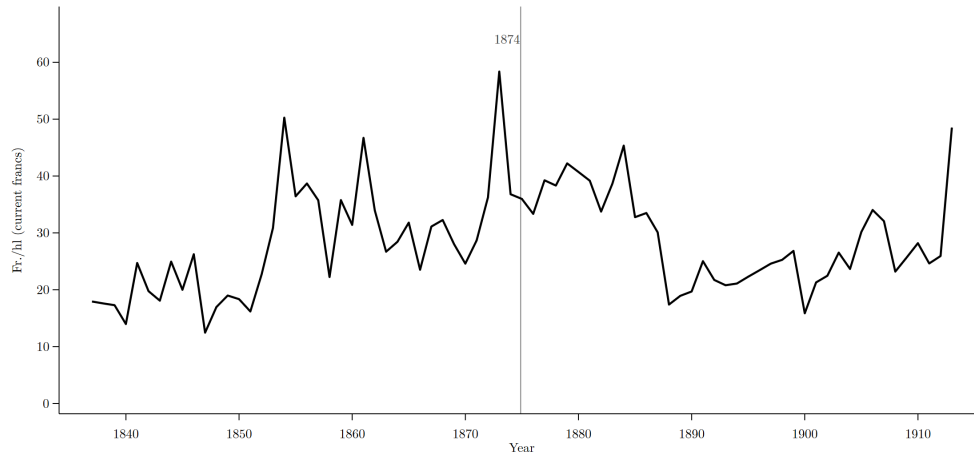


Figure 1: Swiss wine prices, 1837–1913, shown for context. Nominal producer prices in current francs per hectoliter: what growers received, not what drinkers paid. The vertical line marks 23 November 1874, when phylloxera was found in the vine greenhouses at Pregny, near Geneva, in the year the insect first appeared in Switzerland. Source: HSSO Table H.2a, column X ([Historische Statistik der Schweiz Online \(HSSO\)](#), 2012e). Every value in that table is an estimate after Ritzmann rather than an observed price.

2.2 The federal alcohol monopoly

The Council’s 1884 message stated the public-health justification. “The chief evil is the quality of our schnapps,” it wrote; the task was to do away with that drink and put less harmful ones in its place ([Schweizerischer Bundesrat, 1884](#), p. 455). Its general doctrine said the same: since alcoholic drinks could not be condemned outright, the more harmful ones should be displaced by the less dangerous, or their effects reduced by legislation ([Schweizerischer Bundesrat, 1884](#), p. 401). By its first-stated aim, then, the reform set one of its own standards of success: whether it displaced the drinks it had named as most harmful, or reduced their harm.

The tax system the reformers inherited worked against that substitution. The cantonal tax structure was levied on volumes of drink ([Dubler, 2003](#)), and a charge on volume falls more lightly on strong drink per unit of alcohol unless the tariff offsets it. Prior to the establishment of the Federal Alcohol Administration, no Swiss tariff taxed drink by its alcohol content, and the cantons’ tax structures differed from one another. Brandy, which on that principle would have paid some six times the duty on ordinary wine, paid on average only about twice as much, the Council reckoned ([Schweizerischer Bundesrat, 1884](#), p. 443).² Nor did every canton levy an entry duty on spirits ([Schweizerischer Bundesrat, 1886](#), p. 425). By the law of demand, this tipped the scales as industrial alcohol production increased ([Marrus, 1974](#), p. 128). While price is generally positively related to alcohol

²This issue was not unprecedented within the time period (see [Nye, 2007](#), pp. 34–36).

content, the tax ran the other direction, and people had demanded a higher quantity of high-proof cheaper schnapps. The Federal Council attempted to reorient this pricing structure around alcohol content.

The correction came through Article 32bis, adopted on 25 October 1885, and the law of 23 December 1886, confirmed at the polls on 15 May 1887, whose monopoly articles took effect that July ([Schweizerischer Bundesrat, 1886](#), p. 421; [Swissvotes, 1887](#); [Schweizerischer Bundesrat, 1887](#), p. 675). The Council had said what the law was for: dearer and purer spirits for consumption, an end to the abuses of distilling and sale, revenue enough to relieve the drinks that were not distilled, and the fullest regard for the agricultural and industrial interests bound up with distilling ([Schweizerischer Bundesrat, 1886](#), pp. 423–424; Appendix [A.1](#)). Cheap fermented drink was an objective in its own right. The planned abolition of the cantonal entry duties was meant “quite specifically” to cheapen wine, cider and beer, so that in a country rich in wine and fruit “a part of the population should not be dependent on schnapps,” and “in good fruit years,” the Council promised, “cider is to be had throughout the whole country at cheap prices” ([Schweizerischer Bundesrat, 1884](#), p. 473). Revenue entered the design as a means and a requirement rather than a measure, and at least a tenth of what reached the cantons was reserved for combating alcoholism ([Schweizerischer Bundesrat, 1886](#), pp. 423–424; [Schweizerische Bundesversammlung, 1885](#), pp. 478–479).

The settlement was an agricultural bargain. A farmers’ petition of March 1886 welcomed a dearer spirit “provided that not merely fiscal but also sanitary and moral purposes are pursued thereby,” and provided that the tax did not make the agricultural distillery impossible ([Schweizerischer Bundesrat, 1886](#), p. 456). The Council answered with what it called a first concession to agriculture: distilling in Switzerland would not be suppressed ([Schweizerischer Bundesrat, 1886](#), p. 458). The statute had the Federal Council prefer domestic raw materials and agricultural cooperatives when awarding the distilling lots ([Schweizerische Bundesversammlung, 1886](#), p. 1310), referred to even twelve years thereafter by the Federal Council as the compromise of 1886 ([Schweizerischer Bundesrat, 1898b](#), pp. 622–623). The same settlement relieved wine of the entry duties, left the distilling of wine and fruit outside federal law, on the Council’s reasoning that such spirits were too dear to be much abused ([Schweizerischer Bundesrat, 1886](#), pp. 422, 440), and barred special cantonal taxes on fermented drinks ([Schweizerischer Bundesrat, 1886](#), p. 422; Appendix [A.1](#)); these passages establish what wine gained within a broader agricultural bargain, not who secured each provision.

The Council’s draft would have taxed production and made a control meter central to

licensing the ordinary taxable distilleries, provisions under which it expected the smallest household stills to disappear ([Schweizerischer Bundesrat, 1886](#), pp. 444, 473). Parliament enacted a monopoly instead. The Confederation took the exclusive right to make and import spirit from every material federal law did not exempt, which the Council said meant in practice chiefly potatoes and grain, and bought domestic spirit under contracts tendered in lots of at least 150 hectoliters of absolute alcohol ([Schweizerische Bundesversammlung, 1886](#), pp. 1309–1310; [Schweizerischer Bundesrat, 1886](#), p. 479); the meter moved into the contracts, whose conditions of 1888 placed each contract still under official seal from the point where condensation begins and required a control reservoir under official lock or a control meter ([Schweizerischer Bundesrat, 1888](#), p. 901).

Absinthe entered this settlement as a customer rather than a target. The 1886 message singled out the export trade in liqueur and absinthe, an industry working predominantly from foreign fine spirit ([Schweizerischer Bundesrat, 1886](#), p. 468; Appendix A.1). After 1887 that input came through the monopoly, and the Administration later wrote that absinthe liqueur was made wholly from the drinking spirit the Confederation supplied ([Milliet, 1907](#), p. 358). In 1897 the monopoly's domestic sales of spirit per head came to 35.2 kilograms in the Val-de-Travers, against 2.2 for Switzerland as a whole, figures the Administration said described its traffic, not a statistic of consumption, though a rough guide to where drinking was concentrated ([Schweizerischer Bundesrat, 1898a](#), p. 89); its export accounts carried absinthe on a line of its own from the monopoly's first year ([Schweizerischer Bundesrat, 1900](#), p. 1013; [Schweizerischer Bundesrat, 1889a](#), p. 174). The Administration did not claim to see all of the trade: its register of makers omitted those working by the cold method, and purchases through intermediaries went unrecorded ([Milliet, 1907](#), pp. 359, 361). That record made absinthe a feasible object of regulation; by itself it does not explain the later ban, which later sections take up.

The small distillers were the opposite case. The Council's survey counted 1,022 distilleries that made more than two hectoliters a year from starchy materials and sold the product; of the 985 with returns, 937 reported less than 100 hectoliters each, below the smallest lot the law would offer ([Schweizerischer Bundesrat, 1886](#), pp. 425, 428, 435; Appendix A.1). By 1898 the monopoly bought from seventy contracted suppliers ([Schweizerischer Bundesrat, 1899a](#), p. 330). Much of the difference was bought: by the spring of 1890 the Federal Council had settled 1,171 of 1,376 claims for compensation, not all of them from surveyed distilleries ([Schweizerischer Bundesrat, 1889a](#), pp. 185–187; [Schweizerischer Bundesrat, 1890a](#), p. 56; [Schweizerischer Bundesrat, 1892a](#), pp. 811–812). What remained formed a class of its own under the new law, the free distilleries, which might distill fruit, wine or fee-paid materials “without further involvement of the Confederation”

([Schweizerischer Bundesrat, 1889a](#), p. 109). The record counts the distillers who took compensation, names the free distilleries without ever counting them, and sees illicit distilling only where a case was brought, as the Council had admitted in 1886 it could not count the exempt stills either ([Schweizerischer Bundesrat, 1886](#), p. 425).

The Administration's control service was built around the contracts. The decree of 1890 named ten controllers and drew their nine districts as lists of the contract distilleries ([Schweizerischer Bundesrat, 1890b](#), pp. 161–164; Appendix A.1). The free distilleries were the cantons' to supervise ([Schweizerische Bundesversammlung, 1886](#), p. 1312), and parliament's committee found in 1897 that several had not properly carried it out ([Nationalrätliche Alkoholkommission, 1897](#), p. 1016; Appendix A.1). Officials, gendarmes and police were bound to report every offense they found ([Schweizerischer Bundesrat, 1888](#), p. 877), and the general rule gave a third of each collected fine to whoever reported it ([Schweizerische Bundesversammlung, 1886](#), p. 1313). In Becker's account the expected punishment is the probability of conviction times the penalty ([Becker, 1968](#)). A contract distillery was visible before any offense occurred; production concealed outside the contracts had first to be found, and that was the weak step. The few cases on record are consistent with costly detection, weak incentives or offenders' adaptation; they cannot tell us whether illicit production was large or small.

The published enforcement record is small. Over 1887–98 the Administration received 413 complaints and proceeded on 329, of which 265, about twenty-two a year, ended in a penal order ([Schweizerischer Bundesrat, 1899a](#), pp. 301–302, 460–461; Appendix A.1). Fines paid in fell from Fr. 13,883.45 in 1890 to Fr. 537.78 in 1897 ([Schweizerischer Bundesrat, 1899a](#), p. 462). The informer's bounty reached people outside the Administration in the monopoly's first years and very little afterwards: informers were paid Fr. 3,682.35 directly in 1890 but Fr. 73.40 in 1898 ([Schweizerischer Bundesrat, 1899a](#), p. 462). The Administration's informers' fund changed its meaning in between: founded in 1887/88 for the thirds no informer received ([Schweizerischer Bundesrat, 1889a](#), p. 196), it also took from 1895 the third on every offense the Administration's own officials reported ([Schweizerischer Bundesrat, 1895b](#), p. 185), and the committee explained its receipts for 1896 by the complaints having come, with few exceptions, from the Administration's employees ([Nationalrätliche Alkoholkommission, 1897](#), p. 1018). The accounts cannot separate the new rule from the falling number of cases; either way, by the late 1890s the bounty was bringing in few reports from outside.

Meanwhile the committee counted a larger surplus as a success; it welcomed a modest fall in drinking spirit in 1899 on ethical grounds ([Nationalrätliche Alkoholkommission,](#)

1900, p. 732), but reached first for the surplus. “This favorable result,” it wrote of the surplus of 1897, “is explained . . . also by increased consumption of spirits subject to the monopoly” ([Nationalrätliche Alkoholkommission, 1898](#), p. 193), an increase it traced not to more drinking but to poor fruit and wine harvests, which had left less monopoly-free spirit, must and wine to drink ([Nationalrätliche Alkoholkommission, 1898](#), p. 191). On the committee’s own reading, bad harvests had moved drinkers from the fermented drinks the reform meant to make cheap toward the spirit it meant to make dear, and the accounts recorded a good year (Appendix [A.1](#)). The good fruit year the Council had promised in 1884 had become, in the committee’s forecast, a fiscal disappointment ([Nationalrätliche Alkoholkommission, 1900](#), pp. 734–735).

Taken together, the record suggests a low *ex ante* likelihood that production concealed outside the contracts would be detected and punished. A distiller once found was likely to be punished. The fines collected were small in francs, a total that cannot show the severity of individual penalties: the expected sanction was modest because detection was. Outside the contracts the monopoly had no instrument to measure how much escaped, and its accounts could not show whether harmful drinking as a whole had fallen. Absinthe’s principal factory trade stood on the unusually visible side of this divide: concentrated, identifiable and recorded in its purchases and exports, it offered a feasible target on which responsibility for a wider drinking problem could later be placed.

2.3 The public case: science and temperance

The case for singling out absinthe was presumably medical, and older than the campaign. Valentin Magnan built his account of absinthism on a single case he treated at Bicêtre, published in 1864 ([Luauté, 2007](#), pp. 518, 523). By the 1890s, Prestwich writes, his work on absinthism “was considered classic” ([Prestwich, 1979](#), p. 305). The federal government’s experts endorsed the diagnosis: as Auderset summarizes their report, absinthe’s “poison properties” produce a clearly distinct form of alcoholism ([Gaule et al., 1907](#), pp. 5, 11–13, as reported by [Auderset, 2021](#), pp. 149–150, n. 57). The *Journal de Genève* reported Dr Pallard, at a meeting in Vernier, invoking “the unanimous opinion of the physicians and of the experts appointed by the Federal Council” ([Journal de Genève, 1908](#), 3 July 1908, p. 4). Pallard also warned of harm to absinthe drinkers’ descendants, a claim of the degeneration theory, the theory Magnan and Fillassier took to the first eugenics congress in 1912 ([Journal de Genève, 1908](#), 3 July 1908, p. 4; [Prestwich, 1979](#), p. 306; [Luauté, 2007](#), pp. 524, 530).

Contemporary scientists *did* disagree. Most asylum doctors stood behind Magnan, but dissent came from established clinicians, from experimenters and from a government

inquiry (Luauté, 2007, p. 525).³ According to Luauté, in 1907 the French government ordered an inquiry into Pontarlier, a heavy absinthe-drinking district; its prosecutor, Couleru, found the disorders blamed on absinthe less marked there than where other alcohol was drunk heavily, and crime halved between 1876 and 1906 while absinthe consumption rose (Luauté, 2007, pp. 526–527). Though these accounts come from France, a Swiss rapporteur turned the same comparison on wine, while granting absinthe’s dangers. In the National Council in 1908, the physician Müller (Bern), who spoke for the commission majority against the ban, held that all alcoholic drinks do harm in excess, “the Vaud wine just as much as the marc brandies.” He had seen almost more delirium tremens from that wine than from the federal schnapps (Schweizerische Bundesversammlung, 1908, National Council, 2 April 1908, pp. 27, 30).

As Padosch and colleagues review the laboratory record, the seizures in the period’s animal experiments were produced with pure wormwood essence rather than the drink, and *The Lancet* questioned as early as 1869 whether such findings carried over to drinkers (Padosch et al., 2006, p. 7). Modern chemical analysis of thirteen vintage pre-ban absinthes found nothing besides ethanol that could explain the syndrome (Lachenmeier et al., 2008, p. 3073), and a review of the literature concludes that “the so-called absinthism cannot exactly be distinguished from chronic alcoholism” (Padosch et al., 2006, p. 11).

Swiss temperance began with societies that “required from their members total abstinence from distilled beverages only, and moderation in the use of wine” (Cahannes, 1981, p. 42). Trechsel writes that the Blue Cross’s campaign against absinthe was “internally almost uncontested, probably because it was directed against distilled drinks” (Trechsel, 1989, p. 46). Temperance societies and wine producers drew on the same hierarchy of drinks from different sides, which made their interests appear aligned without requiring any coordination. We treat that hierarchy as context, and our design does not test it.

2.4 The campaign and the recorded interests

On 28 August 1905 at Commugny, in canton Vaud, Jean Lanfray killed his wife and two daughters, and the killings, which public opinion blamed on his drinking of absinthe, became the catalyst for a cantonal petition against its sale (Studer, 2024, pp. 178, 181). Blaming absinthe rather than alcoholism was dubious. A newspaper report of the defense

³Professor Joffroy, 1900: “the proportion of epileptics is minuscule compared to the colossal number of absinthe drinkers” (Luauté, 2007, pp. 524, 530). Lalou’s 1903 experiments, reported in the 1905 *Traité de l’alcoolisme*: commercial absinthe owed its acute effects “almost exclusively” to the alcohol (Luauté, 2007, p. 524).

psychiatrist's evidence had him drinking up to four liters of wine a day, besides many absinthes and marc (Studer, 2024, p. 180). Federal Councilor Ruchet, speaking against the initiative, later called the crime "an honest pretext" (Schweizerische Bundesversammlung, 1908, National Council, 6 April 1908, p. 83). Vaud banned the retail sale of absinthe in May 1906 and Geneva in February 1907, and cantonal votes confirmed both laws (Schweizerischer Bundesrat, 1907b, pp. 344, 346; Heimberg, 2000, p. 98; Trechsel, 1989, p. 151). The café-keepers forced the Vaud vote by referendum (Heimberg, 2000, p. 98). Secretan, for the ban, told the National Council that the café-keepers, who said they earned little on absinthe, had at first "seemed to want to stay neutral," until the manufacturers "summoned them to make common cause with them" (Schweizerische Bundesversammlung, 1908, National Council, 3 April 1908, p. 73). Trechsel records that the abstainers, who in principle stood with the ban movement, remarked "not uncritically" that the Vaud initiative had found large majorities above all in the winegrowing regions (Trechsel, 1989, p. 151).

The proposal for a national ban came from the Neuchâtel and Jura branches of the *Ligue patriotique contre l'alcoolisme*, a moderation league founded by the Blue Cross. After a congress at Bern on 17 December 1905 an initiative committee formed, and Robert Hercod's abstinence secretariat took over "the coordination of the signature collection, as later also of the voting campaign" (Trechsel, 1989, pp. 33, 151–152). The secretariat served as the committee's general secretariat and took the name *Zentralstelle zur Bekämpfung des Alkoholismus* in 1913 (Trechsel, 1989, pp. 125, 152), the name under which Swissvotes records the initiator (Swissvotes, 1908). Hercod later judged: "The abstainers, to be sure, did the real work, but the non-abstainers at least helped us morally, immensely" (Trechsel, 1989, p. 153). The ban's opponents charged that it was a first step toward total prohibition: an anti-ban poster for the vote printed, as the words of a "Dr Herco of Lausanne" from a report it cited by page, that the ban would have "only a quite weak effect on alcoholism in general," that the aim for a more distant future was "the abolition of all alcoholic drinks, wine and beer included," and that "it is a beginning" (Gantner, 1908). Section 3 reports the signature count.

The 1908 deliberations record the wine stake in the participants' own words. Müller observed that only the governments of Vaud, Geneva, Fribourg and Valais had declared for the initiative, "although competition with wine seems to play a certain role in this" (Schweizerische Bundesversammlung, 1908, National Council, 2 April 1908, p. 27). Martin, the majority's French rapporteur and deputy for the absinthe-making Val-de-Travers, found it "rather curious" that absinthe drinking "must have developed above all" in the cantons that produce white wine (Schweizerische Bundesversammlung, 1908, National Council, 2 April 1908, pp. 30, 34). Both remarks come from the ban's opponents. Müller's remark

attributes to the ban's side a competition with wine, which is motive whether it was realized or only *perceived*. Brüstlein, for the ban, told the National Council that "in the canton of Vaud the movement is fortunately helped by the fact that the wine-growers see in absinthe a dangerous competitor and would gladly be rid of it, also from a selfish standpoint" ([Schweizerische Bundesversammlung, 1908](#), National Council, 3 April 1908, p. 69). Writing in 1981, Cahannes concedes that the initiative's "passage was also due to the fact that absinthe, particularly popular in the French part of the country, competed with white wine, and the initiative was therefore supported by the winegrowers" ([Cahannes, 1981](#), pp. 44–45). Gobat, for the ban, charged that the Federal Council "has declared itself ready to protect wine, vineyard and winegrowers against raisin wine, an innocent beverage, but it refuses to protect them against a poison which likewise competes with wine" ([Schweizerische Bundesversammlung, 1908](#), National Council, 6 April 1908, p. 81). Section 5 documents the food-law record. Federal Councilor Ruchet, defending the Council, restated the 1885 program: relief "... of the wholesome drinks, wine, cider, beer" and the "raising of the price of alcohol" ([Schweizerische Bundesversammlung, 1908](#), National Council, 6 April 1908, p. 84). The next day, in the Council of States, he said of the voters' rejection in 1903 (#63) of a revision that would have raised the floor for the free sale of wine and beer from two liters to ten that "there was nothing to do but bow" ([Schweizerische Bundesversammlung, 1908](#), Council of States, 7 April 1908, p. 64; for the date, [Schweizerische Bundesversammlung, 1908](#), National Council, 6 April 1908, p. 85). Fazy, for the ban, named the mechanism: "when it is a matter of protecting wine, of protecting anything whatever, everyone agrees and nobody invokes freedom of industry." He then declined it: "we do not have as our pure and simple mandate the suppression of competition with wines or with this or that product" ([Schweizerische Bundesversammlung, 1908](#), National Council, 7 April 1908, pp. 96–97). The deputies recognized a wine interest and disputed it, but the debate cannot measure its weight at the ballot, which Sections 3–5 estimate.

2.5 What was enacted

Whereas the cantonal laws had banned retail sale, Article 32ter prohibited the "manufacture, import, transport, sale and storage for the purpose of sale" of absinthe and its imitations ([Schweizerischer Bundesrat, 1907b](#), pp. 353–354). Auderset notes that Geneva, which already had a cantonal ban, rejected the federal initiative ([Auderset, 2021](#), p. 152). The article left "all other absinthe-containing beverages which constitute a public danger" to later federal legislation ([Schweizerischer Bundesrat, 1907b](#), pp. 353–354). The Alcohol Administration counted "Chartreuse and above all so-called vermouth wine" among them,

and left vermouth out of its reckoning of the ban’s effects as “only optionally foreseen” and too little measured (Milliet, 1907, pp. 358, 364 n. 1). Among absinthe-containing drinks, the ban fell only on the distilled side of the period’s own line between “wholesome” fermented and distilled drink, with the notable asymmetry regarding Chartreuse and vermouth: a distilled liqueur was deferred alongside a wine (Schweizerische Bundesversammlung, 1908, National Council, 6 April 1908, p. 84; Milliet, 1907, p. 358; Schweizerischer Bundesrat, 1907b, p. 348). The government that recommended rejection had a fiscal stake, which we do not claim drove its recommendation: its Alcohol Administration put the lost monopoly profit at up to about 873,000 francs a year if no other monopoly drink took absinthe’s place. The Administration said the loss would fall especially on the cantonal treasuries, and thought it would not arise in full (Milliet, 1907, pp. 364–366).

3 Data

The analysis uses a canton-level cross-section of the $N = 25$ cantons that existed under 1908 boundaries⁴. The outcome is the canton-level share of votes in favor of federal vote #68, the absinthe-ban constitutional amendment of 5 July 1908. Four groups of right-hand-side variables enter: measures of wine-industry presence, which carry the wine-market interest; the absinthe industry’s own purchases of monopoly spirit, taken from the Milliet 1907 firm-level annex and aggregated to canton shares of the national total; cultural and political controls for language and religion; and population density.

Each arm, the vote and the petition, is anchored to the event it describes. The vote arm uses the 1907 harvest and canton population, and the petition arm those of 1906. The ballot fell in mid-year, so the vote arm takes the last complete harvest and the prior year’s population (Appendix A.2). Language and religion, and the land area behind density, come from the 1900 census, for which there are no intercensal estimates; the federal census cycle is decennial (1900, 1910).

3.1 Data sources

3.1.1 Vote and Petition Outcomes

The outcome variables fall into three groups: the canton yes-share on the absinthe ban itself, vote #68, which carries the primary results; the canton’s signature rate on the 1906

⁴Bern absorbs the future Jura, which separated from Bern only in 1979 and is therefore not a distinct unit during our period.

petition for the ban, per hundred eligible voters; and the yes-share on other federal votes of the preceding decade, which serve as falsification and corroborative comparisons.

The vote-share dataset draws on publicly available primary sources; the yes-shares for every referendum come from Swissvotes (Année Politique Suisse, University of Bern), the canton-level archive of every Swiss federal vote since 1848 ([Swissvotes, 2026](#)). Canton-level outcome totals are cross-validated against the Federal Chancellery’s authoritative archive ([Schweizerische Bundeskanzlei, 1908a,b](#)); Appendix A.4 gives the official record’s page.

Five cantons required their citizens to vote in this period (Appendix A.4). Turnout there measures compulsion as much as interest, so it cannot stand in for mobilization. The petition stage gives the cleaner contrast: the same electorate faced the same question eighteen months earlier, and signing was voluntary in every canton.

Swiss federal votes in this period take three legal forms. A constitutional amendment drafted by the Federal Assembly goes to the people automatically, as a mandatory referendum. A constitutional amendment drafted by citizens reaches the ballot as a popular initiative. And an ordinary federal law already passed by the Assembly is put to a vote, as an optional referendum, only if thirty thousand citizens demand one (Appendix A.4).

Both constitutional forms require a double majority, of voters nationally and of cantons, whoever drafted the text. The constitution being the compact among the cantons, changes to it are ratified by the cantons as units. The optional referendum requires only the popular majority, since no constitutional text is at stake. The canton is a decision unit in the one case and not in the other.

The absinthe ban was a popular initiative, filed with 167,814 valid signatures, more than three times the statutory minimum and, by the Federal Council’s own account, more than any initiative had drawn before. Its outcome was therefore determined canton by canton as well as nationally, which is why we analyze it at the canton level and why the votes we use as falsifications are drawn from the double-majority forms.

The dependent variable in our primary specifications is the canton-level yes-share on vote #68,

$$Y_{i,68} = \text{pct_yes_68}_i = \frac{\text{yes votes}_i}{\text{valid votes}_i} \times 100,$$

for $i = 1, \dots, 25$ cantons, drawn from Swissvotes (anr = 68; [Swissvotes 1908](#)).

The cantons enter unweighted. Under the double majority each casts a single vote, the six half-cantons a half each, but reproducing that arithmetic in the regression would distort the interpretation of the coefficient rather than sharpen it. The question is whether wine

production predicts local support for the ban. Weighting by population or by ballots cast would answer a different question, whether it predicts the national result.

The specification is estimated on nine further federal votes alongside the absinthe ban. Six are falsification tests, on which no wine relationship is expected; three are corroborative, on which one is. Three criteria narrow the set. We take only votes held before the ban, so that no comparison is contaminated by behavior the ban itself changed. We restrict the falsification arm to votes decided under the double majority. And we require the falsification votes to carry no wine-related subject matter. The corroborative votes meet the double-majority criterion with one exception, referendum #65, which is included because it is a wine-salient vote named explicitly in the deliberations on the absinthe ban. Table [A.1](#) lists all ten.

Every referendum uses the vote-share construction given above for vote #68.

3.1.2 Petition signatures

The initiative reached the ballot through a signature drive, and that drive leaves a canton-level record of its own, eighteen months before the vote. We measure it as signatures submitted per hundred eligible voters,

$$P_i = \frac{\text{signatures submitted}_i}{\text{eligible voters}_i} \times 100,$$

for the same $i = 1, \dots, 25$ cantons.

The numerator comes from the twenty-five-canton table at printed p. 984 of the Federal Council's report of 22 February 1907, which gives for each canton the signatures submitted, those declared valid, and those rejected, closing at 169,377, 167,814 and 1,563 nationally ([Schweizerischer Bundesrat, 1907a](#), p. 984). We take the submitted count. The canton figures reconcile: each of the three columns sums exactly to its own printed national total (Appendix [A.4](#)).

The denominator wants a count of those entitled to sign, and no such count survives. We use the cantonal electoral roll recorded for the federal vote of 10 June 1906, the federal ballot nearest the drive in its own year. A population denominator is the obvious alternative and it is the worse one: women, minors and non-citizens could not sign as electors, so scaling by population would import cross-cantonal differences in age structure and foreign residence into a measure meant to record who signed.

That same 1906 vote does two jobs here: its electoral roll scales the petition outcome,

and its yes-share is one of the corroborative referendum outcomes (§5.2.1). The two draw on different quantities, and neither ever appears on both sides of the same specification. The petition is an outcome throughout, and enters no specification as a regressor.

Appendix Table A.19 reports the petition-arm variables.

3.1.3 Wine Data

The central variable of interest is canton-level wine-industry presence, standing for the producer interest that a ban on a competing drink would serve. We digitized the canton wine-harvest table series of the *Statistisches Jahrbuch der Schweiz* to measure it. The vote arm uses the 1907 harvest (BFS, 1909, printed p. 51) and the petition arm the 1906 harvest (BFS, 1908b, printed p. 55). Table A.2 gives the volume and printed page behind every referendum in the set.

The table reports two quantities for each canton: the total harvest in hectoliters and its total value in francs. We take both as printed and convert each to a share of the national total,

$$X_{2i} = \frac{Q_i}{\sum_j Q_j} \times 100, \quad X_{3i} = \frac{V_i}{\sum_j V_j} \times 100,$$

where Q_i is canton i 's total harvest in hectoliters and V_i its value in francs. We report both, and for brevity show wine revenue in the main paper where possible and volume in the appendix; §4.2 sets out why revenue is preferred.

The denominator is national by construction. These variables measure the share of Swiss wine production sitting in canton i ; they do not measure how large wine looms within canton i 's own economy. The vote was federal, and what carried weight in it was a canton's standing in the national industry.

V_i values an entire harvest at the average prices returned by the cantonal governments and statistical bureaus, so it is a valuation rather than a receipt. We nonetheless call it wine revenue throughout, and its share of the national total the wine revenue share, to avoid confusion with statistical values; unless a passage says otherwise, wine revenue means this approximation and not realized sales (Appendix A.2).

Cantons with unreported wine production are coded as zero, except in the cases where the publications indicate insufficient reporting. Certain years substitute the yearbook's footnoted estimates where a year's vintage is incompletely recorded.

3.1.4 Absinthe Producer Data

The absinthe-industry measures come from the annex that E.W. Milliet, Director of the Federal Alcohol Administration, prepared for the Federal Council's December 1907 *Botschaft* on the popular initiative ([Schweizerischer Bundesrat, 1907b](#), pp. 341–366), with the firm-level table at pp. 360–361 ([Milliet, 1907](#)). It lists forty houses in eight cantons, each with its purchases of spirit at 95° ABV from the federal monopoly over 1902–1906; the other seventeen cantons have no listed house and enter every measure at zero. The proxy is coarse, but it is unique: Switzerland did not report liquor production by product until the 1930s.

The register is a list of monopoly customers, not a census of absinthe distilleries. The Administration itself warned that the purchase figures give no reliable indication of the scale of absinthe production, because the listed houses also made other products from the same spirit ([Milliet, 1907](#), p. 361). We therefore verify the register house by house and strike, from both numerator and denominator, every house the sources document as a producer of something other than absinthe (Appendix [A.3](#)). Two series bracket the one judgment call. The loose series strikes six houses; the floor series, carried in the main tables, also strikes one Fribourg wholesaler with no production of its own.

The continuous measure, Abs_i , is the canton's share of the retained national total in percent. Under the floor series Neuchâtel holds 63.0 percent, Geneva 20.0, Vaud 11.0 and Basel-Stadt 6.0, and the four other register cantons hold zero. A discrete counterpart, $1\{Abs_i > 0\}$, marks the four cantons with at least one attested absinthe producer. The loose versions keep the Fribourg house; both are reported in Appendix Table [A.7](#).

3.2 Cultural and political controls

3.2.1 The French-speaking share

The first cultural control is the canton-level French-speaking share,

$$\text{French share}_i = \frac{\text{French speakers}_i}{\text{Resident population}_i} \times 100,$$

constructed from HSSO B.32 (1900 federal census, native-language classification; [Historische Statistik der Schweiz Online \(HSSO\) 2012d](#)) over the resident population of HSSO B.1a ([Historische Statistik der Schweiz Online \(HSSO\), 2012a](#)).

The denominator is the whole resident population rather than French and German

speakers alone. A two-language denominator would omit the language most of Ticino speaks (Appendix A.4). The five Romande cantons cluster at the top, with Vaud, Geneva, Neuchâtel, Fribourg and Valais all above 60 percent.

Language is not a nuisance covariate in this cross-section. The Romande cantons concentrate wine production and absinthe production together, so the linguistic cleavage is collinear with the treatment by geography rather than by accident. We therefore treat the French share as co-explanatory rather than as a control to be partialled out.

3.2.2 The Protestant share

The second cultural control is the canton-level Protestant share,

Protestant share_{*i*} = $\frac{\text{Protestants}_i}{\text{Protestants}_i + \text{Catholics}_i} \times 100$, constructed from HSSO B.27 (1900 federal census; [Historische Statistik der Schweiz Online \(HSSO\) 2012c](#)). The denominator is Protestants and Catholics rather than the resident population, since no canton is predominantly Jewish, Eastern Orthodox or unaffiliated (Appendix A.4).

The Protestant share measures religious composition. What it proxies is the density of standing temperance organization, and the two are joined by how the organizations actually spread rather than by assumption. Switzerland's oldest and largest abstinence body, the Blue Cross, was founded in 1877 and declared itself neutral between denominations in Article 2 of its 1884 statutes. It did not spread that way: it reached, in [Trechsel's](#) account, only Protestant areas, with the single exception of the Bernese Jura, and the founding of the Catholic Abstinence League in 1895 ended its cross-denominational ambitions ([Trechsel, 1989](#), pp. 35–36).⁵ The pattern is not one organization's idiosyncrasy (Appendix A.4).

Catholic temperance was not absent, and we do not claim it was (§5.2; Appendix A.4). The claim is one of timing and organizational depth. The Protestant body had been compounding its membership since 1882, while the Catholic League was assembled in 1895 out of three local societies founded in 1893 and 1894 ([Trechsel, 1989](#), p. 98). Reviewing his own membership figures, [Trechsel \(1989, p. 132\)](#) concludes that the abstinence movement was to a large extent a Protestant one.

Religious affiliation is an indirect measure of the organizational capacity the argument runs on. Many Protestants did not support temperance, and some Catholic clergy did, particularly in cantons with large numbers of both. We keep the variable because the mechanism runs through it, and because omitting it would load the Protestant–Catholic

⁵“Obwohl laut Statuten konfessionell neutral, verbreitete sich das Blaue Kreuz mit Ausnahme des ‘Berner’ Juras nur in protestantischen Gebieten. [...] Faktisch wurde so das Blaue Kreuz eine protestantische Organisation.” Our translation.

divide onto the wine coefficient.

3.3 Population density

The structural control is the log of canton population density, $\ln D_i = \ln\left(\frac{\text{Population}_i}{\text{Land area}_i}\right)$, with population in persons and land area in square kilometers. The two arms differ in the numerator alone: the vote arm uses the 1907 canton population and the petition arm the 1906 population, both read from the Statistical Yearbook rather than interpolated. We recover the denominator, land area, from the 1900 census, dividing canton population ([Historische Statistik der Schweiz Online \(HSSO\), 2012a](#)) by canton population density ([Historische Statistik der Schweiz Online \(HSSO\), 2012b](#)). The recovery is exact rather than approximate (Appendix A.4).

We enter density in logs because its spread is very wide, running from 15.4 to 3,510 persons per square kilometer, a factor of more than two hundred. We do not report a levels specification, and the log is a scaling choice rather than a fitted one.

3.4 Summary statistics

Table 1 reports summary statistics for the analysis dataset, each variable on the scale at which it enters the regressions, for all twenty-five cantons and for the German- and French-plurality cantons separately. Ticino, the one Italian-plurality canton, appears only in the all-canton columns.

4 Empirical Approach

4.1 Specification

For canton i and referendum r , the estimating equation is

$$Y_{i,r} = \alpha + \beta_X X_i + \beta_{Fr} Fr_i + \beta_{Abs} Abs_i + \beta_{Pr} Pr_i + \gamma_D \ln D_i + \varepsilon_{i,r}, \quad (\text{OLS})$$

where $Y_{i,r}$ is the yes-share in percentage points, X_i the wine measure of §3.1.3, and the remaining terms are the French-speaking share, absinthe-industry presence, the Protestant share and log population density.

Every specification we report is a variation of equation (OLS), excluding or substituting

Table 1: Descriptive statistics: canton-level analysis variables, 1908 cohort

Variable	All cantons (N = 25)					German-plurality (N = 19)		French-plurality (N = 5)	
	Mean	SD	Median	Min	Max	Mean	SD	Mean	SD
Yes-vote share, referendum #68	64.3	11.7	65.1	35.3	83.3	67.7	9.2	50.7	11.9
Petition signatures (per 100 eligible)	20.5	9.5	17.5	3.4	41.3	20.4	9.1	23.9	9.4
Wine revenue share, 1907	4.0	7.5	1.2	0.0	32.1	1.7	3.4	12.1	13.4
Wine volume share, 1907	4.0	6.9	1.0	0.0	25.6	1.8	3.7	10.9	11.1
Attested absinthe producer (0/1)	0.16	0.37	0.00	0.00	1.00	0.05	0.23	0.60	0.55
Absinthe-sector exposure, 1902–06	4.0	13.1	0.0	0.0	63.0	0.3	1.4	18.8	26.1
French share, 1900	16.5	31.3	0.5	0.1	86.5	1.4	3.7	77.1	9.7
Protestant share, 1900	43.1	35.0	48.2	1.3	90.2	44.1	34.4	47.5	39.3
Tavern employment share, 1905	2.09	1.01	1.94	0.75	5.18	1.91	0.64	2.18	1.31
Tavern and inn employment share, 1905	3.99	1.57	3.49	1.48	6.82	3.93	1.51	3.74	1.69
Population density, 1907 (per km ²)	264	687	108	15	3510	292	782	202	234
Log population density, 1907	4.67	1.12	4.68	2.73	8.16	4.68	1.15	4.77	1.20

Notes: Every variable is summarized on the scale at which it enters the regressions. Percentage term measures include: national shares (wine revenue, wine physical output, absinthe-sector exposure) defined as the canton's share of the Swiss total and sum to 100 across cantons. The French and Protestant shares are within-canton percentages, of the 1900 census population and of Protestants plus Catholics respectively. The tavern employment shares are within-canton percentages of male employment, net of cottage industry, in class 301 of the federal business census of 9 August 1905 — restaurants and taverns not attached to an inn — and in classes 300+301, which add inns, hotels and boarding houses (BFS, 1908a). The yes-share of votes (64.3) is unweighted; it treats half-cantons and full cantons with equal weight. Absinthe-sector exposure is the canton's share of national purchases of 95-degree spirit from the federal monopoly over 1902–1906 (Milliet 1907 annex) after the record-verified exclusion of non-absinthe firms; the attested-producer indicator marks Neuchâtel, Geneva, Vaud and Basel-Stadt.

some selection of the regressor set. The columns of Table 2 enter the controls in a fixed order, holding the not-yet-entered coefficients at zero: the wine measure alone, then the French share, then the Protestant share, then absinthe-industry presence, then log population density. Two further columns repeat the last two with the absinthe conditioner in its continuous rather than its discrete form. The order is fixed in advance and does not vary across tables.

The petition arm re-estimates equation (OLS) with its right-hand side unaltered, putting the signature rate P_i of §3.1.2 per 100 eligible from the 1906 vote eligibility records. We use this in place of the yes-share and read the harvest-year and population covariates at their 1906 values; the census-based shares keep their own dates. The specification changes are annotated in the table, along with which coefficient carries the claim in each arm. The wine measure is focal at the vote, in Table 2, and the Protestant share of §3.2.2, the temperance measure, at the petition. Table 4 reports the four readings and the effective degrees of freedom of the Protestant coefficient in both of its arms, because the cross-arm comparison of that one coefficient is what the table is built for, and the wine coefficient's readings belong to Table 2. In the tavern column the claim coefficient is the tavern employment share, measured as within-canton employment for men, since only men could vote or sign the initiative at the time, net of cottage-industry employment in the category. This gives the maximally strict match of potential initiative signatures to their within-canton employed members. In the petition arm the absinthe regressor enters as the attested-producer indicator; the continuous encoding is excluded for brevity. The tavern employment share, defined in the note to Table 4, enters one column of Table 4 on its own, as a further measure of what a ban put at stake. Our outcomes are what each canton's voters recorded, their signature rate on the petition and their yes-share at the vote, and not the financing of the campaign behind them. We do not test whether any interest financed that campaign, and our argument does not require it.

Wine and tavern never enter a column jointly. Their correlation over the cantons is indistinguishable from zero, at 0.030 by Pearson, 0.163 by Spearman and -0.029 on the broad tavern measure, and at $N = 25$ a joint column would cost a degree of freedom we have little room to spend. We report the canton-level leverage and influence anatomy for the interest coefficient of each arm, the wine coefficient at the vote in Appendix Table A.11 and the tavern coefficient in Appendix Table A.17. Both rest on a few cantons, with effective degrees of freedom of 3.57 and 3.60, and each table shows which cantons carry the estimate and what deleting each one does to it.

4.2 The wine measure, and what price introduces

We report wine revenue share as the focal measure and volume share beside it. The two differ by one term: wine revenue is volume scaled by the canton's average wine price relative to the national average, so the wine revenue share is the volume share weighted by local price levels.

That weighting is the reason to prefer it, on two grounds. The first is substantive. If absinthe competed with wine on price, then scaling by local prices puts more weight on cantons where absinthe was the more available substitute and less where wine was cheap. The second is theoretical. In the Stiglerian account what is at stake in regulation is a transfer of wealth, and the political market distributes to those whose effective demand is highest, denominated in the resources a coalition can mobilize (Stigler, 1971, p. 12; Peltzman, 1976, pp. 212–213). Peltzman (1976, pp. 214–215) formalizes this with beneficiaries paying in both votes and dollars and the per-capita gain entering support directly. The implication for cross-canton producer weight is that the relevant index is the wine revenue at stake rather than the physical magnitude of output: a canton with high revenue per producer can finance a more effective campaign than one with the same output and lower revenue per producer, and higher per-capita stakes make collective organization easier (Olson, 1965).

We add that price is an estimate and carries its own assumptions: that the price measure is locally accurate, that expected revenue approximates realized revenue, and that the within-canton price distribution does not make the average unrepresentative. None is testable here. The volume share, which is not price-scaled, is reported throughout for that reason, and the two measures agree in sign and closely in magnitude.

4.3 Estimator choice

The yes-share is a bounded quantity, and for a bounded outcome the Papke–Wooldridge fractional logit is the better-specified estimator: a share confined to $[0, 100]$ cannot respond linearly across its whole range, and a linear specification can in principle return fitted values outside the bounds (Papke and Wooldridge, 1996, pp. 619, 621).

That advantage is a property of the boundaries, and these data sit away from them. Where the outcome is interior the logistic link is close to linear, so the two estimators should give nearly the same answer, and they do: the fractional-logit average marginal effects track the least-squares coefficients across every specification and both wine measures, always slightly smaller in magnitude and never of a different sign. No fitted value falls outside the bounds on any referendum, and neither specification test rejects at any vote in the table

(Table A.3). The tests corroborate rather than gate, and at twenty-five observations neither carries much weight in either direction. The failure the fractional logit exists to prevent does not occur here.

What differs is what can be done with each estimate afterwards. Least squares admits three finite-sample corrections that a quasi-maximum-likelihood estimator does not: HC3, the Bell–McCaffrey adjustment, which re-weights both the variance and the degrees of freedom for leverage, and the wild bootstrap. At twenty-five observations these are not refinements but the difference between an interval that is honest about the sample and one that is not. The fractional logit has one further reading, a permutation test whose refits do not always converge at this sample size. We report both estimators throughout and take the inference from least squares, where the corrections are available.

Because the fractional-logit coefficients index a log-odds transform rather than the yes-share itself, we report average marginal effects throughout, so that the two estimators are read on one scale.

4.4 Collinearity diagnostics

Because the wine measure and the French-speaking share describe overlapping sets of cantons, the two rows at the foot of Table 2 carry a collinearity diagnostic for each column. Both answer one question: across the twenty-five cantons, how close does some weighted combination of the regressors, the constant included, come to being zero in every canton? When no combination comes close, each coefficient is separately determined by the data. When one does, the data pin down a weighted sum of the coefficients in that combination but not the coefficients themselves, which then trade off against each other almost freely. The variance inflation factor (VIF) asks this of one regressor at a time. For regressor j it is $1/(1 - R_j^2)$, where R_j^2 is the fit from regressing that regressor on the other regressors in the column, so it states how much of the regressor’s variation across cantons the remaining controls already span. We report it centered, meaning that the auxiliary regression carries an intercept and the fit is measured around means, which is the convention O’Brien (2007, n. 1) endorses; it is reported for the wine measure in every column, and the table note names the regressor on which each column’s maximum falls.

The Belsley–Kuh–Welsch condition number asks the same question of every combination at once, which is what lets it see a dependency that involves the constant. Write the regressors of a column, the constant included, as the columns of a matrix with one row per canton, and rescale each column to unit length so that units drop out; what remains of

each regressor is its direction, so the diagnostic depends on the angles between regressors and not on their scale. Some weighted combination of those columns comes closer to zero across the twenty-five cantons than any other. The closer it comes, the more nearly one regressor is a linear function of the rest, and the less the data can tell the coefficients in that combination apart. The condition number is the ratio of the largest spread the data have along any such combination to the smallest, so it equals one when the columns are mutually orthogonal and grows without bound as a near-dependency forms (Belsley et al., 1980). It is a single number and cannot say which regressors form the combination. That is the job of the variance-decomposition proportions of Belsley (1991, pp. 35, 38–39), which allocate each coefficient’s variance across those combinations. On his rule a near-dependency is present only when a combination with a high condition index also carries most of the variance of two or more coefficients, and both parts are required: a poorly determined combination that no coefficient loads on is harmless, and a combination that carries a coefficient’s variance is harmless when the data have ample spread along it. None of this is a test. The customary bands, with indices below ten read as weak and those above thirty as strong, are conventions; their author declines to fix a number and disclaims any reading of the indices as significance levels (Belsley, 1991, pp. 38, 43), and the same holds for the VIF cutoffs, which O’Brien (2007, p. 681 and n. 2) traces to textbook habit rather than to any inferential result.

We follow Belsley (1991, pp. 36–38) in keeping the column of ones and leaving the columns uncentered. Centering a regressor means subtracting its mean, which makes it orthogonal to the constant by construction, so a dependency between a regressor and the intercept is invisible once the columns are centered; Belsley calls that masking the constant’s role rather than measuring it. The VIF, whose auxiliary regressions carry an intercept, never treats the constant as a candidate for a dependency, and the condition number, being one number, cannot name the regressors in one. Each is blind where the other sees, which is why both are reported.

One reading of the VIF does not carry over to this paper. Under the classical variance formula the VIF is the factor by which collinearity raises a coefficient’s sampling variance relative to an orthogonal design, but that formula has no place for leverage, and the heteroskedasticity-robust (HC3) standard errors of the next subsection weight each squared residual by a function of its observation’s leverage. What the VIF states here is a property of the design, the share of a regressor’s variation the controls span, and not the width of any interval in the table. Nor does a low value certify an estimate. The components of a coefficient’s variance are already in its standard error, so a favorable VIF can no more be counted for a result than an unfavourable one against it (O’Brien, 2007, p. 681); collinearity

is one source of a large variance among several, and it is the standard error that carries the conclusion (Goldberger, 1991, §§23.1, 23.5). What the diagnostics can establish is narrower, and it is what we use them for: whether the width of the intervals is a property of the design’s internal geometry or of the information that twenty-five cantons carry.

4.5 Small-Sample Inference

Inference at twenty-five observations cannot rest on a single asymptotic approximation, so we report four readings of every least-squares estimate. None is an extrapolation to a sample this small: the smallest design in the simulations that established HC3 is twenty-five observations with four regressors, tested one coefficient at a time (Long and Ervin, 2000), and the evidence behind the resampling procedures reaches still smaller samples (Anderson and Robinson, 2001; Roodman et al., 2019). Each of the four addresses inferential problems the others do not, which is what makes reporting their agreement paint a more complete picture of the findings’ robustness.

For the referenda studied, ranging from the mid 1890s to 1908, wine production in Switzerland was geographically concentrated. Most cantons produced little or none while a few produced a great deal of wine, with Vaud alone holding roughly a third of the national wine revenue in the 1907 harvest. In brief, a canton’s share of the national total is right-skewed on both measures, the wine revenue share the paper carries and the volume share beside it. The default robust interval is built for a regressor without that shape, and relying on it alone would be conventional rather than wrong. The four readings below are the corrections the study’s data and inference call for, and what they buy is conservative standard error coverage, reducing the chances of false positives at the cost of precision.

Standard errors are heteroskedasticity-consistent (HC3), the most conservative of the MacKinnon–White finite-sample correction family (MacKinnon and White, 1985). At $N = 25$ cantons, the choice is the bare minimum correction, since simulations show that uncorrected robust estimator rejects a true null more than twice as often as it should.⁶

We report three further readings of every wine coefficient, each addressing an inferential problem that HC3 cannot address alone. HC3 corrects the variance and nothing else: the statistic it produces is still compared against a t distribution with the usual degrees of freedom, the observations less the parameters. For a regressor as skewed as the wine share, that comparison is where coverage fails. Intervals built this way cover the true value less often than they claim to, and the fault lies in the reference distribution rather than in

⁶Using HC3 standard errors holds the size distortion below two percentage points (Long and Ervin, 2000).

the sample size ([Imbens and Kolesár, 2016](#)). The Bell–McCaffrey adjustment changes two things to repair it. It estimates the variance with a second heteroskedasticity-consistent estimator (HC2), closely related to HC3 but somewhat smaller. It replaces the usual degrees of freedom with a considerably smaller value computed from the design itself, which is where any skewness in the regressor is carried. In our case, the effective degrees of freedom are lower, because the variance estimate for the wine coefficient rests on a handful of cantons rather than on all twenty-five. Since a t distribution with fewer degrees of freedom has fatter tails, a result must clear a higher bar to reach any given significance level. The two changes pull in opposite directions. The smaller variance estimator lowers the p -value on its own; the fatter-tailed reference distribution raises it; and which dominates varies across specifications, so the Bell–McCaffrey and HC3 readings can fall on either side of one another. That disagreement is informative rather than troubling. It is the difference between correcting the variance alone and correcting the distribution the variance is compared against, and [Imbens and Kolesár \(2016\)](#) recommend computing the adjustment routinely for exactly this case.

A wild bootstrap- t with Rademacher weights and the null imposed dispenses with the reference distribution altogether ([Roodman et al., 2019](#)). Rather than assuming any shape for it, the procedure builds it directly, flipping the sign of each canton’s residual at random and recomputing the statistic many times. Because every residual is flipped in place, each canton keeps its own error variance, so no assumption about the form of the heteroskedasticity is needed. The cost is that the observed residuals stand in for the unobserved errors. A Freedman–Lane permutation test replaces the distributional assumption with a randomization argument ([Freedman and Lane, 1983](#); [Anderson and Robinson, 2001](#)): it asks how often a t statistic this large would arise if the part of the yes-vote not explained by the controls were shuffled across cantons at random. A permutation test is exact only when the errors are exchangeable, meaning they could be reassigned among cantons without changing their joint distribution, and heteroskedasticity breaks that condition. Computing the test on a heteroskedasticity-robust t statistic rather than on the raw coefficient restores its validity ([DiCiccio and Romano, 2017](#)), but the restored guarantee holds in the limit rather than at twenty-five: the test’s rejection rate approaches its nominal level as the sample grows, and at this size it is an approximation like the others. Because no two of the four rest on the same approximation, it is their agreement rather than any single reading that carries the inference.

5 Results

5.1 Absinthe Ban Referendum

Table 2 reports the least-squares cascade for the 5 July 1908 vote. The raw association between a canton's share of national wine revenue and its yes-share, column (1), is negative and indistinguishable from zero. That column mixes two things the design separates: the wine cantons are disproportionately French-speaking, and the French-plurality cantons supported the ban by 17.1 points less, on average, than the German-plurality ones. Column (2) enters the French share, the co-explanatory variable the design holds fixed, and the wine coefficient is 0.547 (0.257), with the French share at -0.316 (0.087), each additional percentage point of French-speaking population lowering the yes-share by about a third of a point. The Protestant share and the producer indicator, columns (3) and (4), move the wine coefficient little, to 0.639 (0.207) and then 0.620 (0.232) in the preferred specification. The Protestant share enters negative and does not reach conventional significance once the producer indicator is in. The indicator itself carries no signal, with an interval that spans both signs, and the same holds for the continuous absinthe measure in columns (6) and (7).

Because the wine measure is a share of a national total, the coefficient prices each canton's stake in the national outcome rather than the intensity of viticulture within it. One percentage point of national wine revenue is associated with 0.620 points of yes-share, holding the French share, the Protestant share and producer status fixed, so between two cantons alike in those three respects, one at the mean share of four percent (Table 1) and one with no wine at all, the specification attributes about 2.5 points of yes-share to the wine share. The conditioning carries the sign: the unconditional comparison in column (1) runs the other way, because the cantons with wine are the cantons that speak French. The two interests are competing coalitions, and that is why neither coefficient is legible alone. The French share predicts lower support for the ban and so does absinthe presence; wine is the only regressor that predicts higher. Because the same cantons carry all three, a specification omitting either of the two that matter leaves the other confounded, and the wine effect is not detectable at all until language is held fixed. The Protestant share is statistically indistinguishable from zero in most specifications at the vote stage, and the remaining terms are limited by precision rather than by sign. All four readings of the preferred column reject at five percent: .015 on HC3, .032 under Bell-McCaffrey, .023 on the wild bootstrap and .022 on the Freedman-Lane permutation test. The Bell-McCaffrey reading is the most demanding of the four because it is referred to 3.57 effective degrees of freedom rather than 20. That value is a property of the design and not of the outcome (Bell

and McCaffrey, 2002, p. 176), and at so few degrees of freedom the adjustment errs on the side of under-rejection (Bell and McCaffrey, 2002, Table 3).

The remaining columns stress the estimate. Adding log population density in column (5) moves the coefficient to 0.549 (0.325), less than a third of a standard error from the preferred value, while the density term itself is indistinguishable from zero and the interval widens; the four readings range from .078 to .107. Columns (6) and (7) replace the producer indicator with the continuous absinthe measure and give 0.512 (0.292) and, with density, 0.438 (0.316), with readings from .032 to .095 and from .109 to .182. No column after the first turns the sign, the French share stays between -0.241 and -0.316 throughout, and the volume measure gives the same picture, with a preferred-column coefficient of 0.582 and readings between .060 and .074 (Appendix Table A.5). Why the density columns widen the interval without moving the estimate is a question about where the wine measure's variation sits among the cantons, and we take it up after the collinearity diagnostics.

The two rows at the foot of Table 2 rule out one explanation for the width of its intervals. In the preferred specification, column (4), the condition number is 4.63 and the wine measure's variance inflation factor is 1.55, which says that roughly a third of the wine measure's variation across cantons is predictable from the French share, the Protestant share and producer status, and two thirds is not. The wine measure's own factor rises to at most 1.90 as controls are added, and the column's maximum sits on the French share, at 2.44, as the overlap between viticulture and the French-speaking cantons predicts. By Belsley's procedure the decomposition step is never reached in column (4): an index of 4.63 lies well inside the range read as weak, so no direction of the design qualifies for inspection, whatever the proportions on it. The two large condition numbers in the table, 17.87 in column (5) and 13.19 in column (7), belong to the columns that add log population density, and they register a dependency between the intercept and the density term rather than one among the substantive regressors.⁷ No near-dependency registers in the preferred specification, and the wine coefficient draws almost none of its variance from the one that registers elsewhere. What the diagnostics leave standing is a narrower question. The intervals in Table 2 are not widened by near-dependence among the regressors; what sets their width is the number of cantons and where among them the wine measure's variation sits, which we take up below.

Where that variation sits is reported canton by canton in Appendix Table A.11. A

⁷Log density has little spread relative to its level, so its unit-scaled column lies close to the column of ones, and a convention that keeps the constant registers exactly that. On the top index of column (7) the constant and the density coefficient draw .947 and .979 of their variance, and the wine coefficient .044 of its own. Centering the density term removes the dependency and changes no coefficient but the constant, which is what identifies it as a property of the reporting convention rather than of the specification.

Table 2: Wine revenue and the 1908 absinthe ban — corrected-absinthe specification

	Raw	+ French share	+ Protestant share	+ producer (binary)	+ population density	+ purchase share	+ population density
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Wine revenue share	−0.196 (0.219)	0.547** (0.257)	0.639*** (0.207)	0.620** (0.232)	0.549 (0.325)	0.512* (0.292)	0.438 (0.316)
French share		−0.316*** (0.087)	−0.315*** (0.073)	−0.280** (0.119)	−0.311 (0.193)	−0.249** (0.116)	−0.241** (0.113)
Protestant share			−0.083* (0.046)	−0.069 (0.052)	−0.048 (0.068)	−0.062 (0.051)	−0.030 (0.068)
Absinthe (binary) [†]				−4.361 (8.095)	1.371 (15.499)		
Absinthe (share) [‡]						−0.198 (0.284)	−0.185 (0.205)
Log population density					−2.417 (3.709)		−2.081 (2.651)
Constant	65.13 (2.58)	67.38 (2.08)	70.56 (2.77)	70.17 (2.80)	80.42 (15.52)	69.86 (2.72)	78.30 (10.95)
R^2	0.02	0.51	0.56	0.57	0.60	0.59	0.62
Adjusted R^2	−0.03	0.46	0.50	0.49	0.49	0.51	0.52
AIC	196	181	180	181	182	181	181
BIC	199	185	185	188	189	187	188
Small-sample diagnostics for wine coefficient (p-values)							
HC3 robust	0.379	0.044	0.006	0.015	0.107	0.095	0.182
Bell–McCaffrey	0.385	0.072	0.025	0.032	0.086	0.080	0.141
Wild bootstrap- t	0.435	0.036	0.031	0.023	0.097	0.032	0.109
Freedman–Lane	0.325	0.059	0.016	0.022	0.078	0.080	0.135
Bell–McCaffrey effective D.O.F.	2.04	3.58	3.74	3.57	4.01	4.09	4.41
BKW condition no. (with constant)	1.68	2.48	3.56	4.63	17.87	4.79	13.19
VIF (wine revenue share)	1.00	1.47	1.53	1.55	1.63	1.82	1.90

Notes: Ordinary least squares estimates; $N = 25$ cantons. The dependent variable is the canton yes-share in the 5 July 1908 referendum. Wine is the canton's share of national wine revenue ($\times 100$), 1907 harvest — the last complete harvest before the vote. *, **, and *** mark $p < 0.10$, 0.05, and 0.01, on heteroskedasticity-robust (HC3) p -values; the diagnostic rows report, and are labeled by, their own schemes (HC3, Bell–McCaffrey, wild bootstrap- t , Freedman–Lane permutation). The variance inflation factor (VIF) is centered and reported for the wine revenue share; the specification maximum falls on the French share (2.44 in column 4, 2.82 in column 7). Condition numbers are Belsley–Kuh–Welsch, constant included. [†] the attested-producer indicator; [‡] the canton's share of verified 1902–06 purchases of monopoly spirit, both defined in §3.1.4. The two encodings never enter the same column; the raw purchase share is in Appendix Table A.6 and loose membership in Appendix Table A.7.

canton's leverage is the weight it carries in its own fitted value, and it is fixed by the regressors alone: it measures how far that canton's combination of wine share, language, confession and producer status lies from the center of the sample, and how it voted plays no part in it. The twenty-five leverages sum to the five estimated parameters, so the average is one fifth, and twice that average is the usual line above which a value is worth noticing (Hoaglin and Welsch, 1978, pp. 18–19). Four cantons sit above it: Vaud at 0.671, Basel-Stadt at 0.551, Fribourg at 0.452 and Valais at 0.435. The effective degrees of freedom reported in Table 2 summarize the same design matrix, though along the wine coefficient's own direction rather than from this column.

An unusual canton is not by itself a problem, and the second diagnostic is what separates the two ideas (Hoaglin and Welsch, 1978, p. 20). A canton with distinctive regressors pulls the fitted surface toward itself only if its vote is also surprising given those regressors; if it votes as the model would predict, its distinctiveness supplies information and distorts nothing. Influence combines the two, weighting how far the canton sits from the center by how far its vote falls from its own fitted value, so a canton must be unusual in both respects at once to move a coefficient. That is why the two columns are read together rather than separately, and the four possible combinations mean four different things. High leverage with low influence is the informative case, a canton that carries much of the regressor's range and behaves as expected. Low leverage with low influence is an ordinary canton. High influence with low leverage would mean a vote that the regressors cannot account for. High leverage with high influence is the only combination that warrants inspection, because there a single canton both occupies an unusual position and pulls the estimate from it. In our design one canton qualifies, and the standardized measure puts it past the conventional threshold of two over the square root of the sample: Fribourg, at -0.691 (Belsley et al., 1980). The sign says which way it works. Fribourg holds almost no wine, and it pulls the wine coefficient down, so removing it raises the estimate from 0.620 to 0.809; the one canton with material influence on the result is working against it. The cell that would signal an unaccountable vote is empty.

Deleting each canton in turn leaves the wine coefficient positive and at least 0.550, against 0.620 in the full sample, with Neuchâtel and Fribourg at the extremes. At the ten percent level a single deletion costs significance, and it is Vaud; the largest p -value among the other twenty-four is .080. That one exception is informative rather than troubling, because Vaud is not one canton in twenty-five for this purpose. It holds 32.1 percent of national wine revenue, so omitting it removes roughly a third of the variation the coefficient is estimated from, and a point that carries the regressor's spread is expensive to lose for exactly that reason and not because it was distorting anything (Hoaglin and Welsch,

1978, eq. 5.5). What the deletion does to the estimate confirms the reading: the coefficient moves only to 0.579 while its interval widens enough to take the p -value to .177. The two diagnostics therefore identify different cantons, which is the useful result. The canton that moves the estimate most is not the canton whose removal costs precision, and neither is the canton the other one flags.

5.1.1 Robustness Analysis

The wine coefficient moves from -0.196 in column (1) to 0.620 in column (4), and the cascade enters the controls one at a time, which invites the question of how much of that movement each one accounts for and whether the answer depends on the order. Table 3 answers both. The decomposition of Gelbach (2016, p. 522) splits the movement into one contribution per added covariate, and the split is invariant to the order in which they enter, so the fixed order of the cascade does not determine the attribution. Contributions are signed so that a positive entry moves the coefficient from its column (1) value toward its column (4) value, which is the reverse of the orientation in Gelbach's own equations; the shares are the same either way. Of the total movement of 0.816 , the difference between the two coefficients, the French share accounts for 0.658 , or 81 percent; the Protestant share contributes 0.079 (10 percent) and the producer indicator 0.079 (10 percent). The continuous reference in column (7) gives the same reading, with the French share at 89 percent of a smaller total movement, and log population density entering with the only negative contribution in the table, -0.014 .

That the language variable carries the movement is the design's premise rather than a finding, and the two coefficients say why. The French share enters negative in every column, between -0.241 and -0.316 , so a canton that is more French-speaking voted less in favor of the ban; the wine share enters positive once language is held fixed, so among cantons alike in language, those with more at stake in wine voted more in favor. The difficulty is that the same cantons tend to be high on both. A specification carrying only one of the two therefore loads it with the other's variation, which is what column (1) shows: without language the wine coefficient is negative, because it is standing in for a linguistic pattern that runs the other way. The decomposition puts a size on that problem rather than solving it. It states nothing beyond that, and in particular it is an accounting identity and not a test of whether a covariate matters: Gelbach (2016, pp. 525–526) works an example in which a block of covariates is jointly significant to four decimal places and yet contributes $+0.11$ with a standard error of 0.90 . Each contribution inherits whatever causal standing its own coefficient has and none of its own, which is why we read the table as a description

of where the adjustment comes from and not as evidence about language.

Table 3: Gelbach decomposition of the bivariate-to-adjusted wine coefficient shift

Added covariate	Column (4), preferred		Column (7), reference	
	To col. (4)	Share	To col. (7)	Share
French share	+0.658	81%	+0.567	89%
Protestant share	+0.079	10%	+0.034	5%
Absinthe producer (attested, binary)	+0.079	10%		
Milliet absinthe purchase share (corrected)			+0.048	8%
Log population density			−0.014	−2%
Total shift	+0.816	100%	+0.634	100%

Notes: Gelbach decomposition of the change in the wine revenue-share coefficient from the raw bivariate model to (i) the column-(4) preferred specification (French share, Protestant share, and the attested-producer binary; the wine coefficient moves from -0.196 to $+0.620$, a shift of $+0.816$) and (ii) the column-(7) continuous reference (corrected Milliet purchase share plus log population density; -0.196 to $+0.438$, shift $+0.634$). Entries are signed adjusted-minus-raw contributions and sum to each column's total; shares are contributions divided by the total shift. This is the reverse of the orientation in Gelbach's (2016) equations (11) and (12); shares are unaffected. The two absinthe encodings are alternative measures of the same variable and never enter together. $N = 25$ cantons.

The cascade shows the estimate surviving the controls we have, and [Oster \(2019\)](#) asks the same question of the control we do not have: how strongly an unobserved variable would have to select, relative to the observed ones, to account for the estimate, and what the estimate would be if it selected exactly as strongly. The check has a precedent in economic history, where what was omitted is often unknowable: [Blum and Dimico \(2018, pp. 386–387\)](#) present it as a way to check “the impact of putative factors which may be imputed to omitted variables”. Appendix [A.5.7](#) sets out the method for a reader who has not met it, and Table [A.10](#) reports it on two baselines, one running from the raw column (1) to column (4) and one that holds language fixed throughout, from column (2) to column (4). The answer is categorical on her first standard and passes her second. Because the controls carry the wine coefficient up through zero rather than toward it, unobserved variables that behaved like the observed ones would raise the estimate rather than erase it, and no degree of same-direction selection produces a zero. Under her calibrated ceiling, and with unobserved variables as strong as the observed ones, the coefficient adjusts from 0.620 to 0.672 on the baseline we rely on and to 0.882 on the other, an adjustment of 0.22 standard errors on the first; the identified set lies within her band of 2.8 standard errors on the first baseline and within it on the second. The exercise is a sensitivity analysis and not an inference procedure. What twenty-five cantons support as inference is the four readings in Table [2](#); the appendix states the assumptions the adjustment rests on, which baseline her results cover and which the design calls for, and what the exercise does not show.

5.2 Absinthe Ban Petition

Table 4 sets the two stages of the campaign beside each other, and the coefficient carrying the claim changes hands between them. At the petition it is the Protestant share, at 0.174 (0.065) in the focal column. All four readings of the focal column reject at five percent: .014 on HC3, .011 under Bell–McCaffrey, .011 on the wild bootstrap and .009 on the Freedman–Lane permutation test, referred to 13.67 effective degrees of freedom. The wine revenue share is not detectable at the petition, entering at -0.177 (0.784), and Appendix Table A.22 puts its four readings at .823, .648, .602 and .697. Its standard error is more than three times what the same coefficient carries at the vote, 0.232, on 2.50 effective degrees of freedom, so the wine coefficient is estimated far less precisely at the petition than at the vote. The standardized panel states the cross-arm contrast in one place: the Protestant share stands at 0.638 standard deviations at the petition against -0.208 at the vote, and the wine revenue share runs the other way, 0.399 at the vote against -0.159 at the petition. In the vote columns the Protestant share is not detectable, at -0.069 (0.052) in the focal column, with every reading between .166 and .199 on 13.32 effective degrees of freedom.

The separation follows the campaign's own calendar. The Blue Cross founded the standing delegation that convened the Bern congress of December 17th, 1905, and the initiative committee, formed after it, ran the collection (Trechsel, 1989, pp. 33, 122, 151–153); the Katholikentag declared for a ban in 1906 and the bishops followed in 1908 (Trechsel, 1989, pp. 105, 107, 152); and the Catholic People's Association called for an emphatic yes on 22 June 1908, with the Methodist pastors endorsing unanimously (Journal de Genève, 1908, p. 3), sixteen months after the signatures were submitted to the Federal Chancellery by 31 January 1907 (Schweizerischer Bundesrat, 1907b, printed 353, PDF p. 13). A reader who sets the two arms side by side can conclude that religion mattered only for getting the petition started, or that temperance had stopped mattering by the time the country voted. Neither is what the coefficients identify. What the Protestant share picks up at the petition is early confessional organization, and the Blue Cross tie is why: Protestants were already organized as a temperance movement when the collection opened, and Catholics, who did not want hard prohibition as uniformly, could unite against a drink with absinthe's reputation more easily. By 1908 both confessions stood on the same side, so at the vote the measure reads the weight of one wing of a coalition the two had closed ranks in. It would be believable that Protestants were more emphatic about the ban, and an emphatic yes and a general yes are the same vote, so no differential is left between the groups for the coefficient to capture, or it is small enough that the null is uninformative about it. Across the two arms the table describes a mobilization in stages: the temperance organization

supplied the signatures and the wine cantons supplied the majority, and we do not find wine to have been a player in the initiative process, at least not in a way our measures capture. The proxy will misread a canton where Catholic organization came early, as it may in Fribourg, which hosted the congress the manifesto invokes.

Appendix Table [A.22](#) rules out one account of the widths in these columns. In the focal column the condition number is 4.66 and the Protestant share's variance inflation factor is 1.25, which says that about a fifth of that measure's variation across cantons is predictable from the wine revenue share, the French share and producer status, and four fifths is not; the column's maximum sits on the French share, at 2.43. In the tavern column the condition number is 6.43 and the tavern employment share's own factor is 1.04, with the maximum again on the producer indicator, at 2.15. By Belsley's procedure the decomposition step is never reached in the focal column or in the tavern column, where both indices lie well inside the range read as weak ([Belsley et al., 1980](#)), so no direction of either design qualifies for inspection, whatever the proportions on it.

The measure in the tavern column is the canton's share of male employment, net of cottage industry, in class 301 of the federal business census of 9 August 1905, restaurants and taverns not attached to an inn ([BFS, 1908a](#), printed 32, PDF p. 208), and Appendix Table [A.19](#) reports it among the petition-arm covariates. The coefficient is -3.894 (1.268), with .006 on HC3, .024 under Bell–McCaffrey, .036 on the wild bootstrap and .010 on the Freedman–Lane permutation test, referred to 3.60 effective degrees of freedom. The estimate therefore rests on a few cantons rather than on all twenty-five, and Appendix Table [A.17](#) shows which: Ticino carries 41.9 percent of its identifying weight (Appendix Table [A.16](#)), and deleting each canton in turn leaves the estimate between -4.643 and -3.355 without turning the sign. Wine and tavern never enter a column jointly, for the reason given with the specification. The column is consistent with a lower signature rate where the consumer-facing drink trade was a larger employer, and we hold it to that; the broad companion that adds the lodging trades is where the measure is weakest, and Appendix Table [A.14](#) sets out both.

Table 4: The 1906 anti-absinthe petition versus the 1908 ban vote

	Petition arm focal (4)	Tavern focal	Petition arm density-in (5)	Vote arm focal (4)	Vote arm share form (7)	Vote arm producer form (5)
Wine revenue share	−0.177 (0.784)		−0.227 (0.740)	0.620** (0.232)	0.438 (0.316)	0.549 (0.325)
French share	0.099 (0.115)	0.057 (0.069)	0.083 (0.158)	−0.280** (0.119)	−0.241** (0.113)	−0.311 (0.193)
Protestant share	0.174** (0.065)	0.166*** (0.053)	0.187** (0.076)	−0.069 (0.052)	−0.030 (0.068)	−0.048 (0.068)
Log density			−1.372 (3.525)		−2.081 (2.651)	−2.417 (3.709)
Absinthe	−3.462 (7.062)	−0.935 (5.149)	−0.081 (13.039)	−4.361 (8.095)	−0.185 (0.205)	1.371 (15.499)
Tavern employment share		−3.894*** (1.268)				
Constant	12.59 (2.67)	20.67 (3.89)	18.36 (14.99)	70.17 (2.80)	78.30 (10.95)	80.42 (15.52)
Standardized β:						
Wine revenue share	−0.159		−0.203	0.399	0.282	0.354
Protestant share	0.638	0.609	0.685	−0.208	−0.088	−0.144
Tavern employment share		−0.413				
R^2	0.39	0.54	0.40	0.57	0.62	0.60
Adjusted R^2	0.27	0.45	0.25	0.49	0.52	0.49
Claim coefficient:						
HC3 robust p	0.014	0.006	0.024	0.199	0.668	0.490
Bell–McCaffrey p	0.011	0.024	0.016	0.175	0.635	0.418
Wild-bootstrap- t p	0.011	0.036	0.014	0.178	0.633	0.427
Freedman–Lane p	0.009	0.010	0.013	0.166	0.629	0.424
Effective df	13.67	3.60	12.41	13.32	11.07	12.54

Notes: Ordinary least squares; $N = 25$ cantons in every column. In the two petition-arm columns and the tavern column the outcome is signatures per 100 eligible voters, a rate whose denominator is the cantonal eligibility roll of the Lebensmittelgesetz federal vote of 10 June 1906 (referendum #65); in the vote columns it is the canton yes-share in the referendum of 5 July 1908. *, **, and *** mark $p < 0.10$, 0.05, and 0.01, on every slope coefficient's heteroskedasticity-robust (HC3) p -value. The diagnostic rows report, and are labeled by, their own schemes (HC3, Bell–McCaffrey, wild bootstrap- t , Freedman–Lane permutation); they and the Bell–McCaffrey effective degrees of freedom belong to the coefficient named in the Claim coefficient row directly above them, the Protestant share in every column except the tavern column, where it is the tavern employment share. The Absinthe row is the attested-producer indicator in the petition-arm focal and density-in columns, the tavern column and the vote-arm focal and producer-form columns, and the canton's share of verified 1902–06 purchases of monopoly spirit in the vote-arm share-form column. The tavern measure is the canton's share of male employment, net of cottage industry, in class 301 of the federal business census of 9 August 1905 — restaurants and taverns not attached to an inn, excluding inns, hotels and boarding houses (BFS, 1908a). The drive ran from late 1905 to late 1907, so 1906 is the only year it spans in full, and the wine and population covariates enter at their 1906 values; the vote columns use the 1907 wine year, the last complete before the ballot — each arm's inputs anchored to its own event. The standardized-coefficient panel restates the wine revenue share, Protestant share and tavern employment share coefficients in standard-deviation units for cross-column comparison. Alternative tithe-based measures of temperance subsidy payouts are given in Appendix Tables A.20–A.21.

5.2.1 Falsification

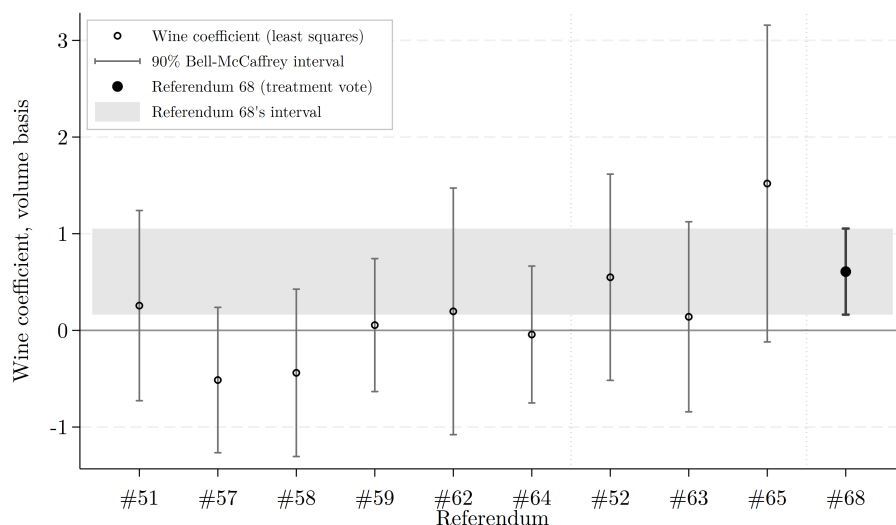


Figure 2: Wine coefficient by referendum, volume basis, at the column (3) specification (the wine, French and Protestant shares; Appendix Table A.5). Points are least-squares estimates with 90 percent Bell–McCaffrey intervals computed on each vote’s own effective degrees of freedom. Dotted lines separate the falsification, corroborative and treatment votes. The shaded band is referendum 68’s interval, drawn across the plot for comparison.

No vote of this period is a true placebo for the absinthe ban, and whether one could be constructed is not the question this set answers. What it can answer is whether the wine coefficient appears only where a wine interest was at stake. Every estimate quoted below for the ten votes is set out in Table A.12 and Table A.13, so that the arms can be read off the tables rather than taken on our word. The two arms and the criteria that fix them are given in §3.1.1, and were settled before any of these votes were estimated. Figure 2 and Table A.13 read them on the volume basis, which is the only measure on which all ten votes exist and is also the less favorable of the two for the treatment vote. Across the six falsification referenda the wine coefficients cluster near zero, the largest in absolute value being -0.514 on referendum #57, and the intervals are too wide to resolve any of them, the least-squares Freedman–Lane readings at the revenue basis running from .181 to .874. A falsification vote reaches the display only if its Bell–McCaffrey, Freedman–Lane and Freedman–Lane average-marginal-effect readings all sit at or above the ten percent line, and the build halts on the first that crosses, so the table exists because none of them did (Table A.3).

On every falsification vote no wine relationship is detectable, and we do not rule out imprecision. The coefficient appears where the design says it should. On the absinthe vote

the wine revenue share returns 0.639 in column (3) of Table 2, the specification every vote in the set shares, at .016 on the least-squares Freedman–Lane test. Of the three corroborative votes, referenda #52 and #63 do not produce significance. Referendum #65 does, at the margin: its least-squares Freedman–Lane reading is .083, at the ten percent line, and the other inference schemes fall on either side of that line (Table A.3), which is why we read it as weakly corroborative and no further. Its own coalition specification, conditioning on cattle and on whether a pre-existing canton-level food control institution existed, is in the appendix (Table A.18); the main text quotes the specification the set shares.

Referenda #52 and #65 are the food-law pair, and they are the two votes on which a wine interest could be engaged, because wine was regulated as food and not as a spirit: the act reserved the federal legislation on distilled spirits and never touched absinthe ([Schweizerische Bundesversammlung, 1905b](#), p. 18). Referendum #52, of July 1897, was the Federal Decree of 26 March 1897 that added Article 69bis to the Federal Constitution and empowered the Confederation to legislate on the trade in foodstuffs and stimulants ([Schweizerische Bundesversammlung, 1897](#), p. 233); the article left execution to the cantons under federal supervision and with federal money and reserved control of imports at the national frontier to the Confederation ([Schweizerische Bundesversammlung, 1897](#), p. 234). Referendum #65, of June 1906, was the act that used that power ([Schweizerische Bundesversammlung, 1905b](#), p. 1). The message that carried the 1897 article had already said that wines were very often adulterated with artificial wine and had listed wines with butter, fats and oils for inspection at the frontier ([Schweizerischer Bundesrat, 1895a](#), pp. 776 and 805). Its precursor, vote #52, has some additional details in Appendix A.9.1.

What the act did was carry, under the name of consumer protection, the demands of a mixed agricultural coalition, and wine and cattle and dairy were the substantive components on that side. The Farmers' Union declared control at the frontier the cardinal point of the law, and a law without it unacceptable, as the committee's rapporteur reported ([Schweizerische Bundesversammlung, 1905a](#), National Council, 9 June 1903, p. 418); the same rapporteur said the customs experts at the frontier were added mainly at its demand, and also at that of other interested circles, the honest wine merchants and the vineyard owners among them ([Schweizerische Bundesversammlung, 1905a](#), National Council, 10 June 1903, p. 442). The frontier laboratories the union had asked for were refused by the Council of States and by the committee, with those experts put in their place ([Schweizerische Bundesversammlung, 1905a](#), National Council, 9 June 1903, p. 419), and the act made even the experts optional ([Schweizerische Bundesversammlung, 1905b](#), p. 9). A second clause, the power to forbid mixtures of natural foodstuffs with substitutes, drew the same protective reading. The Council of States committee's spokesperson, moving to strike that

clause, told his chamber that it was “less a provision [. . .] to assure the sincerity of the goods” than “a measure of protection in favor of natural products”, meant to support them “against an often ruinous competition”, and that “the circles of the farmers and of the wine-growers” counted on it ([Schweizerische Bundesversammlung, 1905a](#), p. 892), and he named “margarine and artificial wine” as the targets ([Schweizerische Bundesversammlung, 1905a](#), p. 893). The same committee’s spokesperson said in October that the stated aim had been public health and the real aim the intermediaries between producers and consumers ([Schweizerische Bundesversammlung, 1905a](#), p. 1117). In the first chamber to debate the bill, a member asked “who can blame the hard-pressed farmers” for the “ulterior motive” of protection against foreign produce, and warned that without it agriculture “would declare that it accepts no law” ([Schweizerische Bundesversammlung, 1905a](#), p. 285). The government’s own statement of purpose openly named the protection of “the genuine producer (farmer and manufacturer)” against “dishonest competition” as an aim of the law ([Schweizerischer Bundesrat, 1899b](#), p. 615). It was secondary to consumer protection in the 1895 message ([Schweizerischer Bundesrat, 1895a](#), p. 803) but openly co-equal with it by 1899 ([Schweizerischer Bundesrat, 1899b](#), p. 615). The 1895 message relayed the food chemists’ expectation that a federal law would protect domestic wine-growing, butter producers and beekeepers against dishonest competition ([Schweizerischer Bundesrat, 1895a](#), pp. 794–795).

Meat originally had the one obligatory inspection regime proposed, only to have it cut to what the farm vote would bear: compulsory inspection was confined to meat destined for sale and home slaughter was left to the cantons ([Schweizerische Bundesversammlung, 1905b](#), pp. 3–4), an exception the Council of States spokesperson called “purely intended to save the law before the popular vote” ([Schweizerische Bundesversammlung, 1905a](#), p. 65) and “a veritable right to poison” ([Schweizerische Bundesversammlung, 1905a](#), p. 33), with the National Council spokesperson calling the meat clause “one of the most dangerous reefs for the little ship of the law” ([Schweizerische Bundesversammlung, 1905a](#), p. 464) and the requirement that the inspector be a veterinarian struck in the Council of States by twenty-six votes to four ([Schweizerische Bundesversammlung, 1905a](#), p. 270); margarine and artificial wine were handled as one class of substitute from the 1895 message ([Schweizerischer Bundesrat, 1895a](#), p. 791) to the National Council’s final reading of the mixture clause, where its rapporteur gave mixed wine, natural with artificial, and butter mixed with margarine among his examples ([Schweizerische Bundesversammlung, 1905a](#), National Council, 29 September 1905, p. 1050).

The Zurich wine merchants’ association explicitly asked for the making of artificial wine to be banned outright ([Schweizerische Bundesversammlung, 1905a](#), National Council, 5 June 1905, p. 393), and the Swiss association repeated the demand at nearly every

stage of the deliberations, only for the commission to refuse each time ([Schweizerische Bundesversammlung, 1905a](#), National Council, 29 September 1905, pp. 1050–1051). Instead, it left the power to prohibit deceptive mixtures to the Federal Council under Article 54 ([Schweizerische Bundesversammlung, 1905b](#), p. 17). The Swiss Farmers' Union (Schweizerischer Bauernverband), founded in 1897 and run from its secretariat by Ernst Laur ([Baumann, 2006b,a](#)), had power as an Olsonian special interest group ([Olson, 1965](#)), able to pork-barrel their producer protections in the textbook sense. The food-safety concerns were the *prima facie* justification, but the producer protections from foreign and inferior goods was explicitly conjoined with this motive. Loaded by an agricultural coalition with a wine protection and a dairy protection riding on it, we interpret this as why the canton wine share loads so uniformly on this vote and on no other in the set. Its precursor, vote #52, is discussed briefly in the appendix.

Only our adjusted model for referendum #65 (Table [A.18](#)) reaches the five percent level for the wine measure, on every reading but Bell–McCaffrey's, which falls just above it. Its precursor, referendum #52, was proposed as a public health solution of adulterated foods (wine included); Additional details on it are in § [A.9.1](#). The other secondary referenda show imprecise nulls. Imprecise nulls cannot rule out a wine relationship on those votes, but if the wine share's significance were an artifact of some omitted variable associated with a canton's share of the national wine revenue or volume, that variable would have to be silent on every other vote in the set, and our results show no significance for it elsewhere.

The falsification and corroboration referenda show an internally consistent track record for the wine measures as predictors for wine-salient referenda. The wine measures are not statistically significant unexpectedly elsewhere. The only significant vote, referendum #65, relates to wine's history as a regulated industry advocating for producer protections.

6 Discussion

Across its two stages the ban drew on two constituencies, temperance at the petition and the wine interest at the vote. Wine's involvement was on record in multiple other instances related to wine interests. The settlement of 1885–87, an agricultural bargain, relieved wine of the entry duties, among other provisions (Section [2.2](#)); the record establishes what wine gained there, not who secured each provision. On the food act of 1905, the Farmers' Union declared control at the frontier the cardinal point of the law, as the National Council committee's rapporteur reported ([Schweizerische Bundesversammlung, 1905a](#), National Council, 9 June 1903, p. 418), and the act did not meet that condition in the form the

union demanded (Section 5.2.1). In referendum #65, which put that act to the voters, the agricultural interests that lobbied for it are strong predictors of vote support on its own coalition specification, albeit with limited effective degrees of freedom (Appendix Table A.18).

Referendum #65 is a wine-salient vote that is not absinthe-salient, and consequently is unconfounded by absinthe producers' interests. The food act it put to the voters regulated wine as a food; it reserved the federal legislation on distilled spirits and nowhere names absinthe (*Schweizerische Bundesversammlung*, 1905b, p. 18). Where it bore on wine, it bore on wine's other competitor, artificial wine (*Kunstwein*). Its own coalition specification, column (3) of Appendix Table A.18, carries three regressors. The wine revenue share carries the wine interest and enters at 0.916 (0.221). Cattle per thousand inhabitants stands for the meat and dairy interests on the same agricultural side of the act (Section 5.2.1) and enters at 0.065 (0.017). We control for whether a canton-level food-control regime already existed in 1895 with a dummy variable, since such a canton could consider its own authority sufficient; it enters at -2.516 (6.792) and is not detectable. When the federal competence was proposed in 1895-96, three such cantons told the federal government as much (*Schweizerischer Bundesrat*, 1896a, pp. 880-882). On the specification every vote in the set shares, the results read #65 as weakly corroborative and no further (Section 5.2.1), with further details in Appendix Table A.18.

Temperance carried the petition to the vote, lowering the cost of supporting the ban for wine-market-salient voters, particularly the agricultural portion. The two-stage inference of our results suggests that the Protestant-leaning temperance movement's organization predicted the signature rate per hundred eligible voters within a canton, whereas the vote shows wine is a significant predictor once the French-speaking share, a major confounder, is held fixed, and the Protestant share is not detectable (Table 4). The wine share is not detectable at the petition, and that is an absence of evidence that wine had a part in the collection, not evidence that they played no part. The American record gives a contrast of venues: *Munger and Schaller* (1997, p. 156) find the drink trade's stake not detectable in the state legislatures' ratifications of 1919 but detectable in the popular votes on repeal in 1933, where the measure of the stake postdates the vote, and they take the change to suggest that producers had effectively organized by then. At the ballot, revealed preference could align cheaply with local economic interest. We read that revealed preference with a caution. Lewis could separate preference from mobilization by counting each side's votes against the eligible voters (*Lewis*, 2008, p. 383). We cannot use turnout that way, because some cantons fined non-voters in 1908 (Section 3), and compulsion pushes voters to a decision even where they hold no strong preference.

Against a murdered family, Trechsel writes, a few possibly unemployed absinthe distillers and wormwood growers counted for little with the broad public, and the ban's supporters had popular emotion behind them, "the more clearly, the farther away from the absinthe-consuming regions" (Trechsel, 1989, p. 153). He judges that a ban on a spirit the great majority of voters had hardly drunk even once was no gauge of the popular mood about alcohol (Trechsel, 1989, p. 153). We read the gradient he describes as placing the moral catalyst of the petition in temperance areas with no stake in absinthe, and because the ban cost most voters so little, it left room for a canton's economic stake to show at the ballot. That premise rests on the contemporary record as well as on Trechsel's modern judgment: Müller, against the ban, relayed the cantons' reports that there was no appreciable consumption of absinthe liqueur in German-speaking Switzerland, nor in Ticino and Graubünden (Schweizerische Bundesversammlung, 1908, National Council, 2 April 1908, p. 27). The French-speaking share turns between the stages: it is positive and not detectable at the petition, and at the vote it is negative, clearing five percent in the focal column on its HC3 reading, the only one the table prints for it (Table 4). Our reading is that, with the wine share and absinthe presence held fixed, the French-speaking share predicts less support for the ban at the vote, despite its overlap with both.

Stigler names four main policies that an industry or occupation may seek of the state: a direct subsidy of money, control over entry by new rivals, powers over substitutes and complements, and price-fixing. Of the third he writes, "Crudely put, the butter producers wish to suppress margarine and encourage the production of bread" (Stigler, 1971, pp. 4–6). On our reading, referendum #65 comes close to that example, and on both of its goods. The act's power over mixtures of natural foodstuffs with substitutes reached margarine beside artificial wine (Section 5.2.1). The commission's other rapporteur said that what the producers aimed at was "in reality, to prevent the manufacture of margarine. They do not hide it" (Schweizerische Bundesversammlung, 1905a, Council of States, 28 June 1905, p. 894). The ban of 1908 is closer to the pairing that Yandle (1999, pp. 5, 7) adds as a footnote to the Stigler–Peltzman theory, in which durable social regulation "evolves when it is demanded by both of two distinctly different groups", one on moral grounds and one that expects to profit from it: here temperance demanded the ban, and the wine-growers were its predicted, indirect beneficiaries. It differs from that statement in two ways that we take up below: wine's thumbprint on viable policy is on the record rather than inferred, and wine need not have paid. On our reading the absinthe ban took Stigler's substitutes route.

The record explains why one spirit could be singled out as public enemy number one (Section 2.2), and that their concession was strategic. The temperance movement, while not

homogeneous, aimed for what was realistically achievable, yet Trechsel's account states most of the movement's societies strove for nothing less than a nationwide ban on alcohol (Trechsel, 1989, p. 171). The anti-ban campaign literature quoted their own argument against them: There Dr Herco[d] of Lausanne says quite plainly: "The prohibition of absinthe will have only a very weak effect on alcoholism in general. What we are striving for, in the more distant future, is the abolition of all alcoholic drinks, wine and beer included. But it is a beginning. . . ." (Gantner, 1908). Still, both of its popular initiatives, against absinthe in 1908 and for local option in 1921–29, were confined to spirits, although the social-hygiene wing stressed the danger of fermented drinks too (Trechsel, 1989, p. 171). We don't infer the gradualist intentions of temperance. Rather, we argue that these attestations are consistent with Peltzman's coalition theory, where a majority-seeker spares the would-be losers who can fight back, so that "some losers will then be made winners when there is a rise in opposition" (Peltzman, 1976, p. 221). We have no evidence that wine interests funded the campaign for the ban, and our results do not attest financial collusion of the kind in Yandle's pairing. We point to the convenient shared goal, which served both the wine interests and the temperance actors' aims, gradualist or otherwise.

What counts is relative, as Becker's corollary has it: a group's political effectiveness "is mainly determined not by its absolute efficiency" at producing political pressure "but by its efficiency relative to the efficiency of other groups" (Becker, 1983, p. 380). While it is true that absinthe can only be coarsely identified, even the strongest version of the data would be unable to overcome the core issue: a four-canton stronghold for absinthe cannot win at the federal level as a mathematical impossibility, all else held equal. A similar case demonstrates this: Poelmans et al. (2022, pp. 215, 221, 223–224) associate larger banned trades with less time under statewide prohibition across the American states, and read national prohibition's passage partly through brewing's concentration in a few states. Their bans were decided state by state; ours was imposed federally, over the cantons. That level of decision is the focal difference: Neuchâtel and Geneva, which rejected the ban, would plainly have preferred to decide the matter themselves.

Our analysis adds details to why absinthe fit the scapegoat role, posited previously by other scholars (Room, 1984, fol. 3; Heimberg, 2000, p. 99; Ben Slimane et al., 2025, p. 13). The central theme, exacerbated by the now-debunked absinthism (Section 1), points to absinthe as a vulnerable target for alcoholism concerns. Crucially, we argue it is economically vulnerable and culturally vulnerable. By contrast, France's petition had passed 400,000 signatures by the spring of 1907, only for the Chamber's hygiene commission to narrowly reject prohibition (Prestwich, 1979, pp. 311–312). Prestwich attributes this to the tax revenue at stake, fifty to sixty million francs a year, to indemnities and to fears of unrest in the

Doubs ([Prestwich, 1979](#), pp. 309, 312). France’s ban only succeeded by wartime decree ([Prestwich, 1979](#), p. 317) and law in 1915 ([Adams, 2004](#), p. 214), and was then “relatively easy”, Prestwich writes, because “the legislation affected only one area of France and compensation was assured” ([Prestwich, 1979](#), p. 318). We read the two-country comparison as a difference in stakes, consistent with Peltzman’s winning-coalition framing.

The 3.57 effective degrees of freedom of the wine coefficient, read against twenty-five cantons, can be read uncharitably as a geographical characteristic dressed up as an interpretable result. In truth, our diagnostics on leverage, influence, and identifying variation suggest this is consistent with the economic theory rather than a damning weakness in inference. Appendix Table [A.11](#) lays these facts bare. The results are indicative of both: wine, absinthe, and French-speaking majorities concentrate in a few cantons, most of which, by no small coincidence, are geographically closer to France.

7 Conclusion

At the petition, cantons with more organized temperance (our proxy is the Protestant share) signed at a higher rate per hundred eligible voters, and the wine share is not detectable. At the vote the two trade places. Among cantons equally French-speaking, a larger share of national wine revenue goes with more support for the ban. Of the other federal ballots, only the food-act referendum of 1906 shows a detectable wine relationship: the Freedman–Lane reading of the specification the ballots share sits at the ten percent line, and the vote’s own specification, conditioning on cattle and on existing cantonal food control, reaches the five percent level on three of four readings.

The record supplies the motive, the ballot the opportunity. Agricultural unions are on the record in both debates: the Swiss Farmers’ Union declared control at the frontier the cardinal point of the food act ([Schweizerische Bundesversammlung, 1905a](#), National Council, 9 June 1903, p. 418), and a deputy for the ban cited the Central Union of French Agriculturists’ call of 1907 for a total ban on absinthe in France ([Schweizerische Bundesversammlung, 1908](#), National Council, 2 April 1908, p. 47). Whether from doctrine or tactics, temperance confined its initiative to spirits and spared wine, aligning the two sides without any need to coordinate (Sections [2.3](#) and [2.5](#)). Once the ban was on the ballot, a vote could follow local economic interest at little cost.

One limit is unavoidable. Attested absinthe producers sat in four cantons, three of them French-speaking, too few for their coefficient to separate from language and too few, on their own, ever to defeat the ban under the double majority. The French ban of 1915 raises

the same question, but France decided it in wartime and without a popular vote.

Our results provide a rare empirical test of the “culprits” of the green fairy’s demise. We show evidence that temperance spearheaded the federal petition, but that the wine market predicted vote support after the petition. In Gantner’s caricature the Blue Cross is the killer, and modern popular accounts, writing of France, name wine an accomplice ([Hicks, 2010](#); [Clifford, 2023](#); [Adams, 2004](#), p. 197). The Swiss record fits the looser arrangement that [Smith and Yandle \(2014, pp. 14–18\)](#) call noncooperative, in which economic interests arrive late to a moral crusade. Our evidence, a canton-level predictor and not a confession, sustains the lesser charge, support at the vote, and is insufficient to convict on the graver one, collusion at the petition. That is not an acquittal. On this evidence wine was less the killer than the ban’s predicted beneficiary or, at most, an opportunistic accomplice.

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A Appendix

A.1 The historical record in full

This subsection gives in full the record that Section 2 states in short form, in that section's order.

The market for drink. The federal record states the cross-sectional regularity as well, for the zone it had already singled out for alcohol consumption:

“In the zone we have already singled out with respect to alcohol consumption, schnapps holds the mastery as the household drink at the between-meals of the agricultural population; only there and then, where cheap wine or cider is abundantly present as a local product, does it share the dominion with it, or is it pushed into second place.” ([Schweizerischer Bundesrat, 1884](#), p. 385)

Industrial alcohol was itself born of a vine disease: when crop diseases struck French vineyards in 1853, distillers turned to beets and cereals as the base for new drinks, and when phylloxera cut the wine crop two decades later that production received “further encouragement” ([Marrus, 1974](#), p. 128).

The settlement of 1885–87. Revenue entered the design as a means and a requirement rather than a measure: the yield had to suffice to relieve the undistilled drinks and, until 1890, to make good the entry duties the cantons were losing, while sparing the Confederation's own finances, and at least a tenth of what reached the cantons was reserved for combating alcoholism ([Schweizerischer Bundesrat, 1886](#), pp. 423–424; [Schweizerische Bundesversammlung, 1885](#), pp. 478–479). Two protections bounded the settlement. The distilling of wine, fruit and their residues, gentian and juniper stayed outside federal law on manufacture and taxation, on the Council's reasoning that such spirits were too dear to be much abused ([Schweizerischer Bundesrat, 1886](#), pp. 422, 440), and from December 1887 only for domestic materials ([Schweizerischer Bundesrat, 1889a](#), p. 185); fermented drinks could bear no special cantonal taxes and no restrictions beyond those needed against adulterated or unhealthy drink, the cantons keeping their powers over public houses and retail sales below two liters ([Schweizerischer Bundesrat, 1886](#), p. 422). The 1886 message singled out the export trade in liqueur and absinthe, an industry working predominantly from foreign fine spirit, and its draft provided on export for refunding the tax and half the duty on that input ([Schweizerischer Bundesrat, 1886](#), p. 468).

The small distillers. The Council's survey counted 1,022 distilleries that made more than two hectoliters a year from starchy materials and sold the product; the 985 with returns reported about 50,000 hectoliters of absolute alcohol in 1885/86, 937 of them less than 100 hectoliters each, below the smallest lot the law would offer, and the Council treated most returns as minima and added 15,000 hectoliters for the medium and small works ([Schweizerischer Bundesrat, 1886](#), pp. 425, 428, 435). By 1898 the monopoly bought from seventy contracted suppliers ([Schweizerischer Bundesrat, 1899a](#), p. 330). Much of the difference was bought. In July 1887 the Federal Council had cantonal officials seal every still that had distilled the newly reserved materials, leaving a few dozen distilleries working under provisional contracts ([Schweizerischer Bundesrat, 1889a](#), p. 123). By the spring of 1890 it had settled 1,171 of 1,376 claims for compensation, not all of them from surveyed distilleries, each owner undertaking, for successors too, never to distill again; and it bought such stills as it could and sold them for scrap ([Schweizerischer Bundesrat, 1889a](#), pp. 185–187; [Schweizerischer Bundesrat, 1890a](#), p. 56; [Schweizerischer Bundesrat, 1892a](#), pp. 811–812).

Supervision and enforcement. The decree of 1890 named ten controllers, charged them first with taking over and controlling the contract distilleries' output and last with penal reports, and drew their nine districts as lists of those distilleries ([Schweizerischer Bundesrat, 1890b](#), pp. 161–164). The free distilleries were the cantons' to supervise ([Schweizerische Bundesversammlung, 1886](#), p. 1312). The Federal Council had to ask the cantons, in 1889 and again in 1892, how they were doing it ([Schweizerischer Bundesrat, 1889b](#), pp. 332–333; [Schweizerischer Bundesrat, 1892b](#), pp. 233–234); parliament's committee found in 1897 that several had not properly carried it out ([Nationalrätliche Alkoholkommission, 1897](#), p. 1016); and some cantonal offices had licensed what the federal law forbade, permits the Administration called unlawful ([Schweizerischer Bundesrat, 1891](#), p. 421). Becker and Stigler add that a rule usually tells no one where violations lie and gives no one a reason to find them, and propose paying enforcers by results ([Becker and Stigler, 1974](#), p. 14). The Council's draft had expected taxed distillers to act as natural and unpaid watchers of the exempt ones ([Schweizerischer Bundesrat, 1886](#), p. 481); the monopoly made them suppliers of the Confederation instead. Over 1887–98 the Administration received 413 complaints and proceeded on 329, of which 265, about twenty-two a year, ended in a penal order; 241 of the 329 concerned unauthorized distilling of monopoly materials and 28 distilling on premises the Confederation had paid to close ([Schweizerischer Bundesrat, 1899a](#), pp. 301–302, 460–461). Fines paid in fell from Fr. 13,883.45 in 1890 to Fr. 537.78 in 1897 ([Schweizerischer Bundesrat, 1899a](#), p. 462); in 1898, when half the year's cases fell to

Solothurn, sixteen penal orders brought in Fr. 3,634.05 (Schweizerischer Bundesrat, 1899a, p. 300; Nationalrätliche Alkoholkommission, 1899, p. 657), though the Council judged the statutory minimum of five times the sum defrauded too high in some cases (Schweizerischer Bundesrat, 1898b, p. 627). At the border, where every lawful import owed its fee, customs fined 553 evasions of the import fee over the same twelve years (Schweizerischer Bundesrat, 1899a, p. 303). The informer's bounty reached people outside the Administration in the monopoly's first years and very little afterwards: informers were paid Fr. 3,682.35 directly in 1890 but Fr. 73.40 in 1898 (Schweizerischer Bundesrat, 1899a, p. 462), and at least one was a policeman (Schweizerischer Bundesrat, 1893, p. 341); the fund's occasional payments to police and other outsiders came to a few hundred francs in a year at most (Schweizerischer Bundesrat, 1896b, p. 366; Schweizerischer Bundesrat, 1897a, p. 312; Schweizerischer Bundesrat, 1898a, p. 99). The Administration's informers' fund changed its meaning in between: founded in 1887/88 for thirds with no particular informer or refused by one (Schweizerischer Bundesrat, 1889a, p. 196), it also took from 1895 the third on every offense the Administration's own officials reported (Schweizerischer Bundesrat, 1895b, p. 185), and the committee explained its receipts for 1896 by the complaints having come, with few exceptions, from the Administration's employees (Nationalrätliche Alkoholkommission, 1897, p. 1018).

The surplus. A majority of the cantonal delegates expected a revision of the law to increase alcohol revenue rather than reduce it (Nationalrätliche Alkoholkommission, 1899, p. 654), and after the good harvest of 1899 the committee foresaw four or five years of reduced consumption of monopoly spirit and, with the larger population of the 1900 census, a lower quota per head and less money for cantons whose population had stayed level (Nationalrätliche Alkoholkommission, 1900, pp. 734–735).

A.2 Properties of the yearbook wine table

The two wine shares differ by exactly one term. Let $\bar{P}_i = V_i/Q_i$ be the average price implied by canton i 's row, and $\bar{P}_{CH}$ its national counterpart. Then

$$X_{3i} = X_{2i} \times \frac{\bar{P}_i}{\bar{P}_{CH}}.$$

The wine revenue share is the volume share scaled by relative price, and that ratio is the only thing separating the two measures. Comparing X_2 against X_3 therefore reads off how much the price term contributes. It is not a comparison of two independently constructed

variables. Wine revenue is concentrated: the five largest wine cantons hold 77.7 percent of national wine revenue (Vaud alone 32.1) in 1907's harvest.

V_i records no transaction, and no harvest is sold entire. We nonetheless call it wine revenue throughout (§3.1.3). The reason is a collision of words: value already carries its statistical sense in every table, and the longer form at each use would be tedious.

Alongside the totals, the yearbook reports harvest, value and average price separately for red, white and mixed wine. On the 1907 harvest the three class values sum to the printed total exactly for every canton and for the national row, and each class price equals that class's value divided by its quantity to within print rounding. The average-price column is therefore a quantity-weighted composite of the class prices and carries the canton's grape mix by construction. Grape class is too coarse to stand in for price, and we do not use it that way.

Two printed cells are internally inconsistent, both on negligible quantities: the Geneva mixed-wine price against a nineteen-hectoliter cell, and the Appenzell Outer Rhodes white-wine price against a half-hectoliter cell. Neither enters the shares, which are built from the total columns.

Three different years attach to every wine observation, and Table A.2 separates them. A referendum's measures are built from the last complete harvest preceding its ballot, and that harvest is published in a later volume, whose own imprint year is later again. The citation year is the imprint year, so vote #68 of July 1908 draws on the 1907 harvest, printed in the volume labeled 1908, which was issued in Bern in June 1909 and cites accordingly. Because the ballot fell in mid-year and wine is the variable of interest, we take the nearest complete harvest rather than extrapolate across a partial year, which would import whatever revenue and yield trends the ban itself set in motion. The rule has one exception. No 1897 yearbook was ever published, so the 1896 harvest first appears in the 1898 edition, and votes #51 and #52 reach two years back rather than one.

Page numbers are printed folios rather than positions in a scanned file, and they are taken from each volume's own subject register. That is a matter of necessity rather than preference: from the 1902 edition the canton table is a single landscape leaf carrying no printed folio at all, so the page can only be established from the register, which names it. In the 1898 and 1900 editions the table instead spans a facing pair and the folios are legible on the pages themselves. Where a volume prints a companion table of harvests by commune, it sits on the following leaf and is not the table used here.

Two rows depart from the one-harvest-per-referendum pattern. The petition arm is measured on the 1906 harvest, the last complete vintage before the signature drive

closed, rather than on the vintage preceding the 1908 ballot. A secondary specification for referendum #65 uses the 1904 harvest, from the 1905 edition at printed page 72.

A.3 Producer Verification

The Milliet register (§3.1.4) lists forty houses that bought monopoly spirit, not forty absinthe distilleries. We therefore traced each house in the sources a firm of the period leaves behind. These are the commercial-register notices of the *Schweizerisches Handelsamtsblatt*, which state a firm's line of business at registration and at each change of ownership; the trade directory of the canton of Vaud; the federal trademark register; advertisements and court reports in the period press; the firm histories of the Zug and Rigi kirsch trade; and, for the Val-de-Travers, the local historiography of the absinthe industry. We call a house *confirmed* when the sources document it as an absinthe producer, *unconfirmed* when they document a working distillery but name no product, and *struck* when they document a producer of something other than absinthe or a merchant with no production. Confirmed and unconfirmed houses stay in; struck houses leave both numerator and denominator. The classification is encoded house by house in the replication code, and the two series of §3.1.4 follow from it: the loose series strikes the six houses documented as kirsch, fruit-brandy or sparkling-wine producers, and the floor series also strikes the Fribourg merchant. The paragraphs below give, canton by canton, the decisive source for every house the code names and the page on which it appears.

Neuchâtel. Thirteen of the canton's seventeen houses are confirmed, and all thirteen lie in the Val-de-Travers: C. Berger, Couvet; Kübler & Romang, Travers; Giovenni, Bovet & Cie, Sandoz & fils and J. Borel-Pettavel, Môtiers; Dornier-Tüller & Cie, J.-F. von Almen & Cie, Th. Henny fils and J. Ammann, Fleurier; Ed. Pernod S.A., Legler-Pernod and A. Fraissard, Couvet; L.-A. Bolle, Les Verrières. The source is the history of the Kübler house in the *Fête de l'absinthe* cahier, which ranks the thirteen as the valley's sister distilleries by their 1902–1906 spirit purchases and prints three of the quantities (*Fête de l'absinthe*, Boveresse, 2002, p. 5). Its list rests on Couleru (1908, p. 244), the contemporary account of the absinthe industry, and on the Kübler family archives. The same chapter reproduces Kübler's partnership deed of 27 December 1875, whose stated object is 'la fabrication et de la vente d'extrait d'absinthe, vermouth, liqueurs fines et autres produits analogues' (*Fête de l'absinthe*, Boveresse, 2002, pp. 4–5), so the Travers house is a documented absinthe producer with a mixed line. The four lakeside houses are not covered by that source. One of them is struck: Bouvier frères of Neuchâtel appears in the catalog of the Swiss section at

the 1897 Brussels exhibition as a maker of 'Champagne suisse; vins mousseux', exhibiting hors concours as a jury member ([Exposition internationale de Bruxelles 1897, Section suisse, 1897](#), pp. 18, 57), which fits its export ratio of 81 percent and not an absinthe still. The Distillerie d'Auvernier, E. Haller fils of Neuchâtel and W. Dubois & Cie of Cormondrèche returned no registration or press notice in the archives searched and are unconfirmed.

Geneva. Eight of the canton's eleven houses are confirmed. Seven are confirmed by the Administration's own document: the vermouth table of the Milliet annex cross-lists L. Guillemaud, Petit-Saconnex; Vve Mirault & Cie, Moillesulaz; A. Floquet, Chêne-Bourg; Muraour & Cie, J. Demont and Bouvier & Fechoz, Geneva; and Vve J. Sauter fils, Plainpalais, each with the note 'voir absinthe' ([Milliet, 1907](#), p. 363). The eighth, J. Laverrière & Cie of Chêne-Bourg, is not in that table; it registered federal trademark no. 2337 on 11 July 1888 as 'distillateurs-liquoristes, Chêne-Bourg. Absinthe et vermouths' ([Schweizerisches Handelsamtsblatt, 1888](#), Bd. 6, p. 653). Two of the seven also attest absinthe in their own registrations: Mirault et Cie registered in 1903 as 'fabrique de vermouths et d'absinthe; exportation', and its successor Vve Mirault et Cie in 1906 as 'fabrique de vermouth et d'absinthe et commerce de vins' ([Feuille officielle suisse du commerce, 1903, 1906](#)); J. Sauter fils was registered from 1902 as 'distillerie d'absinthe, fabrique de vermouth, liqueurs et sirops' ([Feuille officielle suisse du commerce, 1911](#)). The Geneva police court corroborates two more: it tried Muraour and Bouvier & Fechoz on 2 July 1908 as manufacturers of 'l'achinte', the substitute sold after the cantonal ban ([Journal de Genève, 1908](#), p. 5). J. Crépieux and J. Bonnard returned no registration or press notice and are unconfirmed; L. Magnin has no recorded purchases and carries no weight in any series.

Vaud. One house is confirmed and three are unconfirmed. C.-A. Braillard of Yverdon advertised 'Absinthe . . . distillation lente' in 1901 ([Le Conteur Vaudois, 1901](#), vol. 39, no. 31, p. 4). Weber frères and F. Dæppen-Pache of Lausanne and Vve E. Gamboni & Cie of Morges are registered distillers. The 1905 *Indicateur vaudois* lists Weber frères and Dæppen among the five Lausanne distillers and again among its liquoristes, and gives Dæppen-Pache's trade as distillateur-liquoriste in its alphabetical section ([Indicateur vaudois, 1905](#), pp. 289, 313 and 65). It lists Vve Gamboni & Cie as one of the two Morges distillers ([Indicateur vaudois, 1905](#), p. 125). The 1908 edition lists both Lausanne houses again, Dæppen at the same address and Weber frères still on the place Chauderon ([Indicateur vaudois, 1908](#), p. 312). The directory prints a specialty line for distillers who declare one, and none of the three does, so their product stays unnamed and they are kept as unconfirmed.

Basel-Stadt. Weil frères is confirmed: its 1883 registration gives the business as ‘Distillerie und Weinhandel. (Spezialität in Kirschwasser und Absinth.)’ at Hebelstrasse 21 ([Schweizerisches Handelsamtsblatt, 1883b](#), Bd. 1, H. 2, S. 17). Vve Louis Mayer & Cie is unconfirmed: the firm registered in 1883 as ‘Sprit und Branntwein en gros’, a spirits wholesaler ([Schweizerisches Handelsamtsblatt, 1883a](#), Bd. 1), and re-registered in 1896 as ‘Wein und Spirituosen en gros. Dampfbrennerei’, adding a steam distillery ([Schweizerisches Handelsamtsblatt, 1896](#), Bd. 14, H. 253). Production is documented at the second date; the product never is.

Schwyz and Zug. We strike all four houses as kirsch distillers. The Eichhorn distillery at Arth, founded in 1840, is recorded as the first commercial cherry-brandy distillery in the canton of Schwyz, and Emil Eichhorn’s postcard of 1901 advertises ‘Genuine Rigi Cherry Brandy’ ([Zuger Rigi Kirsch, nd](#), entry 4). Franz Holzgang opened his Küssnacht distillery in 1868 after an apprenticeship at a Zug kirsch house, and the firm’s line was Rigi kirsch and Rigi bitter ([Zuger Rigi Kirsch, nd](#), entry 16). C. Landtwing of Zug is the cherry-brandy distillery that Carl Landtwing-Moos took over from Schmidt in 1902 ([Kleeb, 2017](#)), and the Cherry Brandy Society is named for its product. The Historical Dictionary of Switzerland names Zug kirsch and Neuchâtel absinthe as the two spirits that became export industries in the nineteenth century ([Dubler and Maurer, 1999](#)).

Valais. Louis Morand of Martigny is struck as a fruit-brandy house. The period record holds the registration of Morand frères as a distillery in 1888 ([Feuille officielle suisse du commerce, 1888](#), Bd. 6, p. 1004), the 1890 succession notice ([Schweizerisches Handelsamtsblatt, 1890](#), Bd. 8) and a bronze medal in the liqueurs class of the 1896 national exhibition ([Exposition nationale suisse, Genève 1896, 1896](#), pp. 196–198); the fruit-brandy line is documented from the mid-twentieth century, and no absinthe attestation exists for the house in any source searched.

Fribourg. Jean Pellet of Murten is the canton’s only listed house. The house registered on 29 January 1883 as ‘vins en gros’, and the gazette entry of 15 September 1910, made on his death, strikes ‘J. Pellet, vins en gros’ and registers his heirs’ firm as ‘vins et spiritueux en gros’, a wine and spirits wholesaler, with no distilling apparatus recorded at either date ([Schweizerisches Handelsamtsblatt, 1910](#), Bd. 28, H. 240, S. 1642). Because the sources document trade rather than production of another spirit, he falls under the merchant rule rather than the strike rule: he is retained in the loose series and struck at the floor, the only house on which the two series differ, at 1.2 percent of the loose denominator.

Construction. The register’s purchases sum to 5,402,342 kg and its refund-eligible exports to 320,553 kg. The export column is not used as a regressor: one Couvet house accounts for 79 percent of it. The loose series strikes Holzgang, Eichhorn, Landtwing, the Cherry Brandy Society, Morand and Bouvier frères, 317,035 kg or 5.9 percent of the register, leaving 5,085,307 kg; the floor series also strikes Pellet, 378,941 kg or 7.0 percent in all, leaving 5,023,401 kg. For each series the continuous measure is

$$\text{Abs}_i = 100 \times \frac{\sum_{f \in i} \text{purchases}_f}{\sum_f \text{purchases}_f}, \quad f \text{ over retained houses}$$

(cov2_corrected_floor and cov2_corrected_loose in the replication code), and the indicator $1\{\text{Abs}_i > 0\}$ (abs_bin_floor, abs_bin_loose) marks the cantons with a positive share. Under the floor series the shares are Neuchâtel 63.0, Geneva 20.0, Vaud 11.0 and Basel-Stadt 6.0 percent; the loose series moves these to 62.2, 19.7, 10.9 and 5.9 and gives Fribourg 1.2, so the loose indicator adds Fribourg to the four floor cantons. The uncorrected register share (cov2_total_share) enters only the appendix table that re-estimates the main specification on the register as printed (Appendix Table A.6). For houses that also made vermouth, kirsch or liqueurs, purchases overstate absinthe-specific input; every documented absinthe producer counts as an absinthe interest whatever its product mix, and the two series bracket what the sources leave open. The main tables report both encodings but never enter them in the same column.

The Administration itself warned that the purchase figures give ‘keinen sichern Anhalt über die Grösse der Absinthproduktion’ (Milliet, 1907, p. 361). The verification removes the houses that demonstrably were not absinthe producers; the two series bracket what the sources leave open. The purchase register is nonetheless the closest measure linking high-proof inputs to the distilled spirit of interest. It was also the evidence the Federal Council and the Federal Assembly had before them, so in using it we work from the same data the coalitions of 1908 had to read.

A.4 Data sources Appendix

Vote records. We extract from Swissvotes the canton-level yes counts, no counts, turnout, and eligible voters for vote #68 (the absinthe ban; Swissvotes 1908), along with the list of other referenda votes that serve as falsification and corroboration measures. The Federal Chancellery’s archive (Schweizerische Bundeskanzlei, 1908a,b), against which the canton-level outcome totals are cross-validated, paginates the official Erwahrungsbeschluss to

Bundesblatt 1908 Band IV, p. 572 ([Schweizerischer Bundesrat, 1908](#)).

Compulsory voting. Zürich (law of 1869), St. Gallen (duty from 1835, fine from 1867), Thurgau (1870/71), Aargau (1871) and Schaffhausen (constitutional duty 1876, fine under the electoral law of 1904) fined non-voters in 1908, though enforcement was uneven; Vaud's better-known regime dates only from 1925. See [Vutkovich \(1906, pp. 21–28\)](#), [Arnold \(1929, pp. 27–32\)](#), [Kley \(1989, pp. 162–168\)](#) and [Bechtel et al. \(2018, App. Table 6\)](#). [Schwegler \(2009, pp. 5–8\)](#) dates Aargau from its 1937 statute, but [Lambelet \(1893\)](#), as cited in [Kley \(1989, p. 167, n. 2\)](#), already lists these same five cantons.

Legal forms. A popular initiative reaches the ballot once the required signatures are filed and validated; the Federal Council may recommend rejection, as it did for the absinthe initiative, but cannot decline to hold the vote. The name of the optional referendum describes the referendum rather than the vote: the law takes effect unless citizens demand a ballot, and once they do the vote binds exactly as the other forms do.

The petition. The petition measure has no counterpart elsewhere in the set: the popular initiative dated only from 1891, and vote #68 is the only one of these ballots for which a canton-level signature record survives. The canton figures of the petition numerator ([Schweizerischer Bundesrat, 1907a, p. 984](#)) reconcile against the printed source: each of the three columns sums exactly to its own printed national total.

The census shares. A two-language denominator for the French-speaking share would omit the language most of Ticino speaks and put the canton at 11.3 percent instead of 0.3, and it would move Graubünden with its Romansh population for reasons unrelated to how French the canton is. Graubünden's largest language group is German, at 46.7 percent. The two cultural controls are built on different denominators because their excluded categories differ in kind rather than in size. Ticino is predominantly Italian-speaking, so omitting Italian speakers from the language denominator would misdescribe the canton; no canton is predominantly Jewish, Eastern Orthodox or unaffiliated, so omitting those categories from the Protestant share misdescribes none. The largest share held by any other group anywhere is 2.3 percent, in Geneva, and entering these groups separately would consume degrees of freedom already binding at $N = 25$.

The Protestant share. The Blue Cross's confessional pattern (§3.2.2) is not one organization's idiosyncrasy. The Good Templar lodges likewise stood exclusively in Protestant or religiously mixed areas (Trechsel, 1989, p. 68). Nor was the boundary merely observed: as late as 1911 the Catholic Abstinent League was advising Catholics against joining the Good Templars (Trechsel, 1989, p. 72). Catholic temperance was not absent. The Catholic Abstinent League campaigned for the absinthe initiative, the Katholikentag declared for a ban in 1906, and the Swiss bishops followed in 1908 (Trechsel, 1989, pp. 105, 107, 152).

Population density. The recovery of land area for population density is exact rather than approximate: every canton returns a whole number of square kilometers, which follows from how the published density was itself formed, as population over an integer area. Inverting the quotient therefore returns the area the statistical office used, and does not estimate a new one. The land area excludes water, and it does so selectively: the twenty-five cantonal areas sum to 39,934 square kilometers against a national 41,285, the difference being lake surface.

Table A.1: Federal referenda in the analysis, 1897–1908

Vote	Date	Object	Topic	Legal form	N	Arm
#51	11.07.1897	Bundesbeschluss über die Revision des Art. 24 der Bundesverfassung <i>Federal Decree on the revision of Art. 24 of the Federal Constitution</i>	Forestry; environmental policy	Obligatorisch <i>Mandatory</i>	24	Falsification
#52	11.07.1897	Bundesbeschluss betreffend Bundesgesetzgebung über den Verkehr mit Nahrungs- und Genussmitteln <i>Federal Decree on federal legislation over trade in foodstuffs and consumables</i>	Consumer protection; federalism; health policy	Obligatorisch <i>Mandatory</i>	24	Corroborative
#57	04.11.1900	Volksinitiative ‘für die Proporzwahl des Nationalrates’ <i>Popular initiative for proportional election of the National Council</i>	Electoral system; parliament	Volksinitiative <i>Popular initiative</i>	23	Falsification
#58	04.11.1900	Volksinitiative ‘für die Volkswahl des Bundesrates und die Vermehrung der Mitgliederzahl’ <i>Popular initiative for popular election of the Federal Council and an increase in its membership</i>	Government; electoral system; popular rights	Volksinitiative <i>Popular initiative</i>	23	Falsification
#59	23.11.1902	Bundesbeschluss betreffend die Unterstützung der öffentlichen Primarschule durch den Bund <i>Federal Decree on federal support for public primary schools</i>	Schools; federalism	Obligatorisch <i>Mandatory</i>	25	Falsification
#62	25.10.1903	Volksinitiative ‘für die Wahl des Nationalrates aufgrund der Schweizer Wohnbevölkerung’ <i>Popular initiative to elect the National Council on the basis of the Swiss resident population</i>	Electoral system; parliament	Volksinitiative <i>Popular initiative</i>	25	Falsification
#63	25.10.1903	Bundesbeschluss betreffend Abänderung des Art. 32bis der Bundesverfassung <i>Federal Decree amending Art. 32bis of the Federal Constitution</i>	Addictive substances; hospitality; crop production	Obligatorisch <i>Mandatory</i>	25	Corroborative
#64	19.03.1905	Bundesbeschluss betreffend Revision des Art. 64 der Bundesverfassung (Ausdehnung des Erfinderschutzes) <i>Federal Decree revising Art. 64 of the Federal Constitution, extending inventor protection</i>	Competition policy	Obligatorisch <i>Mandatory</i>	25	Falsification
#65	10.06.1906	Bundesgesetz betreffend den Verkehr mit Lebensmitteln und Gebrauchsgegenständen <i>Federal Act on trade in foodstuffs and articles of use</i>	Consumer protection; federalism; health policy	Fakultativ <i>Optional</i>	25	Corroborative
#68	05.07.1908	Volksinitiative ‘für ein Absinthverbot’ <i>Popular initiative for a ban on absinthe</i>	Addictive substances	Volksinitiative <i>Popular initiative</i>	25	Focal

Notes. Vote numbers, dates, official titles, legal forms and topics are from Swissvotes (Année Politique Suisse, University of Bern). The Object column gives the official German title with an English rendering beneath it; the German is authoritative. *Legal form* is the official *Rechtsform*: an *obligatorisches Referendum* is a constitutional amendment originating in the Federal Assembly and a *Volksinitiative* one proposed by citizens, and both require a double majority of voters nationally and of cantons; a *fakultatives Referendum* concerns ordinary legislation and requires only the popular majority. *N* is the estimation sample. Vote data are complete for all twenty-five cantons throughout; coverage varies only where the wine series has no observation for the harvest vintage in question, namely Thurgau in 1897 and Thurgau with Basel-Landschaft in 1900.

Table A.2: Wine-data source for each referendum, 1897–1908

Referendum	Vote	Harvest	Source (edition, page)
#51	11.07.1897	1896	<i>Jahrbuch</i> 1898, pp. 84–85 (BFS, 1898)
#52	11.07.1897	1896	<i>Jahrbuch</i> 1898, pp. 84–85 (BFS, 1898)
#57	04.11.1900	1899	<i>Jahrbuch</i> 1900, pp. 52–53 (BFS, 1900)
#58	04.11.1900	1899	<i>Jahrbuch</i> 1900, pp. 52–53 (BFS, 1900)
#59	23.11.1902	1901	<i>Jahrbuch</i> 1902, p. 65 (BFS, 1902)
#62	25.10.1903	1902	<i>Jahrbuch</i> 1903, p. 69 (BFS, 1904)
#63	25.10.1903	1902	<i>Jahrbuch</i> 1903, p. 69 (BFS, 1904)
#64	19.03.1905	1904	<i>Jahrbuch</i> 1905, p. 72 (BFS, 1906)
#65	10.06.1906	1905	<i>Jahrbuch</i> 1906, p. 67 (BFS, 1907)
Petition	1906–07 drive	1906	<i>Jahrbuch</i> 1907, p. 55 (BFS, 1908b)
#68	05.07.1908	1907	<i>Jahrbuch</i> 1908, p. 51 (BFS, 1909)

Notes. Each row gives the source of the wine measures for one referendum. *Vote* is the ballot date, *harvest* the vintage the measure is computed from, and the source column names the yearbook edition carrying that harvest together with the printed folio of its canton wine-harvest table (*Darstellung der schweizerischen Weinernte*). Citation years are the volumes’ imprint years and differ from the edition labels; see §A.2 for the construction and for the vintage rule.

A.5 Referendum 68 Supplemental Results

A.5.1 Estimator Diagnostics

Interpretation: Two specification tests appear in Table A.3, and they are not read across. The linear column is the Ramsey regression specification error test (Ramsey, 1969): the model is refitted with powers of its own fitted values added, and the test asks whether those powers belong. The fractional column is the Lagrange-multiplier analog for fractional response models (Ramalho et al., 2011), which puts the same question to the logit index. The two differ in which powers they add, in what they assume about the errors, and, in the respect that matters at this sample size, in what they are referred to: the Ramsey statistic carries the number of observations in its degrees of freedom, while the fractional statistic is referred to a chi-square fixed at two degrees of freedom whatever the sample. Neither test rejects at any vote in the table. At twenty-three to twenty-five observations we read that as weak evidence rather than strong, and we therefore settle the functional-form question on two quantities that are counts rather than test statistics: no fitted value falls outside the bounds on any referendum, and the least-squares and fractional-logit estimates agree throughout.

Table A.3: Estimator diagnostics and the fractional-logit inference record, by vote

Referendum	HC3 p	Bell– McCaffrey p	Freedman–Lane p (AME)	Finite draws	Out-of- bounds	RESET (linear)	RESET (fract.)
#51 [‡]	0.611	0.576	0.628	553,997	0	0.193	0.244
#52 [‡]	0.331	0.306	0.325	958,566	0	0.171	0.117
#57	0.243	0.239	0.211	1,000,000	0	0.832	0.390
#58	0.394	0.382	0.302	1,000,000	0	0.542	0.700
#59	0.901	0.883	0.966	276,947	0	0.849	0.389
#62	0.636	0.605	0.464	87,367	0	0.369	0.260
#63	0.712	0.636	0.512	1,000,000	0	0.697	0.279
#64	0.801	0.652	0.806	428,459	0	0.927	0.255
#65 [§]	0.100	0.106	0.019	560,150	0	0.711	0.573
#68 [¶]	0.006	0.025	0.027	1,000,000	0	0.486	0.596

Notes: Estimator diagnostics and the inference record, by vote. Referendum rows are estimated at the Table 2 column (3) specification. Coefficients are in Table A.12, on the same per-vote measure basis as these p -values; standard errors are in Table A.13, which is volume basis throughout and so matches these readings only for the two votes marked [‡]. OOB counts out-of-bounds fitted values from the linear model (zero for every vote). RESET (linear) is the Ramsey test; RESET (fractional) is the Ramalho LM test with powers 2–3. The final columns report the fractional-logit average-marginal-effect Freedman–Lane permutation p -value and the number of completed draws, out of 1,000,000; a p computed on surviving draws is computed on a selected sample. [‡] #51 and #52 are wine volume share only (no value series exists). [§] #65 has its tailored coalition model in Table A.18, naive model here. [¶] #68, the treatment vote: inference columns at column (3) like the referendum rows, out-of-bounds and both RESETs at column (4). The absinthe conditioner rests on records covering 1902–06. To maximize comparability of the other referenda, the referenda set runs at column (3); #68’s specification diagnostics are reported at column (4).

A.5.2 Methods comparison

Table A.4: Functional-form and measure invariance: the wine coefficient under two estimators and two wine measures (descriptive)

	Raw	+ French	+ Protestant	+ producer	+ population	+ purchase	+ population
		share	share	(binary)	density	share	density
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>A. Ordinary least squares, revenue share</i>							
Wine coefficient	-0.196	0.547	0.639	0.620	0.549	0.512	0.438
HC3 <i>p</i>	0.379	0.044	0.006	0.015	0.107	0.095	0.182
Bell-McCaffrey <i>p</i>	0.385	0.072	0.025	0.032	0.086	0.080	0.141
Wild bootstrap- <i>t p</i>	0.435	0.036	0.031	0.023	0.097	0.032	0.109
Freedman-Lane <i>p</i>	0.325	0.059	0.016	0.022	0.078	0.080	0.135
Effective D.O.F.	2.04	3.58	3.74	3.57	4.01	4.09	4.41
<i>B. Ordinary least squares, volume share</i>							
Wine coefficient	-0.165	0.548	0.608	0.582	0.519	0.478	0.406
HC3 <i>p</i>	0.553	0.037	0.029	0.070	0.177	0.097	0.198
Bell-McCaffrey <i>p</i>	0.537	0.053	0.040	0.060	0.123	0.086	0.160
Wild bootstrap- <i>t p</i>	0.498	0.053	0.047	0.074	0.145	0.043	0.134
Freedman-Lane <i>p</i>	0.520	0.055	0.041	0.063	0.128	0.084	0.154
Effective D.O.F.	3.17	4.96	5.00	4.88	5.11	4.88	5.04
<i>C. Fractional-logit average marginal effect (pp)</i>							
Revenue share	-0.192	0.513	0.595	0.580	0.500	0.485	0.407
Freedman-Lane <i>p</i>	0.532	0.060	0.027	0.035	0.078	0.100	0.174
Volume share	-0.163	0.528	0.585	0.563	0.487	0.466	0.388
Freedman-Lane <i>p</i>	0.640	0.078	0.049	0.064	0.112	0.134	0.217

Notes: This table reports each specification under two estimators and two wine measures: ordinary least squares, and fractional-logit average marginal effects, each on the revenue-share and volume-share measures. Stars unshown; each p-value row names its scheme, for comparison across calculation schemes. Effective degrees of freedom are reported for the least-squares rows; average-marginal-effect rows are descriptive. Permutation tests draw 1,000,000 replications throughout. A fractional-logit permutation p-value is shown only where the refit returned a finite average marginal effect on every draw; for referendum 68 it did so on all 1,000,000, in each column and on both wine measures. Per-vote completion across the battery is in Appendix Table A.3.

A.5.3 Volume (Yield) Results

Table A.5: Wine volume (yield) share and support for the 1908 absinthe ban

	Raw	+ French share	+ Protestant share	+ producer (binary)	+ population density	+ purchase share	+ population density
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Wine volume share	−0.165 (0.273)	0.548** (0.247)	0.608** (0.260)	0.582* (0.304)	0.519 (0.370)	0.478* (0.274)	0.406 (0.305)
French share		−0.304*** (0.080)	−0.298*** (0.072)	−0.266** (0.113)	−0.305 (0.196)	−0.227** (0.091)	−0.222** (0.089)
Protestant share			−0.074 (0.049)	−0.062 (0.055)	−0.039 (0.069)	−0.052 (0.052)	−0.020 (0.067)
Log density					−2.738 (3.636)		−2.163 (2.693)
Absinthe (corrected share)						−0.229 (0.246)	−0.210 (0.213)
Absinthe (attested producer)				−4.013 (7.884)	2.473 (15.849)		
Constant	65.00 (2.64)	67.17 (2.17)	70.04 (2.97)	69.71 (3.00)	81.37 (15.20)	69.34 (2.85)	78.19 (11.25)
R^2	0.01	0.50	0.55	0.55	0.58	0.58	0.61
Adjusted R^2	−0.03	0.45	0.48	0.46	0.48	0.49	0.51
AIC	197	182	181	183	183	181	181
BIC	199	185	186	189	190	187	189
Small-sample diagnostics for wine coefficient (p-values)							
HC3 robust	0.553	0.037	0.029	0.070	0.177	0.097	0.198
Bell–McCaffrey	0.537	0.053	0.040	0.060	0.123	0.086	0.160
Wild bootstrap- t	0.498	0.053	0.047	0.074	0.145	0.043	0.134
Freedman–Lane	0.520	0.055	0.041	0.063	0.128	0.084	0.154
Bell–McCaffrey effective D.O.F.	3.17	4.96	5.00	4.88	5.11	4.88	5.04

Notes: Ordinary least squares; $N = 25$ cantons. Wine is the canton's share of national wine volume (yield, $\times 100$), under the same vintage rule as Table 2 (1907 harvest, the last complete before the vote). This table establishes measurement robustness for Table 2: the headline cascade re-estimated on the volume measure. *, **, and *** mark $p < 0.10, 0.05$, and 0.01 , on heteroskedasticity-robust (HC3) p -values; the diagnostic rows report, and are labeled by, their own schemes (HC3, Bell–McCaffrey, wild bootstrap- t HC3-studentized, Freedman–Lane permutation).

A.5.4 Uncorrected Absinthe Results

Table A.6: Wine interests and the 1908 absinthe ban — raw (uncorrected) purchase-share back matter

	Raw (1)	+ French (2)	+ Protestant (3)	+ Density (4)	+ Absinthe (5a)	+ Absinthe (5b)
Wine revenue share	−0.196 (0.219)	0.547** (0.257)	0.639*** (0.207)	0.552** (0.253)	0.438 (0.315)	0.539* (0.267)
French share		−0.316*** (0.087)	−0.315*** (0.073)	−0.302*** (0.079)	−0.241** (0.113)	−0.346*** (0.112)
Protestant share			−0.083* (0.046)	−0.048 (0.064)	−0.031 (0.067)	−0.026 (0.077)
Log density				−2.184 (2.657)	−2.073 (2.647)	−2.891 (2.682)
Absinthe (share)					−0.197 (0.235)	
Absinthe (producer)						4.103 (7.326)
Constant	65.13 (2.58)	67.38 (2.08)	70.56 (2.77)	79.36 (11.03)	78.39 (10.95)	81.21 (10.57)
R^2	0.02	0.51	0.56	0.60	0.62	0.61
Adjusted R^2	−0.03	0.46	0.50	0.52	0.52	0.50
AIC	196	181	180	180	181	181
BIC	199	185	185	186	188	189
Small-sample diagnostics for wine coefficient						
HC3 robust p	0.379	0.044	0.006	0.041	0.181	0.058
Bell–McCaffrey p	0.385	0.072	0.025	0.059	0.141	0.070
Wild bootstrap- t p (HC3-studentized)	0.435	0.036	0.031	0.068	0.109	0.085
Freedman–Lane p	0.325	0.059	0.016	0.046	0.135	0.055
Bell–McCaffrey effective D.O.F.	2.04	3.58	3.74	4.10	4.42	4.15

Notes: Raw (uncorrected) spirits purchase share. This encoding sweeps in kirsch and fruit-brandy distillers; the corrected producer memberships appear in Appendix Table A.7. Alternative wine-industry encodings are never entered together (see the Table 2 note). *, **, and *** mark $p < 0.10$, 0.05 , and 0.01 , on heteroskedasticity-robust (HC3) p -values.

Table A.7: Wine revenue and the 1908 absinthe ban — bounds appendix (loose producer membership)

	Raw (1)	+Fr. (2)	+Prot. (3)	+Disc. (4) ^L	+Dens. (5) ^L	+Cont. (6) ^L	+Dens. (7) ^L
Wine revenue share	−0.196 (0.219)	0.547** (0.257)	0.639*** (0.207)	0.688*** (0.226)	0.688 (0.529)	0.511* (0.295)	0.437 (0.318)
French share		−0.316*** (0.087)	−0.315*** (0.073)	−0.354*** (0.078)	−0.481*** (0.144)	−0.249** (0.117)	−0.241** (0.114)
Protestant share			−0.083* (0.046)	−0.090* (0.052)	−0.041 (0.058)	−0.062 (0.051)	−0.030 (0.068)
Absinthe producer (attested, loose)				3.320 (5.645)	16.537 (12.283)		
Milliet absinthe purchase share (corrected, loose)						−0.198 (0.278)	−0.185 (0.245)
Log population density					−4.815 (2.853)		−2.084 (2.662)
Constant	65.13 (2.58)	67.38 (2.08)	70.56 (2.77)	70.66 (2.83)	90.47 (11.87)	69.89 (2.72)	78.34 (11.00)
R^2	0.02	0.51	0.56	0.57	0.66	0.59	0.62
Adjusted R^2	−0.03	0.46	0.50	0.48	0.57	0.50	0.52
AIC	196	181	180	182	178	181	181
BIC	199	185	185	188	185	187	188
Small-sample diagnostics for wine coefficient							
HC3 robust p				0.006	0.208	0.098	0.186
Bell–McCaffrey p				0.023	0.097	0.083	0.145
Wild bootstrap- t p (HC3-studentized)				0.032	0.145	0.039	0.113
Freedman–Lane p				0.011	0.135	0.081	0.137
Bell–McCaffrey effective D.O.F.				3.55	3.57	4.13	4.45

Notes: ^L marks columns (4)–(7), which use the loose producer membership. Floor membership: Neuchâtel, Geneva, Vaud, Basel-Stadt, the cantons with houses attested in the purchase register as absinthe producers. The loose set adds Fribourg, whose one listed house, a wine and spirits wholesaler, is retained under the merchant rule and struck at the floor (Appendix A.3); the kirsch, fruit-brandy and sparkling-wine houses are struck in both series. *, **, and *** mark $p < 0.10$, 0.05 , and 0.01 , on heteroskedasticity-robust (HC3) p -values.

A.5.5 Subsample Diagnostics

Table A.8: The wine revenue-share effect in the non-French subsample ($N = 20$ cantons, cov. French $< 50\%$)

	Bivariate (1)	+ Prot. (2)	+ Prot., dens. (3)
Wine revenue share	0.614 (0.304)	0.879 (0.385)	0.908 (0.550)
N	20	20	20
R^2	0.06	0.14	0.21
OLS fitted-value range shares in $[0, 100]$	[66.6, 75.5] 20/20	[62.3, 75.7] 20/20	[56.4, 75.0] 20/20
Wine-coefficient inference			
HC3 robust p	0.06	0.04	0.12
Bell–McCaffrey effective D.O.F.	1.90	2.00	2.02
Bell–McCaffrey p	0.17	0.13	0.15
Wild bootstrap- t p (HC3-studentized)	0.12	0.07	0.23
Freedman–Lane p (10^6)	0.08	0.06	0.12
Leave-one-canton-out range (<i>widest swing when dropping</i>)	[0.494, 0.759] NW	[0.768, 1.112] TI	[0.702, 1.263] BS

Notes: Non-French-speaking subsample ($N = 20$), ordinary least squares. This exhibit is read for sign and existence only: with effective degrees of freedom of 2.02 in the restricted columns, significance testing is deferred to the full-sample battery. The quoted cell is the bivariate estimate, 0.614, with no controls.

A.5.6 Pairwise correlations of the regressors

Table A.9: Pairwise correlations among the Table 2 regressors

	(1)	(2)	(3)	(4)	(5)	(6)
(1) Wine revenue share	1					
(2) French share	0.56	1				
(3) Protestant share	0.25	0.15	1			
(4) Producer (attested)	0.36	0.67	0.37	1		
(5) Absinthe share (corrected)	0.15	0.63	0.31	0.71	1	
(6) Log density	-0.05	0.10	0.46	0.58	0.26	1

Notes: Pairwise Pearson correlations among the regressors of Table 2 columns (4) and (7). $N = 25$ cantons; lower triangle. Computed on the post-switch cohort, in which the French share is French speakers as a share of TOTAL canton population (2026-08-22 language-control switch). At $N = 25$ a correlation's standard error is roughly 0.21, so the table RANKS pairs and does not measure them; no value is meaningful to a third decimal, and no significance stars are shown. The two absinthe encodings, (4) and (5), are alternative measures of the same variable and never enter a specification together.

A.5.7 Oster Bounds

Table 2 shows the wine coefficient moving as the controls enter, from -0.196 in the raw column, column (1), to 0.620 in the preferred specification, column (4). A reader who accepts that movement is entitled to ask about the next control, the one we could not measure. If three observed variables moved the coefficient this far, what would a fourth, unobserved, do to it? Oster (2019) gives that question a form that can be answered from the regressions themselves. Her starting point is a practice referees already use: a coefficient that barely moves when controls enter is read as evidence that omitted variables would not move it either. Her point is that the practice is incomplete on its own. A coefficient can sit still because the controls explain nothing about the outcome, in which case its stillness says nothing about what remains unobserved; stillness reassures only when the controls explain a good deal. Her method therefore reads the coefficient's movement together with the movement in R^2 , and asks how much of the outcome the unobserved variables would have to explain, and how strongly they would have to be tied to the wine share, before they could account for the estimate.

The mechanics need two ingredients the reader has already seen and two that must be assumed. The first two are the coefficient and the R^2 before the controls enter and after they do; for the naive baseline they come straight from Table 2, and for the French-adjusted baseline the note to Table A.10 gives them. The first assumption is a ceiling: the share of the variation in the yes-share that could be explained if every relevant variable, observed

and unobserved, were in the regression. Oster writes it $R_{\max}^2$, and the table prints it so; it cannot be estimated, only assumed, and we return below to how she chooses it. The second assumption is a rule tying the unobserved variables to the observed ones. It says that whatever we could not measure is related to the wine share by some multiple of the strength with which the controls we did measure are, and she writes that multiple δ . She calls this proportional selection; we call it the proportionality rule below. A δ of one means the unobserved variables are tied to the wine share exactly as strongly as the observed controls are; a δ of two, twice as strongly. Given a ceiling and a δ , the coefficient that would result with the unobserved variables included can be computed; the computation is the one Oster's own routine, `psacalc`, performs, which is the name in the table's row labels, and she reports two summaries of it. One fixes the coefficient at zero and solves for the δ that would get it there, written δ^* : how much stronger than the observed controls the unobserved ones would have to be to explain the estimate away. The other fixes δ at one and reports the adjusted coefficient, written β^* : what the estimate becomes if the unobserved variables matter exactly as much as the observed ones.

Both summaries depend on the ceiling, so the ceiling is where the method's judgment lives. She collected every randomized result whose authors reported its sensitivity to controls, from the experiments published in five of the field's top journals over six years, sixty-five results in all, and applied the adjustment to each. She then asked at what ceiling ninety percent of them survive on both of the standards described below, on the ground that a randomized result can be taken as causal, so that a standard which failed most of them would be too harsh to be useful. The answer is a ceiling of 1.3 times the R^2 of the regression with controls (Oster, 2019, §5.2). The ceiling is therefore not a matter of taste, and her yardstick follows from it. A result meets the standard that ninety percent of her randomized results meet if the set running from the estimated coefficient to the adjusted coefficient at $\delta = 1$, which she calls the identified set and which Table A.10 reports under that name, excludes zero; for a coefficient the controls move toward zero, the case to which she applies that test, this is the same thing as δ^* exceeding one (Oster, 2019, §5.1 and §5.2). We report that ceiling and, on the French-adjusted baseline, the limiting case in which the ceiling is set to one, so that the observed and unobserved variables together explain the outcome completely. She shows on her own evidence that the limiting case over-adjusts: fewer than half of her randomized results survive it, because at a ceiling of one even a small movement in the coefficient is, in her phrase, blown up into a large adjustment (Oster, 2019, §5.2). We therefore report that row for transparency and do not treat it as the test.

Table A.10 reports both summaries on two baselines, one to each panel; its second column gives the ceiling for each, and its note the controlled R^2 the ceiling is built from.

The naive baseline runs from the raw regression of column (1), where the wine coefficient is -0.196 , to the preferred specification of column (4), treating all three controls as the observed variables. It is the comparison her formal results cover in full, and it starts from a specification the paper treats as a benchmark rather than as an estimate, since the French share enters as the co-explanatory variable the design holds fixed by construction rather than as a control in the ordinary sense. The French-adjusted baseline therefore removes the French share from the outcome and from every regressor by regression first, and runs the same comparison on what remains. Its starting point is the wine coefficient of column (2), 0.547 , which already holds language fixed, and its end point is again column (4), so that the observed variables whose entry it studies are the two controls whose inclusion is a modeling choice, the Protestant share and the producer indicator. Oster’s §3.3.3 describes that arithmetic, and it is her arithmetic we run. She states it for an additional control that is unrelated to the remaining observed variables, and that premise is not met here: the French share’s correlation with the Protestant share is 0.15 and with the producer indicator 0.67 (Appendix Table A.9). The French-adjusted baseline is the one we rely on, because it asks the question the design poses, and its formal footing is weaker for exactly that reason. Whether or not her result certifies it, the verdict does not depend on it: the naive baseline, which her theory does cover, returns the same one, and we report both so the reader can see that.

The first thing the table records is a sign, and here the sign follows the direction in which the controls move the coefficient. Conditioning does not shrink the wine coefficient toward zero. It moves the coefficient up, from -0.196 in the raw column, through zero, to 0.620 : in Oster’s classification an away-from-zero mover, and the direction means the observed controls were holding the estimate down while they were omitted. One further assumption of hers carries that reading from the controls we estimated to the true ones: that what is unobserved is not so large as to reverse the direction in which the observed controls, taken together, are tied to the wine share (Oster, 2019, §3.3.2). Unobserved variables that behave like the observed ones, which is what the proportionality rule assumes, would then hold it down further while omitted, so including them would raise the coefficient rather than erase it. To reach zero, the unobserved variables would have to be tied to the wine share in the opposite direction from every control we have. The negative entries record exactly that: δ^* is -4.676 on the French-adjusted baseline at her ceiling and -1.314 at the limiting one, and -1.148 on the naive baseline. For Oster’s simple approximation that reasoning is exact algebra. For the exact estimator the table reports, it is what the numbers show at every ceiling we run, and we present it as a property of this table rather than as a general result. Oster’s bounding set takes δ to be non-negative; a negative value would

describe unobserved variables that select against the pattern of the observed ones (Oster, 2019, §3.4). Within the range that set entertains, δ between zero and one, no degree of selection on unobserved variables produces a zero. On her first standard, then, the result is categorical. We read the entries for their sign only; their magnitudes say how strong that opposite-direction selection would have to be, and we do not read them against her threshold of one, which she states for positive values. That is also why the first standard cannot come out against a coefficient of this kind, and why the second standard, below, is the one that matters.

A categorical pass on the first standard is not, on its own, reassuring, and Oster says why of the zero-null test that hers extends. A coefficient that grows when controls enter passes such a test automatically, so that, as she says of it, every result of that type counts as robust while the conclusions about magnitude may be radically biased even though the rejection of zero is retained (Oster, 2019, §2). She accordingly applies her first standard only to results the controls move toward zero, and supplies a second for the rest (Oster, 2019, §5.1). Take the identified set again, from the estimated coefficient to the adjusted coefficient at $\delta = 1$, and ask whether it stays inside a band of 2.8 standard errors either side of the estimated coefficient, the bounds of a 99.5 percent confidence interval. If it does, allowing for unobserved variables as strong as the observed ones would not change the conclusion about the coefficient's size (Oster, 2019, §3.4 and §5.1). This is the test we could have failed. The adjusted coefficient is 0.672 on the French-adjusted baseline and 0.882 on the naive one, both above 0.620, so under proportional selection the reported estimate is a floor. The band is 0.620 plus or minus 0.651, the HC3 standard error of Table 2 scaled by 2.8; it is wide, as it must be with twenty-five observations, and it reaches below zero. The identified set on the French-adjusted baseline, from 0.620 to 0.672, lies within it, and the set on the naive baseline, from 0.620 to 0.882, lies within it; within means the set does not leave the band at either end. Because the band is wide, the informative quantity is the size of the adjustment against the standard error: on the French-adjusted baseline the adjustment at $\delta = 1$ is 0.22 standard errors, and on the naive baseline 1.13. In words: the set stays within the band, so unobserved variables that mattered as much as the observed controls would move the wine coefficient by an amount inside its own sampling uncertainty.

The cascade shows the estimate surviving the controls we have. The leave-one-out exercise shows it surviving the removal of any single canton but Vaud at the ten percent level (Table A.11). The decomposition in Table 3 shows that the movement into the preferred specification is largely the language variable's, 81 percent of it, which is what the design says it should be. Oster's bounds address the control we do not have, and say that a control behaving like the ones we do would strengthen the estimate rather than weaken it, by less

than the estimate's own uncertainty. One thing the exercise cannot rule out should be said as plainly. An omitted variable that is more present in wine cantons than elsewhere once language, the Protestant share and producer status are held fixed, and that raises support for the ban for reasons unconnected to the wine interest, would select against the pattern of every observed control. No arithmetic on the observed controls can exclude it. The two candidates that come first to mind, a canton's density of settlement and the strength of organized temperance, are already in the cascade as the population-density and Protestant-share controls, and the coefficient survives their inclusion. The bounds cannot settle whether a third exists. What they settle is what it would have to be: something that runs against every observed control at once.

Four limits belong beside the result, and the first is what the exercise is. Neither this adjustment nor the decomposition in Table 3 is an inference procedure. Each is arithmetic on the sample moments, computable at any sample size, and neither competes with the four readings reported for the wine coefficient in every column of Table 2, which are where the question of what twenty-five cantons can support as inference is answered. Oster's guarantees for the adjustment are asymptotic; her smallest sample runs to several thousand observations, and its behavior at twenty-five is not something she studies, so its statistical properties at our scale are unexamined and we do not claim them. The second limit is the ceiling, which is assumed and never estimated; that is why the table reports her calibrated value and the limiting case rather than a single number, and the calibration is external, drawn from randomized results in the published literature, not a property of our data. The third is the proportionality rule. As Oster herself notes, it always holds for some δ (Oster, 2019, §3), so what is assumed is the bound placed on it, δ at most one, rather than the rule itself; no in-sample diagnostic can check that bound, and she supports it by validation against outside evidence rather than by a test (Oster, 2019, §4). The fourth is scope. The method is stated for linear models and carries nothing to the fractional logit companion in Table A.4. The result is robustness under Oster's two standards, which is narrower than a demonstration that the estimate is unbiased: it speaks to selection on unobserved variables and to nothing else, neither to functional form nor to the measurement of the wine share, which §3.1.3 and Appendix A.2 take up.

Table A.10: Oster (2019) δ^* bounds for the wine revenue-share coefficient

Selection bound	$R_{\max}^2$	δ^*	$\beta^*(R_{\max}^2, \delta=1)$
French-adjusted baseline (French partialled; psacalc-exact)			
$1.3 \tilde{R}^2$	0.341	-4.676	
$R_{\max}^2 = 1$	1.000	-1.314	
Bias-adjusted β ($\delta = 1$)	0.341		0.672
Identified set = [0.620, 0.672], band 0.620 ± 0.651 (HC3): within			
Naive bivariate baseline (psacalc)			
$1.3 \tilde{R}^2$	0.746	-1.148	
Bias-adjusted β ($\delta = 1$)	0.746		0.882
Identified set = [0.620, 0.882], band 0.620 ± 0.651 (HC3): within			

Oster (2019) proportional-selection bound δ^* : the strength of selection on unobservables, relative to the observed controls, that would drive the wine revenue-share coefficient to zero; we read δ^* for its sign; its magnitude is not compared with Oster's threshold of one, which she states for positive proportional selection only. $N = 25$ cantons. Because French share is a suppressor – the raw bivariate wine coefficient is -0.196, flipping to +0.620 once French enters (a Gelbach/Simpson configuration, not a defect) – δ^* is negative in both baselines at the conventional bound $R_{\max}^2 = 1.3 \tilde{R}^2$: the observed controls move the coefficient away from zero, so no same-direction selection of any strength produces zero. Bias-adjusted β ($\delta = 1$) is the coefficient adjusted for unobservables selecting as strongly as the observed controls; values above the estimate mean the estimate is a floor (French-adjusted 0.672; naive 0.882). $\tilde{R}^2$ is the controlled R^2 (residualized 0.263 for the French-adjusted rows; raw 0.574 for the naive). The memo row closing each panel reports Oster's magnitude test: the identified set runs from the controlled estimate to the bias-adjusted β at the $\min\{1.3 \tilde{R}^2, 1\}$ ceiling with $\delta = 1$ (Oster 2019, §3.4); the band is that estimate ± 2.8 HC3 standard errors, her 99.5 percent interval (§5.1). **Within** means the set does not leave the band.

A.6 Leave-one-out stability testing

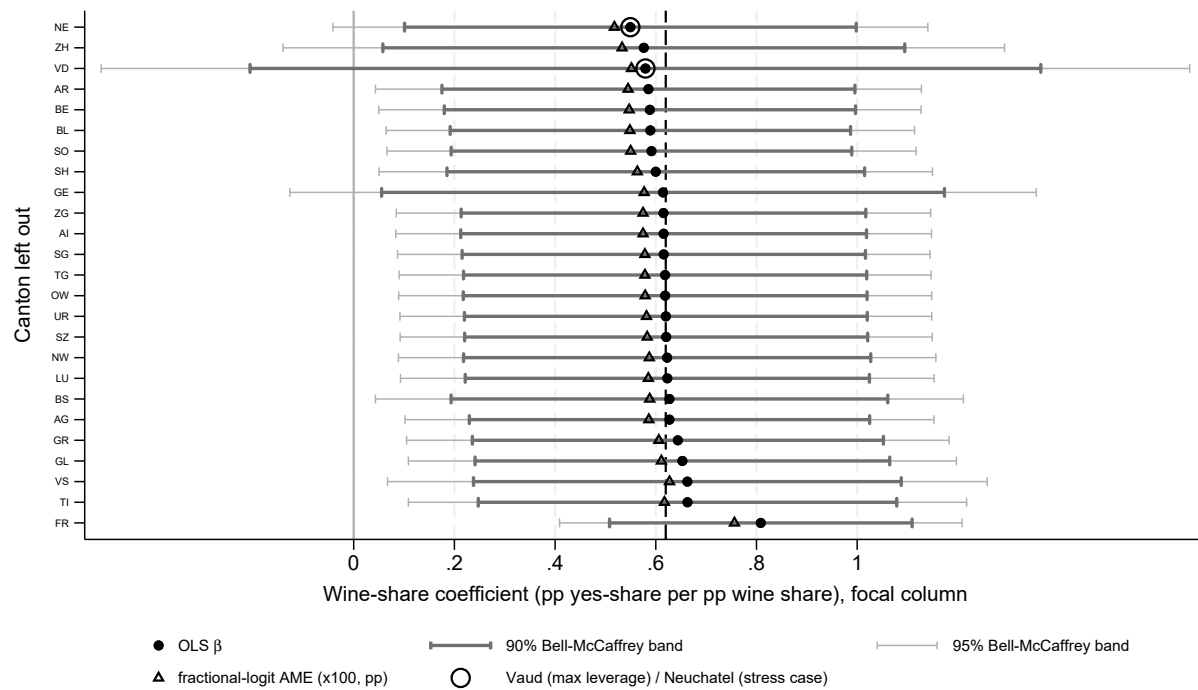


Figure A.1: Leave-one-canton-out stability of the wine coefficient, Table 2 column (4): all 25 single-canton deletions under both estimators on a shared percentage-point axis (A6 companion figure).

Table A.11: Leverage and influence anatomy of the wine coefficient, Table 2 column (4): where the influence sits, and survival under single-canton deletion

Canton	Wine rev. share	Producer (attested)	French share	h_{ii} (leverage)	DFBETA (influence, wine)	Drop-one β (p)
NE	5.43	Yes	82.8	0.367	+0.246	0.550 (0.061)
FR	0.76	–	68.3	0.452	-0.691	0.809 (0.006)
ZH	14.50	–	0.9	0.255	+0.151	0.576 (0.080)
TI	6.16	–	0.3	0.148	-0.151	0.663 (0.031)
VS	19.53	–	65.2	0.435	-0.149	0.663 (0.040)
VD	32.11	Yes	86.5	0.671	+0.140	0.579 (0.177)
AR	0.00	–	0.1	0.175	+0.121	0.585 (0.041)
GL	0.00	–	0.2	0.118	-0.117	0.653 (0.030)
BL	1.21	–	0.9	0.116	+0.111	0.589 (0.036)
BE	2.17	–	16.6	0.175	+0.111	0.588 (0.039)
SO	0.12	–	1.9	0.055	+0.108	0.591 (0.036)
GR	1.21	–	0.5	0.061	-0.093	0.644 (0.030)
SH	5.28	–	0.6	0.129	+0.071	0.600 (0.039)
AG	3.67	–	0.4	0.064	-0.026	0.627 (0.031)
BS	0.17	Yes	2.3	0.551	-0.026	0.627 (0.042)
GE	2.83	Yes	82.8	0.389	+0.019	0.614 (0.080)
ZG	0.00	–	0.6	0.093	+0.016	0.615 (0.033)
AI	0.00	–	0.1	0.096	+0.015	0.616 (0.033)
SG	2.44	–	0.3	0.052	+0.014	0.616 (0.033)
LU	0.03	–	0.5	0.090	-0.011	0.623 (0.032)
NW	0.00	–	0.2	0.109	-0.010	0.623 (0.033)
TG	2.26	–	0.3	0.087	+0.004	0.619 (0.032)
OW	0.00	–	0.2	0.108	+0.004	0.619 (0.032)
SZ	0.12	–	0.5	0.102	-0.003	0.621 (0.032)
UR	0.00	–	0.1	0.101	-0.001	0.620 (0.032)
Full sample (N=25)	100.0	4	16.5	5.00	–	0.620 (0.032)

Yes-vote share (vote #68) on national wine revenue share, the column-(4) preferred specification (+ French share + Protestant share + Absinthe producer (attested, binary)). Neuchâtel (NE) is pinned to the top row as the ratified stress case (disclosed); the remaining 24 cantons are sorted by absolute DFBETA on the wine coefficient. h_{ii} = OLS leverage, the canton's weight in its own fitted value (leverages sum to $k = 5$; the maximum-leverage canton on this design is Vaud). DFBETA = the canton's influence on the wine coefficient (BKW, dropped-one RMSE scaled): a POSITIVE value means the canton pulls the coefficient UP, so deleting it LOWERS β – read each row against the drop-one column beside it. drop-one beta (p) re-estimates the wine coefficient omitting that canton; the parenthetical is the Bell-McCaffrey two-sided p (HC2 with Satterthwaite effective degrees of freedom), $N=24$ for the canton rows and $N=25$ for the full sample. The concentration of the wine coefficient's identification weight in Vaud (Table A.16), amplified by that canton's leverage, is why its Bell-McCaffrey effective degrees of freedom are 3.57. The final row reports full-sample aggregates, each appropriate to its column: wine revenue shares SUM to 100.0 (they are national shares), attested producers are a COUNT (4 cantons), French share is the sample MEAN, and leverages SUM to 5.00 (= k , the identity noted above); DFBETA has no full-sample analog and is left blank. The wine effect stays positive and at least 0.550 under every single-canton deletion; drop-Neuchâtel (0.550) is both the ratified stress case and the leave-one-out minimum.

A.7 Other Referenda

A.7.1 Referenda Full Results – Wine Revenue

Table A.12: Referenda by role: least-squares and fractional-logit estimates per vote

Referendum	OLS coef.	Freedman-Lane p	AME	N
<i>Falsification referenda</i>				
#51	0.256	0.513	0.307	24
#57	−0.452	0.181	−0.462	23
#58	−0.377	0.289	−0.485	23
#59	−0.041	0.874	−0.020	25
#62	0.313	0.517	0.321	25
#64	−0.111	0.669	−0.116	25
<i>Corroborative</i>				
#52	0.550	0.261	0.572	24
#63	0.222	0.605	0.219	25
#65 [§]	1.527	0.083	1.507	25
<i>Treatment vote</i>				
#68 [¶]	0.639	0.016	0.595	25

Notes: Stars unshown. Columns report the least-squares coefficient, its Freedman–Lane permutation p-value, the fractional-logit average marginal effect (descriptive), and N. § marks #65, the wine-related corroborative candidate. Its row carries the universal battery specification. Referendum 65 also has a coalition specification of its own, adding the cattle and existing-regulation controls appropriate to a food-regulation vote, which the 1908 ballot’s covariate set does not contain; it is reported in Table A.18, where the Bell–McCaffrey correction carries 1.72 effective degrees of freedom against 3.36 at the universal specification shown here. Panels group the votes by the role each plays in the argument. The falsification referenda concern proportional representation, primary-school funding and patent protection; their full official titles are in Table A.1. Measure basis, which Table A.2 does not carry: the wine regressor is the canton share of national wine revenue where a revenue series was published for the harvest behind the ballot, and the canton share of national wine volume where none was. Volume basis: #51, #52. Revenue basis: #57, #58, #59, #62, #63, #64, #65, #68. N is the complete-case count on the vote’s own measure and is given in the final column. ¶ marks #68, the treatment vote, shown at the battery specification so that every row of the table is one regression. Its headline estimate is column (4) of Table 2, which adds the absinthe-producer indicator; that row and this one are different numbers. The battery carries no absinthe term because the producer register runs 1902–1906 while three of the six falsification ballots fall outside that window, so conditioning on it across the battery would hold the formula fixed while measurement quality varied from row to row. Per-vote HC3 and Bell–McCaffrey p-values, full estimator diagnostics and per-vote permutation completion appear in Appendix Table A.3.

A.7.2 Other Referenda – Yield Full Results

Table A.13: Volume-basis readings behind the quasi-falsification figure

Referendum	Wine coef.	HC3 (se)	Bell–McCaffrey (se)	Bell–McCaffrey p	Effective D.O.F.	N
<i>Falsification referenda</i>						
#51	0.256	(0.496)	(0.406)	0.576	2.79	24
#57	−0.514	(0.424)	(0.348)	0.217	3.81	23
#58	−0.439	(0.477)	(0.400)	0.337	3.81	23
#59	0.055	(0.397)	(0.298)	0.865	3.16	25
#62	0.197	(0.631)	(0.518)	0.732	2.70	25
#64	−0.043	(0.532)	(0.286)	0.892	2.68	25
<i>Corroborative</i>						
#52	0.550	(0.552)	(0.440)	0.306	2.79	24
#63	0.141	(0.562)	(0.399)	0.750	2.70	25
#65	1.519	(0.939)	(0.730)	0.118	3.40	25
<i>Treatment vote</i>						
#68 [†]	0.608	(0.260)	(0.221)	0.040	5.00	25

Notes: Least squares on the volume basis at the Table A.5 column (3) specification – canton yes-share on the canton share of national wine volume ($\times 100$), the French share and the Protestant share. No absinthe term: the producer register averages 1902–1906 and three of the six falsification ballots fall outside that window, so conditioning on it across the referendum set would hold the formula fixed while measurement quality varied from row to row. The volume basis is used because the 1897 cohort publishes no revenue series, so it is the only measure on which the complete set of 10 referenda exists; it is also the less favorable of the two measures for the treatment vote. Standard errors are shown under both schemes because the gap between them is the small-sample correction: at these effective degrees of freedom, normal critical values would overstate precision. The Bell–McCaffrey p uses the effective degrees of freedom in the final column, never a normal approximation. Sample sizes are complete-case counts on the volume measure, which is missing at source for one canton in 1897 and two in 1900. Across the 6 falsification referenda the coefficients cluster near zero and the intervals are too wide to resolve; the design does not produce significance on any of them. [†] #68, the treatment vote, is the same fit shown in column (3) of Table A.5.

A.8 Secondary Outcomes – Tavern Employment Share

The two cascades below regress each arm's outcome on the tavern employment share, defined in the note to Table 4, entering the controls in the order fixed in §4.1 and reporting the tavern coefficient of every column. At the petition the narrow measure runs -3.384 standing alone, then -3.518 , -3.937 , -3.894 and -3.873 as the controls enter, and in the focal column its standard error is 1.268, with .006 on HC3, .024 on Bell–McCaffrey, .036 on the wild bootstrap- t and .010 on the Freedman–Lane permutation. Its effective degrees of freedom in that column are 3.60, so the estimate rests on a few cantons rather than on all twenty-five, and deleting each canton in turn leaves it between -4.643 and -3.355 without a single change of sign. At the focal architecture the broad measure, which adds the lodging trades the census sets apart as class 300, inns, hotels and boarding houses, returns -1.674 with a standard error of 1.161, no reading below 0.10 and a standardized coefficient of -0.275 ; it recovers to -3.248 with a standard error of 1.739 once log density enters, its four readings running between .037 and .077, with a leave-one-out range of $[-4.30, -2.59]$ and no change of sign. The narrow measure is where the result lives, and the boundary is where it is weakest. At the vote the same coefficient is not detectable. The coefficient keeps its sign and loses magnitude as the controls enter, from -2.296 standing alone to -1.377 with log density entered, and the smallest p -value anywhere in that table's diagnostic block is .138, on the wild bootstrap- t in the column before the producer indicator enters. The Federal Council's *Botschaft* of 20 November 1884 records that many taverns were run as a sideline by wine dealers, wine-growers or other farmers, or by grocers, butchers, bakers or boarding-house keepers (Schweizerischer Bundesrat, 1884, printed 403, PDF p. 36, n. 10), an overlap in one man's occupations that these cantonal aggregates cannot see. Table A.16 sets out, canton by canton and for each coefficient, the squared Frisch–Waugh–Lovell residual that the canton contributes, normalized to sum to 100, which is the object underlying an added-variable plot (Belsley et al., 1980). The two coefficients rest on nearly separate sets of cantons. The tavern coefficient is identified largely in Ticino, which carries 41.9 percent of its weight, with Geneva at 16.9, Nidwalden 7.0, Obwalden 5.1 and Lucerne 4.7; the wine coefficient is identified in the wine cantons, Vaud 36.6, Zurich 13.2, Fribourg 11.6, Valais 10.7 and Geneva 9.5. Geneva is the one canton material to both, and it is also the second-largest attested absinthe producer, at 20.0 percent of the retained national total. Table A.17 sets out where that leaves the petition coefficient. Ticino carries 41.9 percent of the identifying weight, and yet no single deletion moves the coefficient by more than about a fifth of its value: dropping Ticino itself takes the estimate to -4.643 , away from zero, and no deletion takes it above -3.355 . That is leverage without influence. The measure stands

for the consumer-facing side of the drink interest, a different thing from the producer interest the wine measures carry: a tavern-keeper's stake in a prohibition runs through the receipts he loses on a drink he may no longer sell, and a wine-grower's runs through what the removal of a competing drink does to the price of his own. We therefore read the tavern column of Table 4 as suggestive evidence that the petition drew fewer signatures where the consumer-facing drink trade employed more people, and we do not ask the column to carry more: it is one column, on a measure whose variation sits largely in two cantons, with no counterpart at the vote.

Table A.14: Tavern employment and the 1906 anti-absinthe petition (cascade)

	Class 301 (narrow)					Classes 300+301 (broad)	
	Tavern (1)	+ French (2)	+ Protestant (3)	+ Producer (4)	+ Density (5)	Focal (6)	+ Density (7)
Tavern employment share	-3.384*	-3.518**	-3.937***	-3.894***	-3.873***	-1.674	-3.248*
	(1.849)	(1.439)	(1.209)	(1.268)	(1.334)	(1.161)	(1.739)
French share		0.076	0.050	0.057	0.047	0.052	-0.043
		(0.053)	(0.043)	(0.069)	(0.108)	(0.069)	(0.121)
Protestant share			0.164***	0.166***	0.170**	0.126*	0.123*
			(0.047)	(0.053)	(0.065)	(0.072)	(0.069)
Absinthe (attested producer, = 1)				-0.935	0.368	-1.251	11.825
				(5.149)	(9.343)	(5.279)	(10.464)
Log population density					-0.542		-4.460
					(2.796)		(3.299)
Constant	27.540	26.566	20.835	20.667	22.917	21.048	47.744
	(4.364)	(3.812)	(3.692)	(3.885)	(11.654)	(6.672)	(20.498)
Standardized β:							
Tavern employment share	-0.359	-0.373	-0.417	-0.413	-0.410	-0.275	-0.534
R^2	0.129	0.191	0.540	0.541	0.543	0.435	0.511
Adjusted R^2	0.091	0.117	0.474	0.449	0.422	0.321	0.383
AIC	183	183	171	173	175	178	177
BIC	186	187	176	179	182	185	184
Claim coefficient (tavern employment share):							
HC3 robust p	0.080	0.023	0.004	0.006	0.009	0.165	0.077
Bell-McCaffrey p	0.097	0.045	0.023	0.024	0.026	0.127	0.054
Wild bootstrap- t p	0.082	0.083	0.046	0.036	0.031	0.138	0.037
Freedman-Lane p	0.103	0.023	0.010	0.010	0.012	0.116	0.051
Effective df	3.70	3.78	3.37	3.60	3.58	10.16	6.01
BKW condition no. (with constant)	4.45	4.85	5.69	6.43	18.24	9.64	29.13
VIF (tavern employment share)	1.00	1.00	1.01	1.04	1.04	1.32	2.20
N (cantons)	25	25	25	25	25	25	25

Notes: Ordinary least squares; $N = 25$ cantons in every column. Outcome is 1906 anti-absinthe petition signatures per 100 eligible voters. Columns (1)–(5) enter controls cumulatively with population density last; columns (6)–(7) are the broad-measure companion at the focal architecture. The tavern measure is class 301 of the federal business census of 9 August 1905 — restaurants and taverns not attached to an inn (BFS, 1908a); columns (6)–(7) add inns, hotels and boarding houses (classes 300+301). Density and, where applicable, other harvest-year inputs are 1906 vintage. *, **, and *** mark $p < 0.10$, 0.05, and 0.01, on every slope coefficient's heteroskedasticity-robust (HC3) p -value; the diagnostic rows report, and are labeled by, their own schemes (HC3, Bell-McCaffrey, wild bootstrap- t , Freedman-Lane permutation), for the tavern coefficient of each column; the effective degrees of freedom are Bell-McCaffrey's. The variance inflation factor (VIF) is centered and reported for the tavern employment share; the specification maximum falls on the Protestant share (1.03 in column 3) and on the producer indicator (2.15 in column 4, 3.55 in column 5, 2.24 in column 6, 5.67 in column 7); column 1 has a single regressor, and in column 2 the two regressors' VIFs coincide at 1.00. Condition numbers are Belsley-Kuh-Welsch, constant included. Leave-one-out on the column-(4) tavern coefficient: range [-4.64, -3.36], 0 sign changes (Appendix Table A.17 gives the anatomy); on the column-(7) broad-measure coefficient: range [-4.30, -2.59], 0 sign changes.

Table A.15: Tavern employment and the 1908 ban vote (cascade)

	Class 301 (narrow)				
	Tavern (1)	+ French (2)	+ Protestant (3)	+ Producer (4)	+ Density (5)
Tavern employment share	-2.296 (3.593)	-1.875 (1.858)	-1.731 (1.532)	-1.509 (1.546)	-1.377 (1.525)
French share		-0.239*** (0.072)	-0.230*** (0.070)	-0.194*** (0.062)	-0.251** (0.113)
Protestant share			-0.056 (0.055)	-0.042 (0.060)	-0.017 (0.068)
Absinthe (attested producer, = 1)				-4.855 (6.177)	3.190 (13.216)
Log population density					-3.330 (3.169)
Constant	69.152 (7.112)	72.214 (4.387)	74.179 (4.299)	73.304 (4.491)	87.148 (13.338)
Standardized β:					
Tavern employment share	-0.199	-0.162	-0.150	-0.131	-0.119
R^2	0.039	0.448	0.476	0.487	0.535
Adjusted R^2	-0.002	0.398	0.401	0.385	0.412
AIC	196	184	185	186	186
BIC	198	188	190	192	193
Claim coefficient (tavern employment share):					
HC3 robust p	0.529	0.324	0.271	0.341	0.378
Bell-McCaffrey p	0.485	0.291	0.249	0.306	0.323
Wild bootstrap- t p	0.541	0.188	0.138	0.165	0.207
Freedman-Lane p	0.495	0.305	0.251	0.290	0.318
Effective df	3.70	3.78	3.37	3.60	3.58
BKW condition no. (with constant)	4.45	4.85	5.69	6.43	18.24
VIF (tavern employment share)	1.00	1.00	1.01	1.04	1.04
N (cantons)	25	25	25	25	25

Notes: Ordinary least squares; $N = 25$ cantons in every column. Outcome is the 1908 yes-share, Swissvotes #68, in percent. Columns (1)–(5) enter controls cumulatively with population density last. The tavern measure is class 301 of the federal business census of 9 August 1905 — restaurants and taverns not attached to an inn (BFS, 1908a). Density and, where applicable, other harvest-year inputs are 1907 vintage. *, **, and *** mark $p < 0.10$, 0.05, and 0.01, on every slope coefficient's heteroskedasticity-robust (HC3) p -value; the diagnostic rows report, and are labeled by, their own schemes (HC3, Bell-McCaffrey, wild bootstrap- t , Freedman-Lane permutation), for the tavern coefficient of each column; the effective degrees of freedom are Bell-McCaffrey's. The variance inflation factor (VIF) is centered and reported for the tavern employment share; the specification maximum falls on the Protestant share (1.03 in column 3) and on the producer indicator (2.15 in column 4, 3.56 in column 5); column 1 has a single regressor, and in column 2 the two regressors' VIFs coincide at 1.00. Condition numbers are Belsley-Kuh-Welsch, constant included. Leave-one-out on the column-(4) tavern coefficient: range [-2.36, -0.82], 0 sign changes.

A.8.1 Tavern and Wine Identification Weights

Table A.16: Identification weights for the tavern and wine coefficients

Canton	Identification weight (%)	
	Tavern (petition arm)	Wine revenue (vote arm)
TI	41.9	4.0
GE	16.9	9.5
NW	7.0	0.0
OW	5.1	0.0
LU	4.7	0.0
AR	4.6	1.8
BS	3.4	0.0
FR	2.8	11.6
TG	2.5	0.1
BE	2.0	1.9
AI	1.9	0.0
VD	1.8	36.6
ZG	1.6	0.0
AG	1.3	0.1
NE	0.8	7.6
SZ	0.7	0.0
ZH	0.3	13.2
SO	0.2	0.3
SG	0.1	0.0
GL	0.1	1.3
BL	0.1	0.6
UR	0.0	0.0
VS	0.0	10.7
SH	0.0	0.3
GR	0.0	0.2

Identification weights: for each coefficient, the weight of a canton is the squared Frisch–Waugh–Lovell residual of the focal regressor on the other controls in its own arm’s column-(4) architecture (tavern employment share: petition arm; wine revenue share, 1907 vintage: vote arm), normalized to sum to 100 — the object underlying an added-variable plot ([Belsley et al., 1980](#)), tabulated rather than plotted. All 25 cantons are shown, sorted by the tavern-arm weight, descending.

A.8.2 Leverage and Influence Anatomy of the Tavern Coefficient

Table A.17: Leverage and influence anatomy of the tavern coefficient, Table 4 tavern column: where the influence sits, and survival under single-canton deletion

Canton	Tavern emp. share	Producer (attested)	French share	h_{ii} (leverage)	DFBETA (influence, tavern)	Drop-one β (p)
TI	5.18	–	0.3	0.527	+0.504	-4.643 (0.035)
NW	0.75	–	0.2	0.179	-0.375	-3.355 (0.030)
AR	3.09	–	0.1	0.203	-0.337	-3.414 (0.033)
ZG	1.41	–	0.6	0.109	+0.258	-4.248 (0.014)
GE	4.42	Yes	82.8	0.463	+0.246	-4.261 (0.083)
FR	0.98	–	68.3	0.363	-0.240	-3.547 (0.032)
LU	0.98	–	0.5	0.136	+0.142	-4.105 (0.026)
TG	2.81	–	0.3	0.111	-0.137	-3.692 (0.031)
GL	2.21	–	0.2	0.106	+0.101	-4.019 (0.019)
ZH	2.33	–	0.9	0.126	-0.075	-3.786 (0.030)
BS	1.83	Yes	2.3	0.585	+0.063	-3.988 (0.028)
BE	1.31	–	16.6	0.176	+0.055	-3.976 (0.031)
AI	2.71	–	0.1	0.115	+0.050	-3.969 (0.035)
VD	1.78	Yes	86.5	0.323	+0.043	-3.958 (0.026)
AG	1.48	–	0.4	0.076	-0.039	-3.835 (0.025)
SO	1.83	–	1.9	0.054	-0.024	-3.858 (0.026)
SZ	2.45	–	0.5	0.110	+0.018	-3.921 (0.030)
OW	0.94	–	0.2	0.159	+0.017	-3.920 (0.033)
VS	1.75	–	65.2	0.328	+0.015	-3.916 (0.017)
SH	2.01	–	0.6	0.126	-0.015	-3.873 (0.019)
NE	1.99	Yes	82.8	0.300	-0.005	-3.887 (0.025)
BL	1.91	–	0.9	0.111	-0.005	-3.887 (0.024)
SG	2.21	–	0.3	0.053	+0.001	-3.896 (0.025)
UR	1.94	–	0.1	0.102	-0.001	-3.892 (0.025)
GR	2.04	–	0.5	0.059	+0.000	-3.894 (0.024)
Full sample (N=25)	2.09	4	16.5	5.00	–	-3.894 (0.024)

Petition signatures per 100 eligible voters on the class-301 tavern employment share, the tavern column of Table 4 and column (4) of Table A.14 (+ French share + Protestant share + Absinthe producer (attested, binary)). No canton is pinned: this coefficient has no ratified stress case, so all 25 cantons are sorted by absolute DFBETA on the tavern coefficient. h_{ii} = OLS leverage, the canton's weight in its own fitted value (leverages sum to $k = 5$; the maximum-leverage canton on this design is Basel-Stadt). DFBETA = the canton's influence on the tavern coefficient (BKW, dropped-one RMSE scaled): a POSITIVE value means the canton pulls the coefficient UP, so deleting it LOWERS β – read each row against the drop-one column beside it. drop-one beta (p) re-estimates the tavern coefficient omitting that canton; the parenthetical is the Bell–McCaffrey two-sided p (HC2 with Satterthwaite effective degrees of freedom), $N=24$ for the canton rows and $N=25$ for the full sample. The concentration of the tavern coefficient's identification weight in Ticino and Geneva (Table A.16), amplified by their leverage, is why its Bell–McCaffrey effective degrees of freedom are 3.60. The final row reports full-sample aggregates, each appropriate to its column: tavern employment share and French share are sample MEANS, attested producers are a COUNT (4 cantons), and leverages SUM to 5.00 (= k , the identity noted above); DFBETA has no full-sample analog and is left blank. The tavern coefficient stays negative under every single-canton deletion, from -4.643 (drop-Ticino) to -3.355 (drop-Nidwalden).

A.9 Secondary Outcomes – Referendum #65

A.9.1 The Precursor Vote – Referendum #52

The precursor to Referendum #52 was the Federal Decree of 26 March 1897 adding Article 69bis to the Federal Constitution of 1874, and the people and the cantons accepted it on 11 July 1897 ([Schweizerischer Bundesrat, 1897b](#), pp. 89–91). The article empowered the Confederation to legislate on the trade in foodstuffs and stimulants, and on the trade in other articles of use and consumption so far as these may endanger life or health ([Schweizerische Bundesversammlung, 1897](#), p. 233). The health qualifier attached to the second of those alone, so the competence over foodstuffs, and therefore over wine, as the main text sets out, did not depend on a danger to health at all. The message that carried the article opened with the Federal Assembly's instruction of 1889 to draft a federal law on the wine trade ([Schweizerischer Bundesrat, 1895a](#), p. 767), said that wines were very often adulterated with artificial wine ([Schweizerischer Bundesrat, 1895a](#), p. 776), called the extension of frontier control to butter, edible fats, oils and wines, articles often imported adulterated, highly desirable ([Schweizerischer Bundesrat, 1895a](#), p. 805), and relayed the food chemists' expectation of protection for wine-growing, butter producers and beekeepers ([Schweizerischer Bundesrat, 1895a](#), pp. 794–795). Two further paragraphs assigned the work: execution fell to the cantons, under federal supervision and with federal money, and control of imports at the national frontier rested with the Confederation ([Schweizerische Bundesversammlung, 1897](#), p. 234). Neither stipulation existed in the Federal Council's earlier March 8th, 1895 draft, which carried the two grants of competence and nothing further of the article ([Schweizerischer Bundesrat, 1895a](#), p. 808). Parliament added the new stipulations, which the draft validation decree of August 1897 restates ([Schweizerischer Bundesrat, 1897b](#), p. 91). In the recorded deliberations on the Food Act, the Farmers' Union declared frontier control the cardinal point of the law. In doing so, it asked for the use of a power the constitution had carried since July 1897 and not for a new one.

Four cantons voted against, Glarus, Fribourg, Valais and the half-canton Appenzell Innerrhoden, and of those four only Valais grows wine in quantity; Vaud, Neuchâtel and Ticino accepted, and so did Geneva, by a single vote ([Schweizerischer Bundesrat, 1897b](#), p. 83). The rejection was not a wine bloc. The report records no complaint about the vote ([Schweizerischer Bundesrat, 1897b](#), p. 83). The same electorate voted the same day on the forest-police article, referendum #51, and the two ballots did not divide the country the same way: Lucerne, Nidwalden, St. Gallen and Graubünden rejected the forest article and accepted the food article, Glarus did the reverse, and only Fribourg, Valais and Appenzell

Table A.18: Referendum #65 (Food Act 1906): wine-coalition specification ladder, wine revenue share, harvest 1905

	Wine (1)	+ Cattle (2)	+ Regime (3)	+ French (4)	+ Protestant (5)	+ Density (6)
Wine revenue share, harvest 1905	0.551 (0.388)	0.908*** (0.191)	0.916*** (0.221)	1.344** (0.602)	0.979*** (0.332)	0.929** (0.386)
Cattle per 1,000 population (1901)		0.067*** (0.017)	0.065*** (0.017)	0.064*** (0.021)	0.054** (0.026)	0.067 (0.046)
Existing food-control regime, 1895 (=1)			−2.516 (6.792)	−0.456 (7.046)	−2.026 (7.790)	−2.592 (7.830)
French share				−0.182 (0.232)		
Protestant share					−0.112 (0.183)	
Log population density						0.552 (9.090)
Constant	59.83 (4.87)	28.52 (10.12)	30.78 (10.70)	31.31 (12.13)	39.88 (19.00)	27.19 (63.59)
R^2	0.04	0.46	0.47	0.51	0.49	0.47
Adjusted R^2	−0.00	0.41	0.39	0.41	0.38	0.36
AIC	226	214	216	216	217	218
BIC	229	218	221	222	223	224
N	25	25	25	25	25	25
Small-sample diagnostics for the wine coefficient						
HC3 robust p	0.169	<0.001	<0.001	0.037	0.008	0.026
Bell–McCaffrey p	0.225	0.050	0.057	0.064	0.075	0.116
Wild bootstrap- t p (HC3-studentized)	0.371	0.020	0.020	0.053	0.051	0.094
Freedman–Lane p	0.161	0.006	0.010	0.037	0.027	0.042
Bell–McCaffrey effective D.O.F.	1.85	1.70	1.72	3.06	1.82	1.89

Notes: Ordinary least squares; $N = 25$ cantons. Dependent variable: the canton yes-share, in percent, in the referendum of 10 June 1906 on the Federal Act on trade in foodstuffs (Swissvotes #65). Wine: the canton's share of the national wine revenue total, harvest 1905, in percent. Cattle: head per 1,000 population — the 1901 federal-census stocks (HSSO Table I.9) over the 1900 population. The regime indicator marks the fifteen cantons that, in the Federal Council's survey of 1895, had organized food control by an enactment of their own and set up a central examination office (BBl 1895 I, printed 782–784); the other ten had no cantonal laboratory. Cantonal adoptions between 1896 and 1906 are unobserved, so misclassification attenuates the indicator. HC3 robust standard errors in parentheses. Stars use two-sided HC3 p : * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The diagnostic rows read the wine coefficient only and are labeled by their own schemes; the Freedman–Lane and wild bootstrap- t p -values are R-engine computations at 10^6 draws, seed 20260630, and the wine regressor is never permuted. Column (3) is the coalition specification the text refers to.

Innerrhoden rejected both ([Schweizerischer Bundesrat, 1897b](#), pp. 82–85). What separates the two votes is the measure and not the canton’s attitude to federal power, which is why referendum #51 sits in the falsification arm and referendum #52 in the corroborative arm without contradiction. Neuchâtel and Geneva, which accepted the competence in 1897, rejected the act that used it by wide margins in 1906 ([Schweizerischer Bundesrat, 1906](#), p. 121) and were the only two cantons to reject the absinthe ban in 1908 ([Schweizerische Bundeskanzlei, 1908b](#)).

What is known about the campaign is secondary and rests on Bolliger’s reading of the newspapers of that week. He reports that the new federal competence stirred feeling only a little; that the Free Democrats and the Grütli Association recommended acceptance; that a joint appeal was signed by the leaderships of the Swiss Trade and Industry Association, the Swiss Trades Association, the Swiss Agricultural Association and branch bodies of bakers and confectioners, master butchers, brewers, tavern-keepers and chemists; and that the only organized opposition he names was a conservative committee in Geneva, which charged the article with a “protectionist aftertaste” ([Bolliger, 2010](#), pp. 2–3). Agriculture stood in that appeal as one body. The Swiss Agricultural Association was the federation of the cantonal agricultural societies of German-speaking Switzerland, founded in 1863 and recognized and subsidized by the Confederation since 1884 ([Baumann and Moser, 2001](#)); it spoke for the farm interests together, wine-growing among them, it belonged to the Farmers’ Union from that body’s founding in 1897 ([Baumann, 2006b](#)), and in the deliberations on the Food Act its delegates’ assembly pressed beside the Farmers’ League for stricter control at the frontier ([Schweizerische Bundesversammlung, 1905a](#), p. 277).

We estimate referendum #52 on the volume basis alone, because no wine revenue series exists for 1897, and on 24 cantons, because the wine series for the 1896 harvest lacks Thurgau. The gap is in the wine series and not in the vote: the result report returns Thurgau in full, with a cantonal yes ([Schweizerischer Bundesrat, 1897b](#), p. 83). The wine coefficient is 0.550, and on a sample this size we do not rule out imprecision; what the two readings establish is that no wine relationship is detectable at conventional levels, on the Bell–McCaffrey reading or on the Freedman–Lane permutation reading. Table [A.13](#) sets out that coefficient with its two standard errors, the robust and the Bell–McCaffrey, the Bell–McCaffrey p and the effective degrees of freedom behind it; Table [A.12](#) sets out the same coefficient with the permutation reading and the average marginal effect, so that the thinner coverage can be read off the tables rather than taken on our word. Of the corroborative votes we read referendum #52 as the nearest, on its volume basis, to the line referendum #65 reaches on the revenue basis, and it does not reach that line. Referendum #65’s own coalition specification is in the back, in Table [A.18](#) of this appendix;

the main text quotes the specification the set shares and points here for the rest.

We report the precursor here rather than in the main text because its coverage is thinner than the rest of the set's.

A.10 Additional Petition Results

Table A.19: Descriptive statistics: the petition arm at its 1906 vintage

Variable	All cantons (N = 25)					German-plurality (N = 19)		French-plurality (N = 5)	
	Mean	SD	Median	Min	Max	Mean	SD	Mean	SD
Petition signatures (per 100 eligible)	20.5	9.5	17.5	3.4	41.3	20.4	9.1	23.9	9.4
Signatures submitted	6775	8500	4022	286	38337	6327	9001	9558	7135
Eligible voters, June 1906	31391	31682	24888	3021	134354	29386	35327	37318	18047
Wine revenue share, 1906	4.0	8.5	0.9	0.0	40.8	1.4	2.8	13.9	15.7
Wine volume share, 1906	4.0	7.6	0.9	0.0	34.4	1.4	3.0	13.5	12.7
Attested absinthe producer (0/1)	0.16	0.37	0.00	0.00	1.00	0.05	0.23	0.60	0.55
Absinthe-sector exposure, 1902–06	4.0	13.1	0.0	0.0	63.0	0.3	1.4	18.8	26.1
French share, 1900	16.5	31.3	0.5	0.1	86.5	1.4	3.7	77.1	9.7
Protestant share, 1900	43.1	35.0	48.2	1.3	90.2	44.1	34.4	47.5	39.3
Tavern employment share, 1905	2.09	1.01	1.94	0.75	5.18	1.91	0.64	2.18	1.31
Tavern and inn employment share, 1905	3.99	1.57	3.49	1.48	6.82	3.93	1.51	3.74	1.69
Population density, 1906 (per km ²)	260	671	106	15	3430	287	764	199	230
Log population density, 1906	4.66	1.12	4.67	2.73	8.14	4.67	1.15	4.76	1.19
Alkoholzehntel per capita, 1904 (Fr.)	0.30	0.47	0.19	0.15	2.56	0.33	0.54	0.19	0.01

Notes: N = 25; columns as in Table 1. Wine is the 1906 harvest and population the 1906 yearbook estimate over the fixed 1900 land area (the petition drive ran 1906–07; the 1907 harvest is post-outcome). Signatures are the submitted counts of the Federal Council's report of 22 February 1907 (169,377 nationally); eligible voters are the roll of the 10 June 1906 federal vote. The tavern employment shares are within-canton percentages of male employment, net of cottage industry, in class 301 of the federal business census of 9 August 1905 — restaurants and taverns not attached to an inn — and in classes 300+301, which add inns, hotels and boarding houses (BFS, 1908a). The Alkoholzehntel is 1904 spending per 1906 resident, shown for all 25 cantons; Table A.20 excludes Schaffhausen and Vaud.

Table A.20: Tithe-measure robustness (petition arm)

	Raw	Drop SH, VD	Genuine	Temperance
Alcohol-tithe per capita	0.270 [0.053]	−0.052 [0.808]	−0.072 [0.651]	−0.046 [0.708]
Protestant share	0.609 [0.025]	0.636 [0.023]	0.666 [0.031]	0.651 [0.016]
<i>N</i> (cantons)	25	23	23	23

Notes: Ordinary least squares. Cells are standardized coefficients with Freedman–Lane permutation *p*-values in brackets. Columns vary the tithe-based temperance measure: raw Alkoholzehntel spending per capita; the same excluding Schaffhausen and Vaud; the genuine-share measure; and temperance-rubric spending per capita. Schaffhausen and Vaud are excluded from tithe-only specifications on different grounds: Schaffhausen’s recorded spending is inflated by welfare accounting unrelated to temperance effort, while Vaud’s entire tithe was allocated to a single child-welfare institution, so the measure does not capture temperance effort there. Neither canton is dropped as an outlier. Column 1 retains all 25 cantons and is the pre-drop disclosure arm; columns 2–4 are estimated on the 23-canton tithe sample, so the exclusion is applied wherever a tithe quantity enters. The tithe measures are an alternative that does not displace the Protestant-share result.

Table A.21 is estimated on 23 cantons. Schaffhausen and Vaud are excluded from every column because the tithe measure is contaminated for each, on grounds that differ by canton: Schaffhausen’s recorded spending is inflated by welfare accounting unrelated to temperance effort, and Vaud’s entire tithe was allocated to a single child-welfare institution. Neither canton is dropped as an outlier. The columns without the tithe term are re-estimated on the 23-canton sample only so that each pair is a within-sample contrast; they are not the paper’s estimates, which appear at $N = 25$ in Table 4. The pairs follow that table’s order: the petition’s focal and density-in specifications, then the vote’s focal, share and producer forms. The tavern column has no tithe counterpart and remains in Table 4. The three inference schemes are reported side by side and are not compared with one another; the wild bootstrap-*t* is not reported, because its engine carries no 23-canton cells. Alternative tithe measures are in Table A.20.

Table A.21: Tithe pairs on a common sample (petition and vote arms, $N = 23$)

	Pet. focal (4)		Pet. density (5)		Vote focal (4)		Vote share (7)		Vote producer (5)	
	Without	With tithe	Without	With tithe	Without	With tithe	Without	With tithe	Without	With tithe
Wine revenue share	-1.015** (0.418)	-1.009** (0.438)	-1.006* (0.524)	-1.002* (0.539)	0.521 (0.461)	0.511 (0.454)	0.433 (0.586)	0.396 (0.569)	0.547 (0.729)	0.534 (0.724)
French share	0.164* (0.078)	0.165* (0.080)	0.154 (0.128)	0.156 (0.128)	-0.270* (0.132)	-0.270* (0.133)	-0.232* (0.116)	-0.226* (0.115)	-0.310 (0.256)	-0.312 (0.261)
Protestant share	0.162** (0.066)	0.171** (0.080)	0.167** (0.077)	0.175* (0.090)	-0.088 (0.052)	-0.095 (0.063)	-0.047 (0.072)	-0.061 (0.077)	-0.066 (0.076)	-0.075 (0.082)
Log density			-0.653 (3.124)	-0.580 (3.149)			-2.006 (2.944)	-2.219 (3.066)	-2.636 (4.724)	-2.729 (4.966)
Absinthe	-5.430 (6.779)	-5.103 (6.767)	-3.685 (11.360)	-3.579 (11.417)	-3.948 (9.942)	-4.259 (10.038)	-0.174 (0.242)	-0.208 (0.309)	3.167 (22.553)	2.964 (23.234)
Genuine-share tithe (fr. per capita)		-38.569 (96.640)		-35.398 (97.393)		31.870 (109.747)		84.452 (116.433)		46.481 (137.950)
Constant	13.724 (2.386)	14.248 (3.043)	16.434 (13.129)	16.612 (13.487)	70.534 (3.033)	70.110 (3.582)	78.096 (12.269)	77.732 (12.652)	81.485 (19.035)	81.252 (19.582)
Standardized β:										
Wine revenue share	-0.444	-0.441	-0.440	-0.438	0.218	0.214	0.181	0.166	0.229	0.223
Protestant share	0.617	0.651	0.638	0.667	-0.256	-0.277	-0.136	-0.178	-0.191	-0.219
R^2	0.44	0.44	0.44	0.44	0.58	0.58	0.62	0.63	0.61	0.61
Adjusted R^2	0.31	0.28	0.27	0.23	0.48	0.46	0.51	0.49	0.49	0.46
Claim coefficient (Protestant share):										
HC3 robust p	0.025	0.048	0.044	0.069	0.109	0.149	0.524	0.438	0.402	0.371
Bell-McCaffrey p	0.021	0.048	0.030	0.055	0.088	0.128	0.475	0.379	0.301	0.279
Freedman-Lane p	0.015	0.030	0.021	0.035	0.079	0.113	0.471	0.365	0.319	0.284
Effective df	11.18	7.01	10.88	7.39	11.27	7.01	9.83	7.77	11.08	7.42
N (cantons)	23	23	23	23	23	23	23	23	23	23

Notes: Ordinary least squares on the 23 cantons with a usable tithe measure; Schaffhausen and Vaud are excluded throughout (see the text below the table). Each column pair contrasts a specification without the tithe term, re-estimated on the same 23 cantons, with one that adds it; the paper's estimates without the tithe term, at $N = 25$, are in Table 4, whose column order the pairs follow. Stars mark $p < 0.10$, $p < 0.05$ and $p < 0.01$ on HC3 p -values. The diagnostic rows report the Protestant-share coefficient under HC3, Bell-McCaffrey and Freedman-Lane (permutation p -values at 10^6 draws).

A.10.1 Petition-Arm Collinearity and Wine Readings

Table A.22: Design-matrix diagnostics and the wine revenue share's readings, petition arm

	Petition arm focal (4)	Petition arm density-in (5)	Tavern focal
Design matrix:			
BKW condition no. (with constant)	4.66	18.23	6.43
VIF (claim coefficient)	1.25	1.45	1.04
VIF (wine revenue share)	1.78	1.96	
VIF, specification maximum	2.43	3.70	2.15
Regressor with the maximum	French share	Producer indicator	Producer indicator
Wine revenue share (not the claim coefficient):			
HC3 robust p	0.823	0.763	
Bell–McCaffrey p	0.648	0.545	
Wild bootstrap- t p	0.602	0.498	
Freedman–Lane p	0.697	0.608	
Bell–McCaffrey effective df	2.50	2.91	

Notes: Diagnostics for three columns of Table 4, the petition-arm focal and density-in columns and the tavern column; ordinary least squares, $N = 25$ cantons in every column. The claim coefficient is the Protestant share in the two petition columns and the tavern employment share in the tavern column. The variance inflation factor (VIF) is centered and reported for the claim coefficient and for the wine revenue share; the specification maximum and the regressor that owns it are given for each column. Condition numbers are Belsley–Kuh–Welsch, constant included. The lower block reports the wine revenue share, which is not the claim coefficient in any column of Table 4; the diagnostic rows report, and are labeled by, their own schemes (HC3, Bell–McCaffrey, wild bootstrap- t , Freedman–Lane permutation). The producer indicator is the Absinthe row of Table 4. The tavern column carries no wine regressor.